

Master Thesis

Self-Organization Patterns in LLM-based Multi-Agent Systems

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Abstract

Multi-agent debate is promoted as a way to make language models reason better together, but it is evaluated almost entirely by final accuracy, which reveals nothing about how a group arrives at its answer or whether it is deliberating soundly. We study cooperative LLM debate as a dynamical system and define SOMA (Self-Organization Motif Analysis), a suite of detectors for four self-organization motifs, self-reinforcement, limit cycles, bistability, and dominance, each measured from interaction logs applied to 21,000 debates across two benchmarks. Their prevalence varies sharply: self-reinforcement is universal, bistability common, dominance behaviourally present, and genuine limit cycles rare and confined to one topology, an informative pattern in itself.

Our central result is that the four motifs are faces of one mechanism: the debate predominantly amplifies the vote distribution it starts with rather than reasoning a new answer into existence. Self-reinforcement is universal and drives convergence, yet is blind to correctness, strengthening whichever option leads at the outset. A causal seeding experiment on GPQA confirms the amplifier reading: inserting a single correct agent into an otherwise-wrong opening raises accuracy from 1.8% to 23.7%, while the system almost never recovers a correct answer that no agent initially held. The structural design choices we varied, communication topology and memory window, change convergence speed, worst-case runtime, and influence concentration, but not accuracy, which is largely fixed before deliberation begins. Within that envelope, analysing the emergent influence topology within FC runs reveals a complementary lever: effective star formation, where one agent concentrates the majority of conversions, is rare but achieves markedly above-average accuracy because the hub disproportionately started correct; maximally diverse initialisation raises the probability of this emergence from under 1% to 23%. The internal signals a user might trust are misleading: confidence rises regardless of correctness, and early unanimity on biased tasks predicts wrong answers. A persistent adversarial agent exploits the same dynamics in reverse: a single devil’s advocate reduces accuracy from 63.7% to 10.0% even when outnumbered two-to-one by correct agents, because the amplifier cannot distinguish high confidence from adversarial persistence.

Beyond detecting the motifs, we show that each detector, computed from ordinary logs, can serve as a diagnostic of deliberative health, and we distil the findings into design principles graded by evidential strength, as groundwork for a diagnostic suite. All claims are bounded to the tested setting, a homogeneous, cooperative, 4-agent debate on one model family.

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1 Introduction

1.1 Motivation

Large language models (LLMs) are increasingly deployed not as isolated predictors but as interacting agents that reason, argue, and decide in groups. A prominent instance of this shift is *multi-agent debate* (MAD), in which several LLM instances independently propose answers, exchange their reasoning over multiple rounds, and revise their positions before the group commits to a final decision (Du et al., 2024). Across factual and mathematical reasoning tasks, this simple protocol has been reported to improve accuracy and reduce hallucination relative to a single model answering alone (Du et al., 2024, Liang et al., 2024, Chen, Saha and Bansal, 2024), and it now underpins a growing family of collaborative and evaluative systems (Chan et al., 2024). The appeal is intuitive: independent agents make partially independent errors, and deliberation offers a mechanism for the group to identify and correct them.

This optimistic reading, however, rests almost entirely on *outcome* measurements. The literature evaluates multi-agent systems predominantly by final task performance, treating the deliberation itself as an opaque process that either does or does not yield a better answer. Recent critical reassessments show that this outcome-centric view is fragile: debate frequently fails to outperform single-model baselines, collapses under controlled comparison with simpler alternatives, and exhibits failure rates between 41% and 86.7% across seven frameworks (Zhang et al., 2025, Choi et al., 2025, Bertalaníć and Fortuna, 2026, Cemri et al., 2025). More troublingly, the failures that do occur are not random noise but exhibit structure. When two LLMs debate, both tend to grow *more* confident that they have won as the exchange proceeds, with confidence escalating away from a rational baseline even when no new supporting evidence is introduced (Prasad and Nguyen, 2025). Under distributed information, groups of LLM agents converge prematurely on the evidence they already share and fail to surface the private knowledge needed for the correct answer, so that collective accuracy collapses far below what a single agent achieves with complete information (Li et al., 2025). Purely cooperative framings can further collapse the debate into a mutually agreeing echo chamber, a failure mode termed Degeneration-of-Thought (Liang et al., 2024).

What these pathologies have in common is that they are *dynamical*. Rising confidence, premature lock-in, and conformity cascades are properties of how positions evolve and influence one another across rounds, not of any single agent’s competence at a single point in time. An outcome metric, by construction, cannot see them: two systems can reach the same final accuracy through entirely different trajectories, one through genuine error correction and the other through a confident slide into a shared mistake. Understanding *when* and *why* multi-agent deliberation succeeds therefore requires a vocabulary for the interaction dynamics themselves. Miehl et al. (2025) make precisely this argument at a conceptual level, contending that agentic AI must be studied through a systems-theoretic lens because emergent group-level behaviour cannot be predicted from individual agent properties alone. Their position, however, remains programmatic and offers no empirical

instrument for detecting such behaviour in practice.

Complex-systems theory supplies exactly such an instrument. Systems of many interacting units that produce ordered collective behaviour without central control, the defining feature of self-organization (Ashby, 1947), are known to fall into a small number of recurring dynamical regimes. These regimes have precise names and precise signatures: *self-reinforcement*, the positive feedback by which a state amplifies its own persistence (Arthur, 1989); *limit cycles*, self-sustained periodic oscillations that never settle (Strogatz, 2015); *bistability*, the coexistence of multiple stable outcomes selected by initial conditions (Feudel, 2008); and *dominance*, the concentration of influence onto a single node. Such patterns are studied as *motifs*, recurring structural or dynamical building blocks whose presence in a system is both detectable and diagnostic of its behaviour (Milo et al., 2002, Alon, 2007, Driscoll et al., 2024). The pathologies documented in the MAD literature read naturally as instances of this vocabulary: escalating confidence is a self-reinforcement signature, premature consensus is a fixed-point attractor reached through basin selection, and herding to a confident leader is a dominance motif. The connection has begun to be made. Ashery et al. (2025) demonstrate the spontaneous emergence of universally adopted social conventions in decentralised LLM populations, show that strong collective biases arise even when individual agents exhibit none, and find that committed adversarial minorities can impose alternative conventions on the majority. Okawa (2026) derive a formal model predicting that once conformity pressure among agents exceeds a critical threshold, the group is pulled toward a shared bias regardless of what any individual agent believes. What these treatments share is a substrate in which no option is better than any other: conventions, arbitrary labels, or non-empirical propositions. They can show that a group locks in, but not whether it locks in on the right answer, because there is no right answer. And none of them measure whether the detected patterns exceed what chance alignment in a small discrete state space would produce, because no principled null is needed when correctness is not defined. Both gaps remain open, and Mehdizadeh and Hilbert (2026) name them explicitly as a critical open question.

This thesis addresses that gap. We construct a controlled cooperative multi-agent debate system in which homogeneous LLM agents deliberate over multiple rounds on tasks with verifiable ground truth, and we develop a suite of detectors, each grounded in the complex-systems and opinion-dynamics literature, that identify self-reinforcement, limit cycles, bistability, and dominance in the resulting interaction logs. We run this system across a controlled grid of communication topologies and memory configurations on two complementary benchmarks: GPQA-Diamond (Rein et al., 2024), a graduate-level science question-answering benchmark where agents typically start with reasonable priors, and HiddenBench (Li et al., 2025), a hidden-profile benchmark constructed so that agents start systematically wrong and success requires genuine information pooling. The contrast between the two isolates whether observed dynamics depend on the presence of a reliable individual signal or persist even when the group must correct a shared bias. Throughout, we treat motif detection not as a competitor to accuracy but as an explanatory layer beneath it: a way to characterise the trajectory that produces an outcome, and thereby to understand which design choices shape the deliberative dynamics of LLM collectives.

This explanatory layer has a practical purpose. As multi-agent systems are deployed in consequential settings, an operator needs to know not only whether a configuration tends to produce correct answers but whether a given deliberation is proceeding soundly or pathologically, whether the group is genuinely correcting an error or confidently converging on a shared mistake, and whether it will terminate at all. Outcome metrics cannot answer these questions. Instruments are beginning to appear from other directions: online monitors based on conditional transfer entropy over execution traces (Anonymous, 2026b) and learned classifiers that predict from interpretable features whether a query warrants debate at all (Fan et al., 2026). What does not yet exist is an instrument grounded in the dynamical vocabulary itself, one that names which regime a deliberation is in rather than only flagging that something is wrong. A central thread of this thesis is therefore that each motif detector, being computed from ordinary interaction logs, doubles as a candidate *diagnostic* of deliberative health. We do not deliver a validated diagnostic tool; we deliver the empirical groundwork for one, a set of measurable signatures, characterised across a controlled grid, together with tiered, evidence-graded design principles that state honestly how far each is supported by our data.

In one sentence, the objective of this thesis is to understand *how* a cooperative LLM debate arrives at its answer, by measuring the self-organization patterns in its trajectory, and to turn that understanding into evidence-graded guidance for building such systems.

1.2 Research Questions

The overarching goal of this thesis is to establish whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and to characterise the conditions under which self-organization patterns emerge. We decompose this goal into three research questions, ordered by the class of evidence each requires.

- RQ1 Measurement.** Can self-organization motifs, specifically self-reinforcement, limit cycles, bistability, and dominance, be detected in cooperative LLM-based multi-agent debate using detectors grounded in the complex-systems and opinion-dynamics literature and each tested against a principled null model? And which of these motifs actually occur, at what prevalence, and with what structure?
- RQ2 Characterisation.** What governs the presence and strength of each motif, and what do they predict about system outcomes? We examine the two canonical structural axes, communication topology and memory window, together with the information regime of the task and the composition of the opening votes, and ask what each motif implies for accuracy, convergence speed, and worst-case runtime.
- RQ3 Causation.** Is the composition of the opening votes causally responsible for the outcome? Can the system generate a correct answer that no agent initially held, or does deliberation function primarily as an amplifier of the initial vote distribution?

One commitment underpins these questions. Every detector is tested against a null model that destroys the pattern we look for and leaves everything else intact. This matters because our state space is small: with four agents and three or four options, votes repeat and influence concentrates by chance alone. The null tells us what chance looks like, so a near-zero prevalence becomes a measurement rather than a failure. We report the motifs we do not find alongside those we do.

1.3 Contributions & Structure

This thesis makes three contributions, one for each research question.

- **An instrument for measuring debate dynamics.** We define a two-phase synchronous debate protocol that separates public messages from private beliefs and confidences, so that the interaction log exposes exactly the quantities motif detection requires (Section 4). On top of it we specify SOMA (Self-Organization Motif Analysis): detectors for self-reinforcement, bistability, dominance, and limit cycles, each grounded in the complex-systems or opinion-dynamics literature and each tested against its own null model (Section 5). Every detector states what it can and cannot claim, distinguishing a measured signature from a certified attractor. Running $R = 50$ repetitions of every task-config cell is what makes across-repetition attractor statistics measurable at all.
- **An empirical characterisation at scale.** We apply SOMA to 30,000 debates across twelve configurations and two deliberately contrasting benchmarks: a refinement regime in which agents mostly start correct, and a correction regime in which they start systematically wrong (Section 6). Self-reinforcement is universal and drives convergence, bistability is common and is a stable property of the task rather than of the configuration, dominance emerges behaviourally even under a symmetric topology, and genuine limit cycles are essentially absent outside the star, where they arise through a hub-driven mechanism. Reading the emergent influence graph within fully-connected runs reveals a lever the structural topology parameter cannot reach: a rare effective star, formed preferentially from diverse opening positions, is markedly more accurate than the fully-connected mean. Throughout, the structural axes we varied move the dynamics but not the accuracy.
- **A causal test of the initial-condition effect.** A seeded-initialisation experiment manipulates only the composition of the opening votes, leaving the debate itself unchanged. Inserting a single correct agent into an otherwise-incorrect opening raises accuracy from 1.8% to 23.7%, and does so in all 34 tasks tested. In the unseeded arm, where no agent starts correct, the system arrives at the correct answer in under 2% of repetitions, and its ability to do so declines as tasks get harder. A devil’s advocate probe confirms the complementary vulnerability under a controlled composition: a single agent that never updates its position outweighs two agents that start correct, collapsing accuracy from 63.7% to 10.0%. Together these

upgrade the central claim from association to intervention: within the setting we tested, deliberation functions predominantly as a selective amplifier of its opening vote distribution (Section 6).

From these results we derive design guidance graded by how strongly our data support each claim, and show that every SOMA detector, computed from ordinary interaction logs, is a candidate observable for monitoring deliberative health (Section 7). We deliver the measurements and the mapping from principle to observable, not a packaged tool.

Structure of this thesis. The thesis follows the arc of the three research questions. *Background* (Section 2) introduces the vocabulary of self-organization and dynamical motifs and fixes what we mean by an LLM agent and a debate; *Related Work* (Section 3) surveys how multi-agent systems are currently analysed and locates the gap this thesis addresses. The instrument is then built in two steps: *System and Experimental Setup* (Section 4) specifies the debate protocol, the two contrasting benchmarks, and the experimental grid, and *Detection Methodology* (Section 5) defines the four SOMA detectors together with the null model each is tested against. *Results* (Section 6) applies the instrument, reporting accuracy and efficiency, the four motifs in turn, the emergent influence topology, and the seeded-initialisation experiment. *Discussion* (Section 7) reads the motifs together as one amplifier mechanism and distils the findings into evidence-graded design principles. *Limitations and Future Work* (Section 8) states the scope of the claims and the agenda they imply, and *Conclusion* (Section 9) returns to the three research questions.

2 Background

This section provides the minimum conceptual background needed to follow the rest of the thesis. Section 2.1 introduces self-organization and the notion of a dynamical motif, which supplies the analytical vocabulary used throughout. Section 2.2 fixes what we mean by an LLM agent. Section 2.3 describes multi-agent debate and its known failure modes, which are the phenomena the motif lens is meant to explain. Formal definitions of the four detectors are deferred to Section 5; here we only establish the ideas.

2.1 Complex Systems and Self-Organization

A *complex system* is a collection of many components whose interactions produce collective behaviour that cannot be read off from the components in isolation. When such a system settles into ordered, coordinated behaviour without any central controller imposing it, the behaviour is called *self-organizing* (Ashby, 1947). The order is a property of the interaction structure, not of any single unit: no agent “decides” the global outcome, yet a global outcome reliably emerges. This is exactly the regime a multi-agent debate occupies. That LLM populations occupy this regime is now established empirically: decentralised populations of LLM agents spontaneously converge on shared conventions through purely local interaction, without central coordination (Ashery et al., 2025), reproducing the classical consensus-emergence phenomenology of naming games (Baronchelli, 2018). Four homogeneous agents, each following the same local rule (read the others, reconsider, vote), collectively produce a group trajectory, consensus, deadlock, or oscillation, that none of them individually selects. The central premise of this thesis is that such trajectories are amenable to the same descriptive tools developed for physical, biological, and social self-organizing systems.

A recurring theme in that literature is that the space of collective behaviours is not arbitrary: interacting systems tend to fall into a small number of qualitatively distinct *regimes*, each with a characteristic signature. It is useful to separate two senses in which a recurring pattern is called a *motif*. A *structural* motif is a pattern in the wiring: a small subgraph, such as a feed-forward loop, that appears in a network more often than chance and that generates characteristic input–output behaviour (Milo et al., 2002, Alon, 2007). A *dynamical* motif is a pattern in the trajectory rather than the wiring: a recurring feature of how the system’s state evolves over time, such as an attractor or a periodic orbit, a notion recently formalised for recurrent neural networks (Driscoll et al., 2024). This thesis is concerned with dynamical motifs: we hold the communication structure fixed as an experimental variable and ask what patterns appear in the temporal evolution of the group’s votes and confidences.

Four dynamical regimes are central to this work, each classical in nonlinear dynamics (Strogatz, 2015). *Self-reinforcement* is positive feedback: a state that, once entered, amplifies its own persistence. The canonical example is lock-in through increasing returns, where an early lead becomes self-perpetuating independently of intrinsic merit (Arthur,

1989). *Limit cycles* are self-sustained periodic oscillations: the system never settles to a fixed point but instead revisits the same states in a repeating order. *Bistability* (more generally multistability) is the coexistence of several stable outcomes for the same system, with the initial condition selecting which one is reached; the set of starting states leading to a given outcome is its *basin of attraction*. In social systems the corresponding question is how large a committed minority must be to move the collective from one basin to another. Centola et al. (2018) establish experimentally that this critical mass is roughly 25% of the population, well below the majority that equilibrium-stability reasoning would require, and that the threshold rises with the length of individuals’ interaction memory. *Dominance* is the concentration of causal influence onto one component, so that a single node disproportionately drives the collective state. These four were chosen because each has a precise, testable signature and because each corresponds directly to a pathology already reported informally in the debate literature (Section 3).

The propagation and concentration of influence, the mechanism shared by self-reinforcement and dominance, has its own formal home in *opinion dynamics*, which studies how agreement and social power propagate through a network of interacting agents. The classical DeGroot model treats each agent’s opinion as a weighted average of its neighbours’ opinions and asks when this process reaches consensus (DeGroot, 1974); later work shows how the same dynamics let social power concentrate on particular nodes across repeated interactions (Jia et al., 2015). Our agents differ from the classical setting in one important respect: they hold a latent confidence, elicited as an integer from 1 to 10, but cast a discrete vote. This continuous-opinion, discrete-action structure is itself a studied variant of opinion dynamics (Martins, 2008), and it is what lets us measure self-reinforcement on the continuous channel (confidence) while detecting cycles and attractors on the discrete channel (votes).

2.2 Large Language Models and Agents

A large language model (LLM) is a neural network, in current practice a Transformer (Vaswani et al., 2017), trained to predict the next token in a sequence and subsequently aligned to follow instructions through supervised fine-tuning and reinforcement learning from human feedback (Ouyang et al., 2022). The result is a model that maps a textual prompt to a distribution over continuations, from which text is sampled. Prompting the model to produce intermediate reasoning steps before an answer, *chain-of-thought* prompting, substantially improves performance on multi-step reasoning tasks (Wei et al., 2022, Kojima et al., 2022), and modern instruction-tuned models such as those in the GPT-4 family are capable of following structured, multi-part instructions of the kind our protocol requires (Achiam et al., 2023).

An *LLM agent*, in the sense used here, is an LLM placed in a loop: it is given a state (the task, its own prior position, and messages from other agents), it produces an action (a written argument and a vote), and the resulting state is fed back to it on the next step. Sampling is stochastic: at a positive temperature the model does not return a single deterministic answer but samples from its output distribution, so two agents given the

identical prompt can produce different responses. This stochasticity is the source of the initial diversity that multi-agent debate relies on, and, as later chapters show, it is also what makes repeated repetitions of the *same* task on the *same* configuration land on different outcomes, the raw material for a bistability analysis. Because we use a single model family throughout, all four agents share the same priors and the same competence; any diversity in the group is a product of sampling, not of heterogeneous models. This is a deliberate design choice (Section 4) that isolates the dynamics of a *homogeneous* collective, and it is also a scope limitation we return to in Section 7.

2.3 Multi-Agent Debate

Multi-agent debate (MAD) is a protocol in which several LLM agents independently propose an answer to a question and then exchange and revise their reasoning over multiple rounds before the group commits to a final decision (Du et al., 2024). The motivating intuition is an error-correction argument: if agents make partially independent mistakes, deliberation gives the group a chance to surface and correct them, much as an ensemble of independent voters can outperform its average member (Grofman et al., 1983). Reported gains on factual and mathematical reasoning (Du et al., 2024, Liang et al., 2024, Chen, Saha and Bansal, 2024), and the use of debate panels as more reliable evaluators (Chan et al., 2024), follow from this idea.

Debate designs vary along a few structural axes that recur throughout this thesis. The *communication topology* fixes who can read whom: a fully-connected group in which every agent sees every other, versus sparser arrangements such as a star, in which spokes communicate only through a central hub. Sparser topologies can match fully-connected debate at lower token cost (Li et al., 2024), so topology is a genuine design decision rather than a cosmetic one. The *memory window* fixes how much interaction history each agent retains from one round to the next. The *decision protocol* fixes how individual positions are aggregated into a group answer, and whether the group is asked to vote or to reach explicit consensus; this choice alone can change which tasks benefit from debate (Kaesberg et al., 2025). We treat topology and memory window as controlled experimental axes and hold the protocol fixed.

Crucially for this thesis, the same interaction that enables error correction also admits systematic failure. A purely cooperative framing can collapse the debate into mutual agreement that entrenches an initial mistake, the *Degeneration-of-Thought* failure mode (Liang et al., 2024). When two models debate, both tend to grow more confident that they are winning as the exchange proceeds, even absent new evidence (Prasad and Nguyen, 2025), and agents frequently abandon a correct answer to conform to a confident peer (Wynn et al., 2025). This behaviour has since been formalised as inter-agent sycophancy (Yao, Shang, Du, He, Lian, Zhang, Su, Swamy and Qi, 2025), the multi-agent extension of the single-agent sycophancy induced by preference-based alignment (Sharma et al., 2023), and has been shown to drive a measurable oracle gap: teams generate the correct answer somewhere in their pool but discard it during consensus formation, with gaps of up to 32.3 percentage points (Bertalaníć and Fortuna, 2026). When the information needed for the

correct answer is distributed across agents rather than shared, groups lock onto the shared evidence and fail to pool the private pieces, so collective accuracy collapses far below what a single fully-informed agent achieves (Li et al., 2025); this is the LLM analogue of the hidden-profile effect long documented in human groups (Stasser and Titus, 1985). These pathologies are not accidents of a particular prompt but properties of *how positions evolve and influence one another over rounds*, that is, they are dynamical. They are precisely the behaviours the motif vocabulary of Section 2.1 is designed to name, and providing an empirical instrument for detecting them is the task of the following chapters.

3 Related Work

3.1 Multi-Agent Debate: Design and Critique

The multi-agent debate (MAD) paradigm was introduced by Du et al. (2024), who let several instances of a language model independently propose answers and then critique and revise them over multiple rounds before committing to a joint response. This “society of minds” setup was reported to improve factual accuracy and mathematical reasoning while reducing hallucination, treating the underlying model as a black box. The core intuition, that partially independent agents make partially independent errors which deliberation can surface and correct, has since motivated a broad family of systems, from confidence-weighted round tables across heterogeneous models (Chen, Saha and Bansal, 2024) to evaluation panels that judge text more reliably than a single evaluator (Chan et al., 2024).

Two design axes recur throughout that family and are the ones we vary. Communication topology fixes who reads whom: sparse structures can match fully-connected debate at substantially lower token cost (Li et al., 2024), and across arbitrary networks the placement of a hub node shifts performance appreciably (Regan et al., 2024). Memory fixes how much interaction history each agent retains. The two were recently crossed in a single factorial by Mehdizadeh and Hilbert (2026), who find no main effect of topology on coordination success but a strong interaction with memory: longer memory slows settling in decentralised networks and accelerates it in centralised ones, so the same parameter pushes the system in opposite directions depending on where an agent sits. Faster settling in the centralised case, they note, means locking into a fragmented plateau rather than reaching agreement. Beyond these two, the decision protocol itself matters, with voting favouring reasoning tasks and consensus favouring knowledge tasks, and adding agents helping where adding rounds can hurt (Kaesberg et al., 2025).

Against this largely optimistic backdrop, a critical literature has emerged. Zhang et al. (2025) find that MAD frequently fails to outperform chain-of-thought (Wei et al., 2022) or self-consistency (Wang et al., 2023) despite far greater compute, and argue that model heterogeneity rather than debate is what reliably helps. Choi et al. (2025) show that majority voting alone accounts for most of the attributed gains, and prove that under their formalisation debate induces a martingale over belief trajectories and so cannot improve expected correctness on its own. Estornell and Liu (2024) supply the mechanism: when agents are similar in capability or in the responses they produce, debate dynamics become static and the procedure converges to the majority opinion, so that where the majority reflects a misconception shared through common training data, debate converges to the misconception.

Two 2026 results push this further and are the closest published statements of what we measure. Oh et al. (2026) decompose the failure into a biased static initial belief and homogenised dynamics that amplify the majority view regardless of correctness. Bertalanič and Fortuna (2026) show that unguided homogeneous debate matched or underperformed

isolated self-correction at several times the token cost, and that a control injecting rationales from unrelated problems performed no worse than genuine peer exchange. They name the resulting loss an *oracle gap*, the distance between the accuracy achievable from the team’s own generation pool and the accuracy the vote actually returns, a quantity we measure directly in Section 6.1.

What unifies the affirmative and the critical strands is that both evaluate debate almost exclusively by its *final outcome*. The deliberative process that produces that outcome, and whether two systems reaching the same accuracy do so through genuinely different dynamics, remains largely outside the scope of these evaluations.

3.2 Interaction Dynamics in LLM Collectives

A distinct and more recent line looks inside the debate at how positions evolve and influence one another. Each of the four motifs we detect has a correspondingly documented pathology, and we take them in turn.

Rising confidence. Prasad and Nguyen (2025) stage repeated adversarial debates and find pervasive miscalibration: models begin overconfident, grow *more* confident that they are winning as the exchange proceeds even absent new supporting evidence, and frequently both claim victory at once, a logical impossibility. That this drift is independent of correctness has since been confirmed directly, with pre-communication confidence a near-worthless predictor of whether an agent is right, and peer exchange further eroding what little discrimination confidence carries (Bertalaníč and Fortuna, 2026). Upward, anti-Bayesian and correctness-blind confidence drift is exactly the kind of positive-feedback signature that motivates a dynamical treatment.

Conformity and influence concentration. Wynn et al. (2025) show that debate can *decrease* accuracy over time even when stronger models outnumber weaker ones, as agents abandon correct answers rather than challenge flawed arguments; they attribute this to sycophancy (Sharma et al., 2023). Cho et al. (2025) find the gap between an agent’s own confidence and its perceived confidence in peers governs how likely it is to conform, and Mehdizadeh and Hilbert (2025b) characterise the dose-response as a sigmoid with a model-specific threshold. Cau et al. (2026) identify an aggregate directional asymmetry they call *agreement drift*, which survives in balanced populations and so cannot be reduced to a majority effect. Where influence concentrates rather than diffusing, outcomes come to depend heavily on the competence of whichever agent occupies the centre (Anonymous, 2026c), a dependency our star topology is designed to expose.

Premature lock-in. Chuang et al. (2025) report that role-playing LLM groups converge far more strongly and rapidly than human groups do, indicating a systematic bias toward premature agreement. A structurally sharper version is captured by HiddenBench (Li et al., 2025), which adapts the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012): when the evidence needed for the correct answer is distributed across agents, collective accuracy collapses far below what a single fully-informed agent achieves, because agents lock onto shared information and

fail to pool their private knowledge. We adopt HiddenBench as our second benchmark for exactly this reason.

The opening distribution. That round-zero answers govern the outcome is stated in several forms. Hu et al. (2025) make it a formal premise, proving debate improvement over majority vote only under the assumption that at least one initial response is generated under the correct concept. Zhu et al. (2026) report that initial answer diversity correlates significantly with final accuracy and propose a diversity-aware initialisation to raise the chance that a correct hypothesis is present at the start. Empirically, Wu et al. (2025) find that an agent starting inside a correct majority is almost perfectly stable while one facing an incorrect majority rarely recovers. In every case the opening distribution is observed as it arises rather than constructed, leaving its effect confounded with task difficulty and agent confidence, which is what paragraph 6.6 is designed to separate.

The vocabulary needed to name these phenomena precisely comes from complex-systems theory. Recurring interaction structures are studied as *network motifs*, small subgraphs appearing more often than in a randomised network and generating characteristic behaviour (Milo et al., 2002, Alon, 2007); the complementary notion of a *dynamical motif*, a recurring pattern in the trajectory such as an attractor or an oscillation, has recently been formalised for recurrent networks (Driscoll et al., 2024). The individual regimes are classical: positive feedback and lock-in (Arthur, 1989), and limit cycles, fixed points, and multi-stability in nonlinear dynamics (Strogatz, 2015). On the influence side, opinion-dynamics models formalise how agreement and social power propagate and when influence concentrates on a single node (DeGroot, 1974, Jia et al., 2015), and the continuous-opinion, discrete-action framework (Martins, 2008) matches agents that hold latent confidence but cast discrete votes. The structural topology a designer imposes is moreover not the influence structure that forms, a distinction long standard in network neuroscience as structural versus effective connectivity (Friston, 2011) and one we take up in Section 6.7.

That this vocabulary applies to LLM populations is established. Ashery et al. (2025) demonstrate the spontaneous emergence of universally adopted conventions in decentralised populations, show that strong collective biases arise even when individual agents exhibit none, and find that committed minorities can impose alternative conventions on the majority, the LLM analogue of the critical-mass threshold measured in human groups (Centola et al., 2018). That it can be made analytic is established too: Okawa (2026) derive a mean-field treatment of debate from spin models, predicting a conformity threshold above which a collective norm emerges, smoothed into a crossover at the small group sizes typical of debate. Neither setting, however, has a correct answer for the collective to converge on, a limitation we return to in Section 3.4. At the level of principle, Miehl et al. (2025) argue that agentic AI must be analysed through a systems-theoretic lens precisely because group-level behaviour cannot be predicted from individual agent properties, but their contribution is conceptual and supplies no empirical instrument.

3.3 Ways of Analysing Multi-Agent Systems

Before stating the gap, it is worth making explicit *why* a complex-systems treatment is the appropriate instrument, by situating it against the alternatives. Each lens answers a different question and carries different costs.

Outcome evaluation. The dominant approach scores a system by final task performance (Du et al., 2024, Zhang et al., 2025, Choi et al., 2025). It is objective, comparable across systems, and measures what practitioners care about. But an outcome is a scalar summary of a trajectory: two systems reaching the same accuracy, one by genuine error correction and the other by a confident slide into a shared mistake, are indistinguishable under it. It says *whether* a configuration worked, not *why*.

Mechanistic interpretability. Opening the models themselves operates at the wrong level of abstraction for our question. Consensus, oscillation, and influence concentration are properties of the *group* trajectory rather than of any single forward pass (Miehling et al., 2025), and internals are inaccessible for the API-served models typical of applied multi-agent systems.

Qualitative transcript analysis. Reading transcripts directly is rich and interpretable, and we draw on it informally when reading the mechanism behind flagged limit-cycle cases (Section 6.5). On its own it is subjective, does not scale to tens of thousands of repetitions, and yields no falsifiable structure testable against a null.

Influence attribution. A fourth approach quantifies directed influence between agents, either information-theoretically through transfer entropy over execution traces (Anonymous, 2026b,d), or by treating influence as a cooperative game and estimating it through agent removal (Anonymous, 2026a, Cui et al., 2025). These yield causal or information-theoretic estimates rather than the adoption-based signature we use, at the price of model access or repeated re-execution; we return to the trade-off in Section 6.4.

Learned diagnostics from logs. A fifth and very recent approach trains a predictor on log-derived features, either to decide whether a query warrants debate at all (Fan et al., 2026) or to decide when a majority vote should be overturned (Anonymous, 2026e). These establish that ordinary logs carry enough signal to support useful decisions. They are, however, predictive rather than explanatory: they learn that a feature configuration precedes failure without naming which dynamical regime the system is in, and they require labelled outcomes to train.

The dynamical lens. The approach taken here treats the debate as a dynamical system and measures recurring patterns in its state trajectory (Strogatz, 2015, Driscoll et al., 2024). Its advantages match the gaps above: it operates at the group level, scales to the full dataset, needs only interaction logs and no labelled outcomes, and gives each motif a falsifiable signature testable against a principled null. Its disadvantages must be stated as plainly. Transferring motif concepts from physical and biological systems to an LLM collective is a modelling choice, not a law; the detection thresholds are partly conventional; a four-agent group is small for some of these measures; and, absent intervention, the detectors identify *signatures* and *associations* rather than certified attractors

or causes. We adopt this lens because it is the only one that can name and measure the pathologies documented above while remaining applicable to black-box models, but we carry its limitations into the interpretation of every result (Section 8.1).

3.4 Research Gap

The literature surveyed above has converged on a diagnosis without converging on an instrument. Outcome-centric evaluation, from the proponents of debate and its critics alike, measures where a deliberation ends and leaves how it got there unexamined. The dynamics literature does examine the trajectory, but documents one pathology at a time, overconfidence here, herding there, premature consensus elsewhere, and relates them to one another through a shared cause such as sycophancy rather than through the formal dynamical regimes they resemble. None of these signatures is tested against a null that says what the same statistic would look like by chance, so a raw detection rate cannot be distinguished from the ordinary regularity of a small, discrete state space.

More consequentially, where the complex-systems vocabulary *has* been applied to LLM collectives, it has been applied on substrates from which truth was deliberately removed. Ashery et al. (2025) study arbitrary conventions where any label is an acceptable equilibrium. Mehdizadeh and Hilbert (2026) verify that their convention symbols carry no intrinsic model preference, precisely so that dynamics are not confounded by correctness. Cau et al. (2026) choose a non-empirical proposition explicitly to avoid convergence driven by factual correctness, and Okawa (2026) uses symbols with no correct answer. This is a principled choice, and it is what makes those studies clean: if the goal is to understand how a group settles on *a* convention, having no better convention available is exactly right. It also leaves them silent on the question a practitioner faces, which is not whether the group locks in but whether it locks in on the right answer. Mehdizadeh and Hilbert (2026) say as much themselves, naming as a critical open question whether their findings survive in epistemic settings where the symmetry between options breaks because some claims are better supported than others.

This thesis takes up that question. The gap it addresses is the absence of an empirical, literature-grounded detection of self-organization motifs in *truth-valued* deliberation, and our response follows the three research questions of Section 1.2. As an *instrument*, we define detectors for self-reinforcement, bistability, dominance, and limit cycles, each grounded in the corresponding literature and each equipped with its own null model, and we run them over a repetition structure that makes across-repetition attractor statistics computable at all; prior work runs a single simulation per configuration (Cau et al., 2026) or analyses run-level means without across-run endpoint distributions (Mehdizadeh and Hilbert, 2026), and so cannot measure multistability even in principle. As a *characterisation*, we apply the suite where every dynamical quantity can be conditioned on correctness, across two deliberately contrasting information regimes. As a test of *causation*, we construct the opening vote composition rather than observing it, separating its effect from the difficulty and confidence it co-varies with elsewhere. And throughout, we

treat the *absence* of a motif as a finding rather than a failed measurement, a stance the outcome-centric literature has no natural place for.

4 System and Experimental Setup

The premise of this thesis is that a multi-agent debate is a dynamical system whose trajectory carries measurable structure (Section 3.3). Turning that premise into empirical claims requires a system built as a *measurement instrument*, not merely as a high-performing debate pipeline. Two design commitments follow. First, the protocol must expose the quantities the motif detectors need, a discrete vote and a latent confidence, per agent and per round, with private belief cleanly separated from public message, so that the interaction log is a faithful record of the dynamics rather than only of the final answer (Section 4.2). Second, the two structural axes most often cited as design levers in the debate literature, communication topology and memory window (Section 3.1), must be exposed as *controlled perturbation axes* so that we can ask which dynamics they cause rather than merely observe. This section specifies that instrument: the system model and notation (Section 4.1), the two-phase protocol and stopping rule, the two deliberately contrasting benchmarks, and the experimental grid that the results of Section 6 are computed over.

4.1 Notation and System Model

We study a fixed group of N language-model agents that repeatedly answer the same multiple-choice question over a sequence of discrete rounds. All agents are instances of a single frozen model f_θ ; they differ only in the private context each one has seen. Because the same symbols recur throughout the detection methodology of Section 5, we fix the notation here and use it consistently thereafter. Table 1 summarises it.

Three units of analysis must be kept apart, since they nest inside one another and are easy to confuse. A *round* is a single synchronous time step t within one debate, at which every agent (re)casts its vote; a debate runs for up to T rounds. A *repetition* (indexed by r) is one complete debate on a given task from round 0 to termination; because sampling is stochastic, the same task run twice yields different repetitions. A *task-config cell* is a single question paired with one configuration (a topology and a memory window), and we execute R repetitions of each cell. In short: many rounds make up one repetition, and many repetitions make up one task-config cell.

Symbol	Meaning
N	number of agents ($N = 4$ throughout)
i, j	agent indices, $i \in \{1, \dots, N\}$
t	round index, $t \in \{0, 1, \dots, T\}$
T	hard cap on the number of rounds
r	repetition index (an independent re-run of a task-config cell)
R	number of repetitions per task-config cell
M	number of answer options a task presents, $M \in \{3, 4\}$
v_i^t	vote of agent i at round t , one of the M options
c_i^t	private confidence of agent i at round t , $c_i^t \in \{1, \dots, 10\}$
priv_i^t	private reasoning of agent i (never transmitted)
pub_i^t	public message of agent i (broadcast to neighbours)
draft_i^t	<i>Argue</i> draft of agent i (defense, challenge, question)
$\mathbf{A}$	adjacency matrix of the communication topology, $A_{ii} = 0$
$\mathcal{N}(i)$	neighbourhood of agent i , $\{j \mid A_{ij} = 1\}$
W	memory window (rounds of history each agent may see)
$\mathcal{H}_i^{\text{self}}(t, W)$	self history window of agent i (own recent states)
$\mathcal{H}_i^{\text{peer}}(t, W)$	peer history window of agent i (neighbours' recent states)
u	early-stopping threshold (consecutive unanimous rounds)
s_t	system macrostate at round t (the vote composition, see below)
f_θ	the shared frozen LLM used as the update function

Table 1: Notation used throughout this thesis. Rounds are indexed by t and repetitions by r ; the two must not be confused.

Communication topology. The interaction structure is a binary adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, where $A_{ij} = 1$ means agent i receives agent j 's messages. Self-loops are excluded ($A_{ii} = 0$), so no agent is fed its own output as a neighbour signal, and the neighbourhood of agent i is $\mathcal{N}(i) = \{j \mid A_{ij} = 1\}$. Any topology can be instantiated by supplying a different matrix without changing the agent or round logic. We use two topologies:

- **Fully connected (fc):** $A_{ij} = 1 - \delta_{ij}$, so every agent observes all others each round. This maximises information flow and serves as the baseline. With $N = 4$ agents, each round involves $N(N - 1) = 12$ messages.
- **Star:** one hub agent is connected to all $N - 1$ spokes, and the spokes are connected only to the hub. The hub is drawn uniformly at random per repetition, independently of agent indices, so that any observed hub effect is structural rather than an artefact of indexing. Each round involves $2(N - 1) = 6$ messages, half the fully-connected cost. Star therefore trades per-round token cost for slower convergence: the hub is the sole information bottleneck, so agreement typically takes more rounds to propagate.

System macrostate. Individual agents are exchangeable: only how many of them hold each option matters for the system-level analyses, not which agent holds which. We

therefore summarise the group at round t by its *vote composition*, the count of agents on each option,

$$s_t = (n_1^t, n_2^t, \dots, n_M^t), \quad \sum_{k=1}^M n_k^t = N, \quad (1)$$

where n_k^t is the number of agents voting for option k at round t . The number of agents N and the number of options M are independent: $N = 4$ throughout, whereas M varies by task. The composition retains the information a plain set of options would discard, since a 3–1 split and a 2–2 split are distinct macrostates. This macrostate is the object on which the system-level motif detectors of Section 5 operate.

4.2 Debate Protocol

The protocol specifies three things: the persistent state each agent carries across rounds, the within-round message passing that produces a new state, and the memory window controlling how much past history each agent may see. We follow the multi-agent debate paradigm of Du et al. (2024), in which agents independently answer and then revise over several rounds, but instrument it so that the interaction log exposes exactly the quantities the motif detectors require.

4.2.1 Agent State and Two-Phase Rounds

Per-agent state. At each round t , agent i produces four fields:

- v_i^t : the discrete **vote**, one of the task’s M options;
- $c_i^t \in \{1, \dots, 10\}$: a private **confidence**, never transmitted;
- priv_i^t : private chain-of-thought **reasoning**, never transmitted;
- pub_i^t : a **public message**, broadcast to neighbours.

The separation of the public message from the private reasoning and confidence is deliberate: it lets us measure whether agents say what they privately hold. The discrete-vote / continuous-confidence split follows the continuous-opinion, discrete-action template of Martins (2008), and the public/private channel separation follows the DEBATE design of Chuang et al. (2025).

Argue draft. For rounds $t \geq 1$ each agent also emits an ephemeral *draft* before committing to a belief:

$$\text{draft}_i^t = (\text{defense}_i^t, \text{challenge}_i^t, \text{question}_i^t), \quad (2)$$

consisting of a **defense** (one concrete reason the current vote is correct), a **challenge** (a dispute of a specific peer claim, or a concession), and a **question** (a targeted query to a named peer). The draft is visible to neighbours at the start of the *Update* phase

but does not itself change any vote. These fields structure the deliberation but are not themselves used quantitatively; the influence graph for dominance detection is built from vote adoptions during the *Update* phase (Section 5), not from the draft text.

Round structure. Each round consists of two phases executed in parallel across all agents; *Argue* phase completes before the *Update* phase begins. Round 0 is asymmetric, as no communication has yet occurred.

- **Round 0 (initialisation, *Update* phase only).** Each agent receives the question and options and produces an initial belief with no peer input:

$$(v_i^0, c_i^0, \text{priv}_i^0, \text{pub}_i^0) = f_\theta(\text{question}, \text{options}).$$

- **Round $t \geq 1$, *Argue* phase.** Agent i reads its self and peer history windows (defined in Section 4.2.2) and emits a draft. Votes are *not* updated in this phase:

$$\text{draft}_i^t = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W)).$$

- **Round $t \geq 1$, *Update* phase.** Agent i additionally reads its own and its neighbours' current-round drafts, then updates its full state; if it changes its vote it is instructed to cite the specific argument that caused the change:

$$(v_i^t, c_i^t, \text{priv}_i^t, \text{pub}_i^t) = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W), \text{draft}_i^t, \{\text{draft}_j^t\}_{j \in \mathcal{N}(i)}).$$

The two-phase structure compresses a question–answer exchange into a single round; a one-phase synchronous protocol could not achieve this without a two-round lag. For the dynamical analysis, only the committed *Update* state is treated as the system state at round t ; the drafts function as message memory, not as a separate time step. Peers' private reasoning and confidence are never transmitted, and neither judge agents nor any consensus-pressure framing is used, as such framing would reweight the very motif distributions we aim to detect. All outputs are elicited as structured JSON with schema enforcement via function calling, guaranteeing the required fields in every response; the rare malformed response triggers a single retry before the round is logged as a skip. The complete prompt templates are reproduced verbatim in Section A.9. Figure 1 illustrates the full message flow for two agents over the first three rounds.

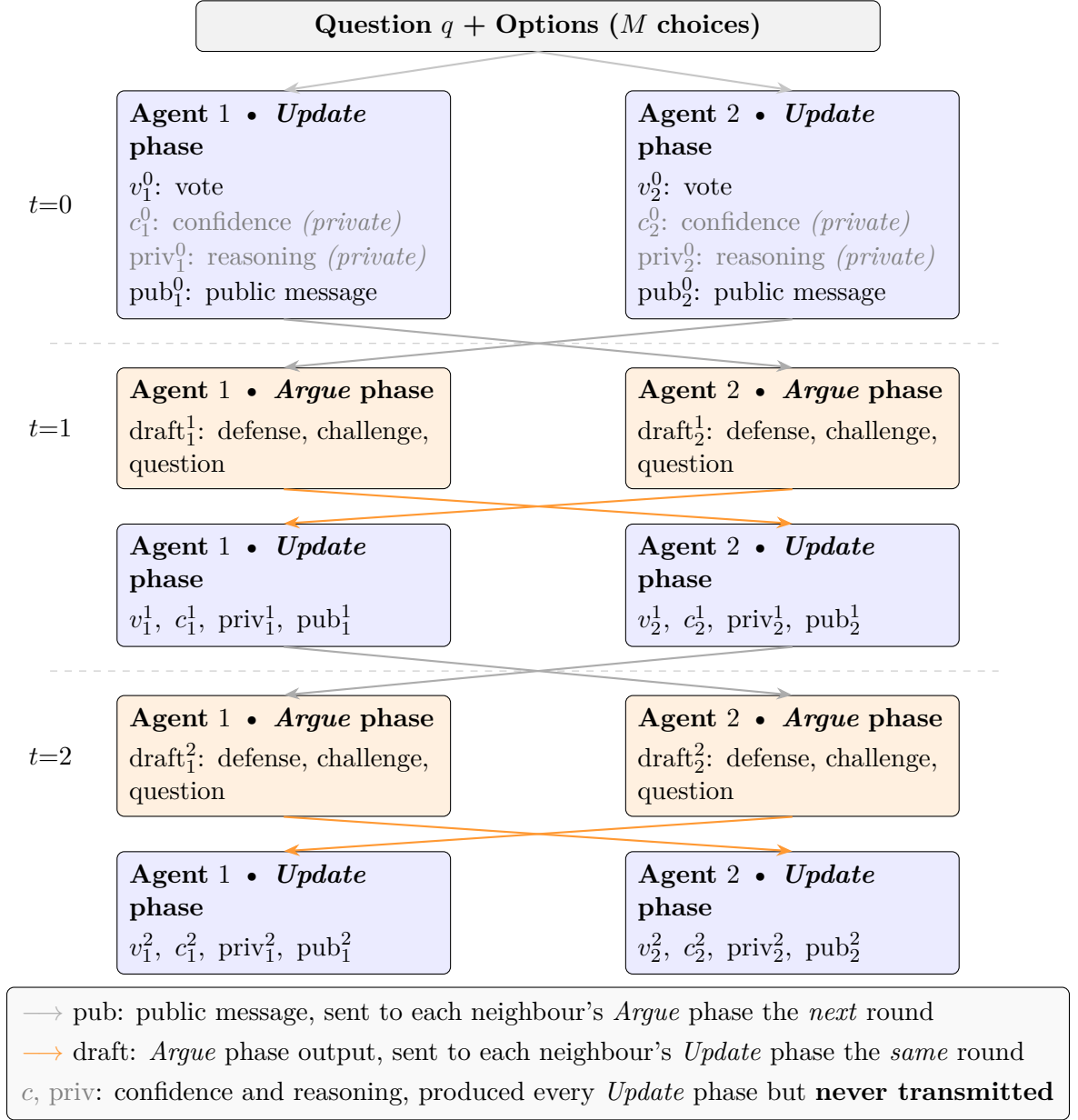


Figure 1: Two-phase synchronous protocol for two agents over rounds $t=0, 1, 2$. Round 0 has no *Argue* phase: agents read the question directly. Each *Update* phase commits the four state fields ($v, c, priv, pub$); the confidence c and reasoning $priv$ stay private (grey). A public message reaches neighbours in the *next* round's *Argue* phase; an *Argue* draft reaches neighbours in the *same* round's *Update* phase, which is what lets one round carry a full question–answer exchange.

4.2.2 Memory Window and Stopping Criteria

Memory window. The memory window $W \in \mathbb{Z}_{\geq 1}$ controls how many past rounds each agent may condition on. At round t , agent i sees a *self* window of its own recent states

and drafts, and a *peer* window of its neighbours’ recent votes, public messages, and drafts (peer private reasoning and confidence are excluded):

$$\mathcal{H}_i^{\text{self}}(t, W) = \left\{ \left(v_i^{t-k}, c_i^{t-k}, \text{pub}_i^{t-k}, \text{draft}_i^{t-k} \right) \right\}_{k=1}^{\min(t, W)}, \quad (3)$$

$$\mathcal{H}_i^{\text{peer}}(t, W) = \left\{ \left(v_j^{t-k}, \text{pub}_j^{t-k}, \text{draft}_j^{t-k} \right) \right\}_{j \in \mathcal{N}(i), k=1 \dots \min(t, W)}. \quad (4)$$

Peer entries are presented in randomised order at each call to mitigate position bias (Zheng et al., 2024), and round-0 entries omit the draft field, which does not exist. A larger W exposes more history but also interacts with documented long-context degradation (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024), which is why we treat W as an explicit perturbation axis rather than fixing it implicitly. $W = 1$ is the near-Markov baseline in which each agent sees only the immediately preceding round.

Early repetition stopping. Running every repetition for the full T rounds is wasteful once agents have converged: pilot transition matrices show a self-loop probability of 0.97–1.00 for the unanimous states, so escaping consensus after reaching it is effectively impossible. We therefore stop a repetition as soon as u consecutive rounds are unanimously agreed. The threshold u trades efficiency against reliability: too small and a transient consensus may be cut short; too large and the savings shrink. Analysing pilot repetitions (see Figure 20 in the appendix), we find that the harmful case, in which stopping would have prevented a later recovery to the correct answer, stays negligible while coverage remains high, and we set $u = 3$ with a hard cap of $T = 15$. Terminating debate at consensus is standard practice in the MAD literature (Yin et al., 2023, Yang et al., 2025, Kaesberg et al., 2025).

Final answer. The system’s answer for a repetition is the *plurality* of the agents’ votes at the final round, the option held by the most agents. Because a repetition stops only after $u = 3$ consecutive unanimous rounds, every converged repetition ends with all N agents on one option, so the plurality is unambiguous and equals that option. A tie can therefore arise only in the rare repetitions that reach the hard cap $T = 15$ without converging (for instance the two-way 2–2 split of a limit cycle); such ties are broken deterministically towards the first option in canonical order. The resulting label is a low-confidence artefact of wherever the unresolved run happened to land rather than a genuine group decision, and Section 6.5 recommends flagging these repetitions as unresolved instead of trusting the reported majority.

4.3 Datasets

Because this work studies motif emergence rather than task-completion performance, our benchmark selection criteria differ from those usual in the MAD literature. A benchmark suitable for motif detection must provide (i) a **discrete shared decision variable**, so that all agents commit to a position in the same finite set each round; (ii) **verifiable ground truth**, so dynamics can be related to correctness; (iii) **calibrated difficulty**,

since saturated tasks remove the disagreement that drives deliberation; and (iv) **sufficient initial disagreement**, without which no motif can be distinguished from trivial consensus.

Applying these criteria rules out most existing benchmarks. Saturated reasoning sets such as GSM8K and MMLU leave no headroom for deliberation dynamics; native multi-agent benchmarks are largely incompatible, being either role-specialised (no shared decision variable) or defined over spatial rather than categorical positions. A full survey of the candidates we considered and the reason each was rejected is given in Section A.11. Two benchmarks satisfy all four criteria and are complementary, so we adopt one as primary and one as a generality check.

4.3.1 GPQA-Diamond

GPQA-Diamond (Rein et al., 2024) is a graduate-level science question-answering benchmark of 198 questions in biology, physics, and chemistry, written by domain experts and validated through a “Google-proof” protocol that resists retrieval-based shortcuts. It satisfies our criteria directly:

- **Discrete decision variable:** four-option multiple choice ($M = 4$), on which vote composition, agreement, and flip rate are immediately defined.
- **Verifiable ground truth:** expert-validated and unambiguous.
- **Calibrated difficulty:** frontier models fall well below ceiling (Rein et al., 2024, Tian et al., 2025), leaving genuine headroom for debate.
- **Initial disagreement:** agents genuinely disagree at round one, as documented in multi-agent studies on this benchmark (Tian et al., 2025).

GPQA-Diamond also has direct multi-agent precedent (Tian et al., 2025, Kenton et al., 2024, He and Feng, 2025, Wang et al., 2025). We adopt it as the **primary** substrate. The main caveat is that it accrued less cumulative MAD literature than MMLU or GSM8K, simply because it is more recent; the migration is driven by benchmark saturation, not a methodological pivot. A second caveat concerns comparability of the accuracy numbers: published figures refer to strong models on the full benchmark, whereas we use a mid-tier model (`mistral-medium-instruct`) on the promoted Tier-3 subset only. Our results are therefore not comparable to published frontier GPQA-Diamond numbers, and are not intended to be; the Tier-3 selection deliberately keeps the hardest, most dynamically contested tasks, where accuracy is low by design.

4.3.2 HiddenBench

We use HiddenBench (Li et al., 2025) as a structurally complementary **secondary** substrate for generality testing. It formalises the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012): the evidence needed

for the correct answer is split across agents, so that no single agent can solve the task alone and success requires genuine information pooling. Its value here is as a control for a specific confound. Motifs seen only on GPQA-Diamond might be artefacts of cooperative reasoning over a reliable individual signal, in which RLHF-induced truth-seeking (Ouyang et al., 2022) amplifies convergence. Under the hidden-profile condition, by contrast, pre-discussion accuracy is very low by construction (Li et al., 2025), so any post-discussion improvement must come from social information transfer rather than from individual competence. Motifs that appear on both substrates are therefore robust to this confound, while motifs that appear on only one are themselves informative.

One structural detail of HiddenBench matters for the analysis: $N = 4$ agents throughout, while the number of answer options is task-dependent, $M \in \{3, 4\}$. The agent count N and the option count M are therefore independent quantities and should not be conflated. This is why every option-indexed quantity in Section 5 is defined over the task’s actual M options. Extension to mixed-motive negotiation settings is left to future work.

4.4 Experimental Design

The system exposes many tunable components. We classify each as **scan** (a primary perturbation axis), **fix** (held constant), or **defer** (out of scope). Table 2 summarises the classification and the literature supporting it; the detailed per-knob justification is deferred to Section A.12.

Scan axes. Two structural axes are varied, chosen because they are the canonical levers for network-motif analysis:

- **Memory window** $W \in \{1, 2, 5\}$, with $W = 1$ the near-Markov baseline. Memory content is an established driver of debate behaviour (Du et al., 2024, Li et al., 2024).
- **Topology** $\in \{\text{FC}, \text{STAR}\}$. Fully-connected probes the symmetric baseline; the star probes hub-induced dominance, following evidence that centralised structures concentrate influence (Li et al., 2024, Chen et al., 2023, Regan et al., 2024).

Their product gives $3 \times 2 = 6$ configurations per task.

Fixed choices. The remaining components are held constant to isolate the effect of the two scan axes:

- **Agents** $N = 4$: the smallest value at which the two topologies are structurally distinct and multi-way splits can occur.
- **Rounds** $T = 15$ (hard cap) with early stopping $u = 3$: T is an observation budget rather than a structural knob, since trajectories can be cropped post hoc but not extended.

- **Model** `mistral-medium-instruct` (homogeneous): a single mid-tier model isolates topology and memory effects from capability-asymmetry effects (Wynn et al., 2025).
- **Sampling** temperature 0.7, reasoning disabled: diverse yet focused outputs, avoiding the degenerate identical-sample case (Zhu et al., 2026).
- **Framing** cooperative, with the full *Argue/Update* structured output as the measurement instrument, and synchronous execution to remove agent-order confounds.

Knob	Verdict	Value	Key references
Memory W	scan	$\{1, 2, 5\}$	(Du et al., 2024, Li et al., 2024)
Topology	scan	fc, star	(Li et al., 2024, Chen et al., 2023, Regan et al., 2024)
Agents N	fix	4	(Du et al., 2024, Li et al., 2025)
Rounds T	fix	15 (cap), $u = 3$	(Kaesberg et al., 2025, Yang et al., 2025)
Output structure	fix	full <i>Argue/Update</i>	(Liang et al., 2024, He and Feng, 2025)
Framing	fix	cooperative	(Liang et al., 2024, Du et al., 2024)
Synchrony	fix	parallel-synchronous	(Du et al., 2024, Liu, Wang, Huang, Xu, Zeng, Jiang, Yang and Li, 2024)
Model	fix	<code>mistral-medium-instruct</code>	(Zhang et al., 2025, Wynn et al., 2025)
Temperature	fix	0.7, no reasoning	(Zhu et al., 2026)
Information cond.	scan (dataset)	GPQA ($M=4$), Hidden-Bench ($M \in \{3, 4\}$)	(Li et al., 2025, Stasser and Titus, 1985)
Tool access	defer	none	(Rein et al., 2024)

Table 2: Classification of tunable components. Full justification per knob is given in Section A.12.

Difficulty-based task selection. Motifs are per-task properties, and running the full grid at high R on every question is infeasible: at roughly 25s per repetition, the full GPQA-Diamond set across six configurations would take on the order of ten days at $R = 30$ alone. We therefore use a two-phase procedure. A shallow *screening* pass runs every task once at a single baseline configuration (fc, $W = 1$, $R = 3$) and assigns each task a difficulty tier from its observed dynamics:

- **Tier 0 (trivial):** unanimous at $t = 0$ with no flips; excluded.
- **Tier 1 (easy):** converges almost immediately (mean trajectory ≤ 4 rounds).

- **Tier 2 (moderate):** debate occurs but always converges to the same answer (no outcome variance across repetitions).
- **Tier 3 (hard):** debate occurs *and* the final answer varies across repetitions, so the dynamics, not the model prior, decide the outcome.

Tier 3 tasks are precisely those where self-organization dynamics can be observed at all, so they are promoted to a deep *characterisation* pass on the full six-configuration grid. This yields the final dataset used throughout Section 6: **50 tasks for each of the two benchmarks, six configurations each, $R = 50$ repetitions**, for a total of $50 \times 2 \times 6 \times 50 = 30,000$ repetitions. GPQA-Diamond yields more than 50 Tier-3 tasks in the screening pass, so we retain 50 of them to match the number of eligible HiddenBench tasks, giving a balanced design with equal task counts per benchmark; the 50 are drawn uniformly at random from the eligible Tier-3 GPQA tasks, so that no further selection is made on any outcome-derived quantity within the tier. Because $R = 3$ in the screening pass introduces some noise, tier assignments near a boundary are mildly unstable, which we accept as a screening cost rather than a final claim.

One consequence of this design should be made explicit for the interpretation of RQ2. Because the deep pass runs only Tier-3 tasks, raw task difficulty is held roughly constant by construction: it is a *selection criterion* for the study rather than a factor we vary within it. The “difficulty regime” contrast that RQ2 refers to is therefore operationalised at the benchmark level, GPQA (a refinement regime, agents mostly start correct) versus HiddenBench (a correction regime, agents start systematically wrong), and, within GPQA, through the single-agent-accuracy strata examined in the seeded analysis (paragraph 6.6); it is not a within-study manipulation of per-task difficulty.

5 Detection Methodology

Each of the four motifs introduced in Section 2.1 is a hypothesis about the debate dynamics: if the system self-organizes in the way a complex-systems reading predicts, then confidence should reinforce within a held position, repeated repetitions should settle into a small number of initial-condition-selected attractors, influence should concentrate on particular agents, and in some topologies vote sequences may oscillate persistently without converging. This section turns each hypothesis into a *detector*: a statistic computed from the interaction log together with a principled null model against which it is tested, so that a positive result reflects genuine structure rather than the chance regularities of a small, discrete state space. Together the four detectors form SOMA (Self-Organization Motif Analysis), the detector suite this thesis contributes. Two properties are common to all four and are worth stating up front. Each detector reads only from ordinary interaction logs, the vote and confidence channels defined in Section 4, which is what later lets us propose the same detectors as diagnostic instruments (Section 7.3). And each detects a *signature*, not a certified dynamical object; every “Limitations” paragraph below makes explicit what the corresponding statistic can and cannot claim, a distinction we carry into the interpretation of every result.

5.1 System Macrostate and Motif Vocabulary

We detect four self-organization motifs in the debate logs. Before defining them individually, this section fixes the two ideas they share: what we mean by a *motif*, and the *macrostate* the system-level motifs are read from.

Two notions of motif. We use the word *motif* in two complementary senses and are explicit about which we mean. A **structural motif**, in the original sense of Milo et al. (2002) and Alon (2007), is a recurring *wiring* pattern: a small subgraph of who-influences-whom that occurs more often than in a randomised network. It is a statement about interaction structure and can live at the node level (positive autoregulation, a self-loop) or span several nodes (hubs, feed-forward loops). A **dynamical motif** (Driscoll et al., 2024) is instead a recurring pattern in the *trajectory*, such as an attractor, a sustained oscillation, or a separatrix between basins; it is a statement about system behaviour rather than wiring. The two are linked, since a structural motif is studied precisely because it generates a characteristic dynamical regime, and they map onto the two levels at which our phenomena live, the agent (node) level and the system level.

The four motifs. We do not enumerate motifs exhaustively but preselect a small set, each motivated by a documented failure mode of LLM-based debate and spanning both senses and both levels:

- **Self-reinforcement** (agent level): node-level positive feedback, also a structural motif in the strict sense (positive autoregulation).

- **Bistability** (system level): a dynamical motif (coexisting attractors) selected by initial conditions.
- **Dominance** (system level): a structural motif in the *emergent* influence graph (a concentration of directed influence, an emergent hub), keyed to topology.
- **Limit cycle** (agent and system level): a dynamical motif (a periodic orbit), read both as individual non-convergence and as cycling of the collective configuration.

A broader list of motifs that the same lens admits but that we leave to future work is given in Section A.10.

System macrostate. The system-level motifs are read off a single agent-anonymised macrostate. As introduced in Section 4.1, the macrostate at round t is the vote composition

$$s_t = (n_1^t, n_2^t, \dots, n_M^t), \quad \sum_{k=1}^M n_k^t = N, \quad (5)$$

where n_k^t is the number of agents voting for option k at round t , M is the number of answer options (4 for GPQA, 3 or 4 for HiddenBench), and $N = 4$ is the number of agents. Agents are exchangeable, so only their counts enter; the options stay labelled, since one is correct and they are not interchangeable. The composition retains what a plain set of options would discard: a 3–1 split and a 2–2 split are distinct macrostates. Every option-indexed quantity below ranges over the task’s actual M options, so an option the task does not present can never be voted for or become an attractor. Bistability is read from the locked-in endpoint; dominance is relational, built on the emergent influence graph, and requires within-repetition agent identity; the system-level limit-cycle test operates on the full sequence $s_0, \dots, s_T$.

Shared data hierarchy. All four detectors respect the same nesting of the data,

$$\text{dataset} \supset \text{config} (W \times \text{topology}) \supset \text{task} \supset \text{repetition} \supset \text{agent}, \quad (6)$$

with $W \in \{1, 2, 5\}$, $\text{topology} \in \{\text{fc}, \text{star}\}$, $N = 4$, and $\text{dataset} \in \{\text{GPQA}, \text{HiddenBench}\}$. Two conventions hold throughout and are not repeated for each motif. First, the **repetition** is the independent unit of replication: agents interact within a repetition, so their trajectories are *not* independent and must never be treated as separate data points (pseudoreplication). Second, all analysis is **conditioned on the config and dataset**: GPQA and HiddenBench are analysed separately and never pooled, and the memory-window comparison is always made within a fixed topology and dataset. Each detector below only states its own *measurement primitive*, the lowest-level object it scores.

Statistical conventions. Three conventions govern every inferential test in Section 6, chosen to be simple and transparent rather than maximally powerful. (i) *The task is the unit of analysis for comparisons across tasks or configurations.* Because the R repetitions of a cell are re-runs of the *same* question, they are not independent across tasks; wherever we test whether a quantity differs between configurations or covaries with accuracy, we first aggregate each task to a single value (its mean or proportion over repetitions) and test across the ≈ 35 tasks with a paired Wilcoxon signed-rank test (Wilcoxon, 1945), a

Spearman correlation, a task-blocked Friedman test (Friedman, 1937), or a sign test, or else resample whole tasks in a cluster bootstrap. Repetition-level counts are reported descriptively only. (ii) *Multiplicity is controlled per family*. Where a motif is tested across the twelve configuration cells or several outcome variables, we control the false-discovery rate with the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) within each pre-defined family and report which results survive. (iii) *Negative claims are stated as bounded effects, not equivalence*. For the design axes that do not move accuracy we report the effect size with a task-level confidence interval and conclude only that it is small and contains zero, i.e. that we find no evidence of a meaningful effect, rather than asserting strict statistical equivalence.

A final property is shared by all four detectors and stated once here: each claim is **phenomenological, not causal**. We detect a trajectory or influence *signature*, not a certified attractor, cycle, or causal hub; separating the signature from its underlying cause would require dedicated perturbation experiments, which we leave to future work. Under the cooperative, convergence-forced design, we also expect several detectors to return **sparse or near-null** results; a near-null result is a valid finding, and we compare observed prevalence against a principled null rather than against zero.

5.2 Self-Reinforcement

Concept. In complex systems, self-reinforcement is *positive feedback*: the current level of a state variable raises the drive toward more of the same, so deviations are amplified rather than damped (Arthur, 1989). At the motif level this is positive autoregulation, a node that augments its own production (Milo et al., 2002, Alon, 2007). For us the state variable is an agent’s expressed *confidence*, and self-reinforcement is the tendency of an agent to push its own confidence upward while it holds a position, driven by accumulating commitment rather than by what the incoming information warrants. The pattern is documented in the substrate: in multi-turn LLM debate, confidence tends to rise across rounds in an anti-Bayesian fashion, even where no new information justifies the rise (Prasad and Nguyen, 2025). A plain upward confidence trajectory is therefore *necessary but not sufficient*, since rational updating and peer conformity also push confidence up; the definition below targets the signatures that go beyond “the line goes up.” The measure is option-agnostic, reading confidence within a held position, so it does not depend on M .

What we measure. Self-reinforcement has three observable signatures. We measure the first and second and treat the third as only partially identified:

- **Directional drift (measured).** While a position is held, confidence ratchets up more often than down.
- **Strength scaling (measured).** The drift grows with how strongly the agent is re-exposed to its own prior commitments, which the memory window W controls directly.

- **Endogeneity (not isolated).** The rise is driven by the agent’s own commitment rather than by new information; this cannot be cleanly separated from peer effects with trajectory data alone.

Measurement primitive: the vote-stable segment. The primitive is the *vote-stable segment*. For agent i in repetition r , this is a maximal block of consecutive rounds $[t_s, t_e]$ (where t_s is the first round of the block and t_e is the last) over which the vote is constant, $v_i^t = v_i^{t_s}$ for all $t \in \{t_s, \dots, t_e\}$ (the vote differs, or the trajectory ends, immediately before t_s and after t_e). A segment is retained only if its length $L = t_e - t_s + 1 \geq 3$; the $+1$ counts both endpoints, so $L = 3$ means three rounds and yields at least two confidence steps. An agent may contribute several segments if it flips, or a single long segment if it locks in early. Agents are exchangeable, so segments are pooled within a config and never matched across repetitions.

Within a retained segment we fit an ordinary-least-squares line of confidence ($c_i^t \in \{1, \dots, 10\}$, agent i ’s integer confidence at round t) on the round index,

$$c_i^t = \beta_0 + \beta_{\text{seg}} t + \varepsilon_t, \quad t \in \{t_s, \dots, t_e\}, \quad (7)$$

and keep the slope β_{seg} , in units of confidence per round, which is comparable across segments of different length without normalisation. Two things are read off it: its *sign*, which feeds the prevalence test below, and its *signed value*, which feeds the mean slope $\bar{\beta}$ used in the W -scaling comparison; the value is kept signed, not as $|\beta_{\text{seg}}|$, because the direction of drift is what the test is about.

Detection. We read the segments in two complementary ways, both conditioned on the config.

The **presence** test measures directional drift. For each (config, task) pair, the prevalence of upward segments is

$$\hat{p}_{\text{SR}} = \frac{|\{\text{seg} : \beta_{\text{seg}} > 0\}|}{|\{\text{seg} : \beta_{\text{seg}} > 0\}| + |\{\text{seg} : \beta_{\text{seg}} < 0\}| + |\{\text{seg} : \beta_{\text{seg}} = 0\}|}, \quad (8)$$

so flat segments ($\beta_{\text{seg}} = 0$) enter the denominator but never the numerator. Under the no-feedback “wobble” null an upward and a downward segment are equally likely, so with a fraction f of flat segments $P(\beta_{\text{seg}} > 0) = \frac{1}{2}(1 - f) \leq 0.5$, and 0.5 is a conservative reference. Self-reinforcement is indicated when $\hat{p}_{\text{SR}} > 0.5$, which is a sign test against a *given* baseline. The confidence ceiling (a segment already at the top of the scale cannot drift further up) makes a positive result conservative.

Identifying signature: W -scaling as a mechanism discriminator. Letting $\bar{\beta}$ be the mean segment slope in a config, the direction of W -scaling discriminates between two candidate mechanisms. Under an *endogenous* reading, re-reading one’s own prior commitments across a longer window amplifies the internal drive, predicting

$$\bar{\beta}(W=5) > \bar{\beta}(W=2) > \bar{\beta}(W=1). \quad (9)$$

Under a *peer-conformity* reading, a larger window increases the amount of peer content visible to each agent; if peer expression moderates the agent’s own drive, the ordering may be absent or reversed. We report the observed W ordering per topology (Section 6.2), which speaks to which mechanism dominates. Prevalence is reported per (config, task) and then summarised across tasks (for example, “above 0.5 in k of 35 tasks”), so that the effect is shown to be broad rather than driven by one task. Uncertainty is attached by a cluster bootstrap over repetitions, stratified by task (Efron, 1979): repetitions are resampled with replacement within each task and all segments of a resampled repetition are kept together, so the within-repetition dependence is respected; a signature is read as present when the resampled interval excludes its baseline (0.5 for prevalence, 0 for slope differences across W).

Worked example. In our running scenario ($N = 4$ agents, options A – D , A correct), suppose agent 0 holds option A for five consecutive rounds with confidences $c = (5, 6, 6, 8, 9)$ at $t = 0, \dots, 4$. This is one vote-stable segment of length $L = 5 \geq 3$. Regressing confidence on the round index gives a slope of $\beta_{\text{seg}} = +1.0$ confidence points per round, a positive (upward) segment that counts toward $\hat{p}_{\text{SR}}$. Had the same agent instead drifted $c = (7, 7, 6, 6, 5)$, the slope would be $\beta_{\text{seg}} = -0.5$, a downward segment counting against it. A segment that never moved, $c = (6, 6, 6, 6, 6)$, has $\beta_{\text{seg}} = 0$ and is a flat segment: retained in the denominator but not counted as upward, which is what makes the 0.5 reference conservative. Under the strength-scaling reading, self-reinforcement predicts that the average of such slopes grows as the memory window widens from $W=1$ to $W=5$.

Limitations. Beyond the phenomenological caveat stated in Section 5.1, three points bound the claim. *Endogeneity is not isolated*: an upward segment is equally consistent with peer copying and with a peer surfacing genuinely new information in a later round, and rising confidence alone cannot distinguish these. *The W ordering provides mechanistic evidence but does not fully resolve the ambiguity*, since a larger window also re-exposes more peer content; the observed direction of the W ordering informs which mechanism dominates without cleanly separating them. *Convergence is a confound*: on easier tasks fast agreement can itself raise confidence, which is why prevalence is reported per task and never as a single pooled figure.

5.3 Bistability

Concept. In dynamical systems, *multistability* is the coexistence of two or more stable attractors for one fixed parameter set; which attractor the system settles into is decided by the basin of attraction containing the initial condition (Feudel, 2008, Pisarchik and Feudel, 2014), and bistability is the two-attractor case. At the motif level this is the dynamical signature of positive feedback with a symmetry: the same interaction rules admit several self-consistent outcomes (Milo et al., 2002, Alon, 2007). For us the state variable is the macrostate s_t , and bistability is read off the *locked-in endpoint*: for a fixed task, the debate converges to different consensus answers on different repetitions, and which one is reached is selected by the initial vote configuration rather than by the task’s correct option. This is the structural reading of unreliable correction: on a task where

the model carries a systematic bias, a wrong-consensus and a right-consensus attractor can coexist, and the group’s fate is basin competition decided at round 0, which self-reinforcement then amplifies and locks in. Seeing more than one final answer is necessary but not sufficient, since a task may simply be noisy; genuine multistability additionally requires that the initial condition *predicts* the endpoint (Pisarchik and Feudel, 2014), and the definition tests both.

Measurement primitive and eligibility. The primitive and independent unit is the repetition, summarised by its initial vote composition $s_0 = (n_1^0, \dots, n_M^0)$ (the macrostate defined in Section 5.1) and its locked-in endpoint attractor. For repetition r we take the per-round composition $s_0, \dots, s_T$ and let k be its maximal trailing constant segment; the repetition has *settled* if $k \geq u = 3$. A settled repetition is **converged** (eligible) only if its locked-in macrostate is *unanimous*, in which case its endpoint attractor $A^{(r)}$ is that option; so attractors are consensus states and there are at most M of them. Repetitions that never settle, and those that settle on a stable non-unanimous split (a frozen disagreement, not a consensus answer), are excluded; bistability is measured on the unanimous-converged subset only. Scoring is per (task, config) cell; let n_c denote the number of converged repetitions in that cell. The cell’s *dominant attractor* is the most frequent endpoint option, and each repetition’s start is summarised by

$$g(s_0) \in \{0, 1, \dots, N\}, \quad (10)$$

the number of agents whose round-0 vote is on the dominant option. This coarse summary keeps the basin-selection table dense enough to have power at $n_c \approx 30$; the full initial composition has too many distinct values to be informative at this sample size.

Signature 1: coexistence. The empirical basin distribution over endpoint attractors is $\hat{P}(A = k) = \frac{1}{n_c} \sum_r \mathbf{1}[A^{(r)} = k]$, a basin-stability estimate of relative basin volumes under the model’s own initial distribution (Menck et al., 2013). The effective number of coexisting attractors is the inverse Simpson (Hill) index (Hill, 1973),

$$N_{\text{eff}} = \frac{1}{\sum_k \hat{P}(A = k)^2} \in [1, M]. \quad (11)$$

$N_{\text{eff}} \approx 1$ is a single attractor, ≈ 2 two coexisting attractors, and so on, so N_{eff} reports *how many* attractors coexist. Because attractors are unanimous options, it is bounded by M . Since M varies across tasks, we report N_{eff} raw and compare it only within a fixed M ; normalising by M would make a bistable three-option task (2/3) look more multistable than an identically bistable four-option task (2/4), an artefact of option count. In practice GPQA is entirely $M = 4$; only HiddenBench mixes $M = 3$ and $M = 4$, so its N_{eff} summaries are reported separately by option count.

Signature 2: basin selection. Coexistence alone does not establish multistability; the initial condition must select the attractor (Feudel, 2008, Pisarchik and Feudel, 2014). We measure the association between $g(s_0)$ and the endpoint A with Cramér’s V (Cramér, 1946) on their contingency table,

$$V = \sqrt{\frac{\chi^2/n_c}{\min(R_{\text{tab}} - 1, C_{\text{tab}} - 1)}} \in [0, 1], \quad (12)$$

where χ^2 is the Pearson statistic and $R_{\text{tab}}, C_{\text{tab}}$ are the table’s numbers of rows (distinct $g(s_0)$, at most $N + 1$) and columns (distinct endpoints, at most M). $V \approx 0$ means the start is irrelevant; V near 1 means the start deterministically selects the endpoint. The null shuffles the pairing of $g(s_0)$ and A across the n_c repetitions, preserving both marginals while destroying the start-to-endpoint link (Theiler et al., 1992); for $B = 1000$ surrogates the add-one one-sided p -value flags basin selection when $p < 0.05$.

Cell label. Each cell is labelled

$$\text{label}(c) = \begin{cases} \text{monostable} & N_{\text{eff}} < 1.5, \\ \text{multistable} & N_{\text{eff}} \geq 1.5 \wedge p < 0.05, \\ \text{stochastic (not bistable)} & N_{\text{eff}} \geq 1.5 \wedge p \geq 0.05, \end{cases} \quad (13)$$

with the degree of multistability read from N_{eff} and capped at M . The cutoff 1.5 is a reporting threshold, not a test; the inferential claim rests on p . The third case is the essential guard: coexistence without basin selection is scatter, not multistability. We attach a bootstrap over the cell’s repetitions (Efron, 1979) carrying N_{eff} as a descriptive interval on the attractor count, and, because per-cell power is limited, lean on the across-task distribution of N_{eff} and V rather than any single cell.

Worked example. In our running scenario, take a cell with $n_c = 30$ converged repetitions; the dominant attractor is option A (correct). The coexistence reading depends only on the endpoint distribution.

- **Monostable.** Endpoints $28 \times A$, $2 \times B$ give $\hat{P} = (0.933, 0.067)$ and $N_{\text{eff}} = 1/(0.933^2 + 0.067^2) \approx 1.14 < 1.5$: one valley.
- **Bistable.** Endpoints $15 \times A$, $15 \times B$ give $N_{\text{eff}} = 2.0$. If repetitions starting with more agents on A (high $g(s_0)$) end on A and those starting with few end on B , then $V \approx 1$, $p < 0.05$: two basins selected by the start.
- **Tristable.** Endpoints $10 \times A$, $10 \times B$, $10 \times C$ with the start predicting each give $N_{\text{eff}} \approx 3.0$, $p < 0.05$ (the maximal degree for $M = 3$).
- **Stochastic.** Endpoints $15 \times A$, $15 \times B$ again give $N_{\text{eff}} = 2.0$, but if $g(s_0)$ is unrelated to the endpoint then $V \approx 0$, $p \gg 0.05$: coexistence without basin selection, correctly *not* labelled bistable.

Limitations. Two points beyond the shared phenomenological caveat. First, classical basin stability draws initial conditions uniformly (Menck et al., 2013), whereas our starts come from the model’s own round-0 distribution, so $\hat{P}(A)$ estimates *effective* basin volume under that distribution, not geometric volume. Second, because $g(s_0)$ counts agents already on the eventual dominant option, a high V is consistent both with genuine basin selection and with ordinary majority-vote persistence; the test detects start-to-endpoint dependence but does not by itself separate these two mechanisms, so we read the association as necessary but not mechanistically decisive.

5.4 Dominance

Concept. Dominance is the concentration of directed influence onto a single node, an *emergent hub*: a structural motif (Milo et al., 2002, Alon, 2007) read at the system level, since it is a property of the whole influence distribution. Its formal home is opinion dynamics: in the DeGroot model (DeGroot, 1974) and its social-power extension, an agent’s influence equals its network centrality, and the group is *autocratic* when these are maximally nonuniform and *democratic* when uniform (Jia et al., 2015); dominance is the autocratic pole. It is the structural substrate of the debate pathologies we target, herding to a confident leader (Chen et al., 2023, Cho et al., 2025) by which one agent can degrade collective accuracy (Wynn et al., 2025, Prasad and Nguyen, 2025). A single converted vote or fast agreement can manufacture apparent concentration, so we measure the concentration of influence *sources* against a topology-respecting baseline that removes the wiring’s own contribution. What we detect is a concentration *signature*, not a causal hub: an adoption edge cannot separate “*j* followed *i*” from coincidental agreement.

Measurement primitive: the emergent influence graph. The primitive and independent unit is the per-repetition influence graph, which requires within-repetition agent identity (agents remain unlabeled across repetitions, respected because the per-repetition score is a single scalar). Dominance is defined over agents and their conversions, so it is independent of the option count M . A *conversion* is the basic event: agent j adopts, at round $t+1$, a vote that a disagreeing neighbour i held at round t . The directed edge $i \rightarrow j$ accumulates the confidence-weighted conversions of j onto i ’s vote, with credit split among the neighbours already holding the target vote,

$$w_{ij} = \sum_t \frac{1}{m_j^t} \mathbf{1}[v_j^{t+1} = v_i^t \neq v_j^t] c_j^t, \quad m_j^t = |\{k \in \mathcal{N}(j) : v_k^t = v_j^{t+1}\}|, \quad (14)$$

where c_j^t is the converting agent’s confidence and $\mathcal{N}(j)$ its topological neighbourhood; the factor $1/m_j^t$ shares one conversion equally among all eligible sources. Per-agent influence is the weighted out-strength $\sigma_i = \sum_j w_{ij}$. We use out-strength rather than eigenvector centrality (Bonacich, 1972), which is undefined or degenerate on non-strongly-connected and acyclic graphs (Newman, 2010) and hub-localises (Martin et al., 2014); our graphs are tiny, sparse, disconnected, and near-acyclic, so out-strength is the only always-defined node score.

Concentration statistic. Dominance is the concentration of the influence-share vector $p_i = \sigma_i / \sum_k \sigma_k$, measured by the normalised Herfindahl index (Hirschman, 1964),

$$D = \frac{\sum_i p_i^2 - \frac{1}{N}}{1 - \frac{1}{N}} \in [0, 1]. \quad (15)$$

$D = 0$ is democratic (equal influence), $D = 1$ a single pure hub. Normalisation fixes the scale and makes D comparable across configs, unlike Freeman centralization (Freeman, 1978), whose maximum is defined only for an unweighted star. D is undefined when $\sum_k \sigma_k = 0$ (no conversions), and such repetitions are excluded, as with the fixed-point exclusion for limit cycles.

Detection. The null is keyed to topology: a constrained shuffle in the spirit of Theiler’s “Algorithm 0” (Theiler et al., 1992) that, per conversion event, keeps the converting agent, round, weight, and number of sources but randomises the source *identity* uniformly among $\mathcal{N}(j)$. In **fc** the null is symmetric, so any concentration is emergent; in **star** the null already concentrates on the centre by wiring alone, so only the *excess* over it is behaviourally meaningful. In the star this excess is moreover often undefined: for a leaf, the only neighbour is the centre, so a leaf conversion has a single possible source and the shuffle cannot randomise it. As a result the star null is degenerate for a large majority of star repetitions (73–87%, reported in Section 6.4), and the emergent-excess test is effectively identified only for the fully-connected topology; raw D is still reported for both. For $B = 1000$ surrogates the add-one one-sided $p = \frac{1 + \#\{b: D^{(b)} \geq D\}}{1+B}$ flags a repetition when $p < 0.05$. Because raw D is mechanically larger in the star, we never compare it across topologies: across topologies we report the standardised $z_r = (D_r - \mu_{\text{null}})/\sigma_{\text{null}}$ or the flag rate, and within a topology we report raw D . As for self-reinforcement, an entrenching hub predicts a directional ordering $\bar{D}(W=5) > \bar{D}(W=2) > \bar{D}(W=1)$ at fixed topology. As a secondary descriptor, not part of D , we also report whether the hub’s vote equals the final consensus and its capitulation rate, separating a followed leader from an overruled one. Uncertainty is attached by a cluster bootstrap over repetitions stratified by task (Efron, 1979), carrying D_r (or z_r) as the unit.

Worked example. In our running scenario (**fc** topology), agent 0 opens at D_8 , with agents 1–3 at A_6, B_5, C_5 —an illustrative case where the hub drives the group to the wrong answer. The *Update* vote table (confidence subscripts):

t	ag ₀	ag ₁	ag ₂	ag ₃
0	D_8	A_6	B_5	C_5
1	D_9	D_6	B_5	C_5
2	D_9	D_7	D_5	C_5
3	D_9	D_7	D_6	D_5

gives three conversions: $0 \rightarrow 1$ gains 6 ($m = 1$); $0 \rightarrow 2, 1 \rightarrow 2$ each 2.5 ($m = 2$); $0 \rightarrow 3, 1 \rightarrow 3, 2 \rightarrow 3$ each $\frac{5}{3}$ ($m = 3$). Then $\sigma = (\frac{61}{6}, \frac{25}{6}, \frac{5}{3}, 0)$, $p \approx (0.635, 0.260, 0.104, 0)$, and $D = (0.482 - 0.25)/0.75 \approx 0.31$. Agent 0 is a clear hub (64% of influence), yet D is far from 1: the $1/m_j^t$ split makes each early adopter a co-source for later conversions, so staggered convergence never yields a pure hub. $D = 1$ needs all credit on one node (a single conversion, or a simultaneous mass flip onto a sole prior holder); a lone dissenter folding into a strong majority is the opposite, $D = \frac{1}{9} \approx 0.11$.

Limitations. With four agents ($N = 4$) the four shares and a discrete null bound the minimum p , so per-repetition power is low and the signal lives in the aggregate. A single-conversion repetition gives $D = 1$ trivially, which is controlled by aggregation and the topology null rather than by discarding. As with every detector, certifying a causal hub would require perturbation experiments.

5.5 Limit Cycles

5.5.1 Agent-Level Detection

Concept. In dynamical systems a limit cycle is an isolated closed orbit: a self-sustained periodic oscillation that requires no external forcing (Strogatz, 2015). At the motif level, sustained oscillation is the dynamical signature of feedback loops, the complementary case to the positive autoregulation behind self-reinforcement (Milo et al., 2002, Alon, 2007). For us the state variable is the agent’s vote $v_i^t \in \{1, \dots, M\}$ (the discrete choice defined in Section 4), one of the task’s M options, and a limit cycle is an agent that keeps cycling through votes (for example $A, B, A, B, \dots$) without ever settling. This matters because convergence to consensus is one system fate, whereas a sustained vote oscillation is a qualitatively different one, periodic disagreement that never resolves; detecting it tests whether cooperative debate can become dynamically trapped in a cycle rather than a point. With only a handful of options, votes repeat by chance and a transient wobble that later converges is not a cycle, so the definition requires both that the oscillation is *self-sustained* (the agent does not converge) and that its periodic structure *beats a chance baseline*. We measure both; the isolation and stability of a true limit cycle cannot be certified from one observed trajectory and are not claimed.

Measurement primitive and end-state classification. The primitive is the per-agent vote sequence within a repetition, $v = (v_0, v_1, \dots, v_T)$, taking the committed *Update* vote at each round. Let k be the length of the maximal *trailing constant segment* of v . If $k \geq u$ (with $u = 3$, the early-stopping window) the agent has settled: it is classified as a **fixed point** and excluded. Otherwise it is a **limit-cycle candidate**, subject to an eligibility guard $L \geq 4$ (room for one repeat of the shortest, period-2, cycle). We deliberately *do not trim* a trailing block: it cannot be separated into cycle-internal repeats and genuine convergence, so any keep- k rule would be phase-dependent and unsound. Excluding converged trajectories wholesale is the definitionally correct move, since a transient that decays into a fixed point is not a limit cycle.

Recurrence quantification. We quantify periodicity with categorical recurrence quantification analysis, using exact matching (radius 0) and no embedding, since the vote is itself the state (Coco and Dale, 2014). For sequence positions a, b the recurrence matrix is

$$R(a, b) = \mathbb{I}[v_a = v_b], \quad a \neq b, \quad (16)$$

with the line of identity excluded (Theiler window 1) (Eckmann et al., 1987). A periodic orbit appears as diagonal lines parallel to the main diagonal, equally spaced by the period; a fixed point fills a solid block; scattered points indicate no periodicity (Marwan et al., 2007). *Determinism* is the fraction of recurrence points on diagonal lines of length $l \geq l_{\min} = 2$ (Zbilut and Webber Jr., 1992, Webber Jr. and Zbilut, 1994),

$$DET = \frac{\sum_{l \geq l_{\min}} l P(l)}{\sum_{l \geq 1} l P(l)}, \quad (17)$$

where $P(l)$ is the histogram of diagonal-line lengths, and the *dominant period* is the lag of the densest off-diagonal,

$$RR(\tau) = \frac{1}{L - \tau} \sum_{a=0}^{L-1-\tau} R(a, a + \tau), \quad \hat{P} = \arg \max_{\tau \geq 1} RR(\tau). \quad (18)$$

Crucially, DET alone cannot separate persistence from periodicity: a solid block (a near-constant sequence) and equally-spaced diagonals (a genuine cycle) both inflate it. A single stray vote in an otherwise constant sequence can therefore attain high DET while having dominant period $\hat{P} = 1$, i.e. no oscillation. We guard against this by requiring $\hat{P} \geq 2$, with the upper bound $\hat{P} \leq \lfloor L/2 \rfloor$ the largest period that can recur at least twice within the window.

Detection. The null uses random-shuffle surrogates (Theiler’s “Algorithm 0”): random permutations of the agent’s own vote multiset (Theiler et al., 1992). Shuffling preserves the symbol frequencies, so the M -option chance-match rate is built into the null automatically, while destroying temporal order. For $B = 1000$ surrogates we compute $DET^{(b)}$ and the add-one one-sided p -value (Phipson and Smyth, 2010)

$$p = \frac{1 + \#\{b : DET^{(b)} \geq DET\}}{1 + B}, \quad (19)$$

and flag a limit cycle when

$$p < 0.05 \quad \text{and} \quad 2 \leq \hat{P} \leq \lfloor L/2 \rfloor, \quad (20)$$

that is, when the observed DET exceeds the 95th percentile of its shuffle null *and* the dominant period is a genuine, resolvable oscillation rather than persistence or a sparse high-lag chance match. We report a binary flag per candidate, prevalence per (config, task) as the flagged share among eligible candidates, and summaries across tasks per topology and W .

Worked example. In our running scenario, suppose agent 2 fails to settle, cycling $v = (A, B, A, B, A, B)$ across $L = 6$ rounds. Its trailing constant segment has length $k = 1 < 3$, so it is a candidate, and $L \geq 4$. The recurrence matrix (main diagonal excluded) is

$$R = \begin{pmatrix} \cdot & 0 & 1 & 0 & 1 & 0 \\ 0 & \cdot & 0 & 1 & 0 & 1 \\ 1 & 0 & \cdot & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdot & 0 & 1 \\ 1 & 0 & 1 & 0 & \cdot & 0 \\ 0 & 1 & 0 & 1 & 0 & \cdot \end{pmatrix}.$$

The ones lie on diagonals equally spaced by 2: $RR(1) = 0$, $RR(2) = 1$, $RR(3) = 0$, so $\hat{P} = 2$, satisfying the guard $2 \leq \hat{P} \leq \lfloor 6/2 \rfloor = 3$. The offset-2 diagonal carries a line of length 4 and the offset-4 diagonal a line of length 2, so $DET = (4 + 2)/(4 + 2) = 1.0$. For the null, the multiset $\{3A, 3B\}$ has $\binom{6}{3} = 20$ arrangements, of which only the two perfectly alternating ones reach $DET = 1$, giving $p = 2/20 = 0.10$, which does *not* clear 0.05: at

$L = 6$ even a perfect period-2 cycle lacks the power to be called significant. The same pattern over a full-length candidate ($L \approx 16$) has an exact combinatorial null probability of $\approx 2/\binom{16}{8} \approx 2 \times 10^{-4}$; with $B = 1000$ shuffle surrogates the add-one p -value saturates at its floor $1/(B+1) \approx 10^{-3}$, so such a case is flagged decisively at that floor, which is exactly why only the non-converged, full-length repetitions are informative. By contrast (A, B, A, B, B, B, B) has $k \geq 3$, is a fixed point, and is excluded; and the near-constant (A, B, B, B, B, B, B) attains high DET but $\hat{P} = 1$ and is rejected by the period guard as persistence, not oscillation.

Limitations. Beyond sparsity and the phenomenological caveat, the alphabet is small ($M \in \{3, 4\}$), which inflates the chance recurrence rate, and the sequence length caps the detectable period at $\lfloor L/2 \rfloor$. The shuffle null controls the chance-recurrence inflation, but the period guard makes the test conservative, so a small nonzero flag rate remains under a chance baseline; we compare observed prevalence against this floor rather than against zero.

5.5.2 System-Level Detection

The agent-level test asks whether an *individual* is trapped oscillating. The system-level reading asks the complementary question, whether the *collective configuration* cycles, by applying the identical recurrence machinery to the macrostate sequence instead of a single agent’s votes. The primitive is now the per-repetition composition sequence $s = (s_0, s_1, \dots, s_T)$, and the repetition is both the measurement primitive and the independent unit.

End-state classification carries over with one addition. Let k be the length of the maximal trailing constant segment of s ; if $k \geq u$ the configuration has **settled** and is excluded. This covers two cases: unanimous consensus, and a *stable non-unanimous split* (a persistent 3–1 or 2–2 composition). The split is a fixed point, not a cycle, and is precisely the polarisation regime we flag for future work; here it is correctly excluded. Classification is done on the sequence, not on whether the debate early-stopped, since early stopping triggers only on unanimity and a stable split runs to the cap T while still being a fixed point. The recurrence matrix, DET , dominant period, surrogate null, and flagging rule are identical to the agent-level case, with $R(a, b) = \mathbb{K}[s_a = s_b]$. The period guard is equally necessary: a stable split produces a solid recurrence block with $\hat{P} = 1$ and is correctly rejected.

The two readings answer different questions and neither implies the other. If two agents swap $A \leftrightarrow B$ each round while the other two hold, every s_t is identical: the agent-level test flags two period-2 cycles while the system-level test sees a fixed point. Conversely a rotating majority can cycle s_t with no single agent being periodic. We therefore report both. The macrostate space has $\binom{N+M-1}{M-1}$ compositions (15 for $M = 3$, 35 for $M = 4$) against M raw vote symbols, so chance recurrence is lower and DET cleaner at the system level, but there is only one sequence per repetition rather than per agent, so candidates are far fewer; as at the agent level, a sparse or near-null result is expected and reported

honestly.

6 Results

This section reports the empirical results of applying the SOMA detectors to 30,000 debates across two benchmarks, GPQA and HiddenBench. The two benchmarks behave so differently that every result below is reported per benchmark rather than pooled: as Section 6.1 shows, debate plays a fundamentally different role on each, and the motif analyses inherit that split throughout. We first characterise overall accuracy and efficiency (Section 6.1), then measure each motif in turn: self-reinforcement (Section 6.2), bistability (Section 6.3), dominance (Section 6.4), and limit cycles (Section 6.5). Outcome-distribution evidence and a causal test for the amplifier reading follow (Section 6.6), and we close by extracting the effective influence topology that emerges within fully-connected runs (Section 6.7). Each section reports detection statistics, structural drivers, and relationship to outcomes; null and near-null results are reported as findings.

6.1 Accuracy and Efficiency

This section reports overall system accuracy and efficiency across all 600 task–configuration combinations ($50 \text{ tasks} \times 2 \text{ datasets} \times 3 \text{ memory windows} \times 2 \text{ topologies}$, $R = 50$ repetitions each, $N = 30,000$ total). Two metrics are reported throughout:

- **Accuracy:** fraction of repetitions in which the final answer (the plurality of the agents’ votes at the final round; see Section 4.2) matches the ground-truth answer.
- **Efficiency:** mean number of rounds until the early-stopping criterion fires or the hard cap $T = 15$ is reached.

Single-agent baseline and debate gain. To isolate the contribution of debate, we compare the multi-agent outcome against two baselines defined at round 0, before any discussion takes place:

- **Per-agent accuracy:** the fraction of individual agent answers that are correct at round 0, averaged across all agents and all repetitions.
- **Majority-vote baseline:** the accuracy of the plurality vote over all agents at round 0 (ties broken arbitrarily). This is the best a voting ensemble can do without any interaction and represents the fair single-agent comparison point.

Table 3 summarises the three accuracy levels together with an oracle upper bound (fraction of repetitions in which at least one agent holds the correct answer at round 0).

Dataset	Per-agent	Majority vote ($r=0$)	After debate	Oracle ($r=0$)
GPQA	38.2 %	40.5 %	42.2 %	70.6 %
HiddenBench	14.9 %	7.6 %	26.1 %	46.3 %

Table 3: Accuracy decomposition at round 0 and after debate. All configurations pooled ($N = 15,000$ per dataset). Oracle: fraction of repetitions where at least one agent is correct at round 0.

The two datasets are different in kind, not merely in difficulty, and cannot be pooled; we therefore analyse them separately throughout. They tell qualitatively different stories:

- **GPQA.** The per-agent accuracy at round 0 is 38.2 %, and majority voting raises this modestly to 40.5 % by aggregating four independent guesses. Debate adds a further +1.8 percentage points (pp), reaching 42.2 %. This marginal gain is consistent with findings across the multi-agent debate literature, where homogeneous agents on knowledge-intensive tasks tend to show small absolute accuracy improvements over single-agent baselines (Du et al., 2024, Liang et al., 2024, Zhang et al., 2025). The model has reasonable prior knowledge on GPQA questions; debate refines but does not transform the outcome. Notably, the oracle bound of 70.6 % reveals that the correct answer is present among the agents in roughly two thirds of repetitions: the limiting factor is not knowledge but the group’s ability to identify and amplify the correct voice. The discrepancy between oracle and achieved accuracy is a documented property of plurality aggregation under peer influence rather than an artefact of our setting (Bertalanic and Fortuna, 2026).
- **HiddenBench.** Here the dynamics are strikingly different. The per-agent accuracy at round 0 is 14.9 %, already well below random. This is expected rather than surprising: HiddenBench is constructed to exploit a shared bias in the underlying model, so individual agents are drawn towards the same attractive-but-wrong option instead of erring independently. Majority voting does not help, it *reduces* accuracy further to 7.6 %. This is a direct consequence of correlated errors: agents do not fail randomly across the options but share a systematic bias, all gravitating towards the same wrong answer. When agents are drawn from a single model, response similarity produces static debate dynamics that converge to the shared misconception rather than cancelling it out (Estornell and Liu, 2024), an LLM analogue of the biased information sampling documented in human hidden-profile experiments (Stasser and Titus, 1985). Debate reverses this partial collapse: final accuracy reaches 26.1 %, a gain of +18.5 pp over the majority baseline.

Interpretation. The contrast between GPQA and HiddenBench is not simply a difficulty difference. It reflects a qualitative shift in the role of debate. On GPQA, debate acts as a *refinement*: agents start mostly correct, and the system consolidates. On HiddenBench, debate acts as a partial *correction*: agents share a systematic bias that drives the majority vote well below chance, and structured discussion recovers most of that deficit without

pushing the group clearly above chance. The self-reinforcement, bistability, dominance, and limit-cycle analyses in the following sections explain when and why this correction succeeds or fails.

Effect of memory window. Figure 2 shows accuracy and mean rounds to consensus broken down by memory window $W \in \{1, 2, 5\}$ for each dataset and topology.

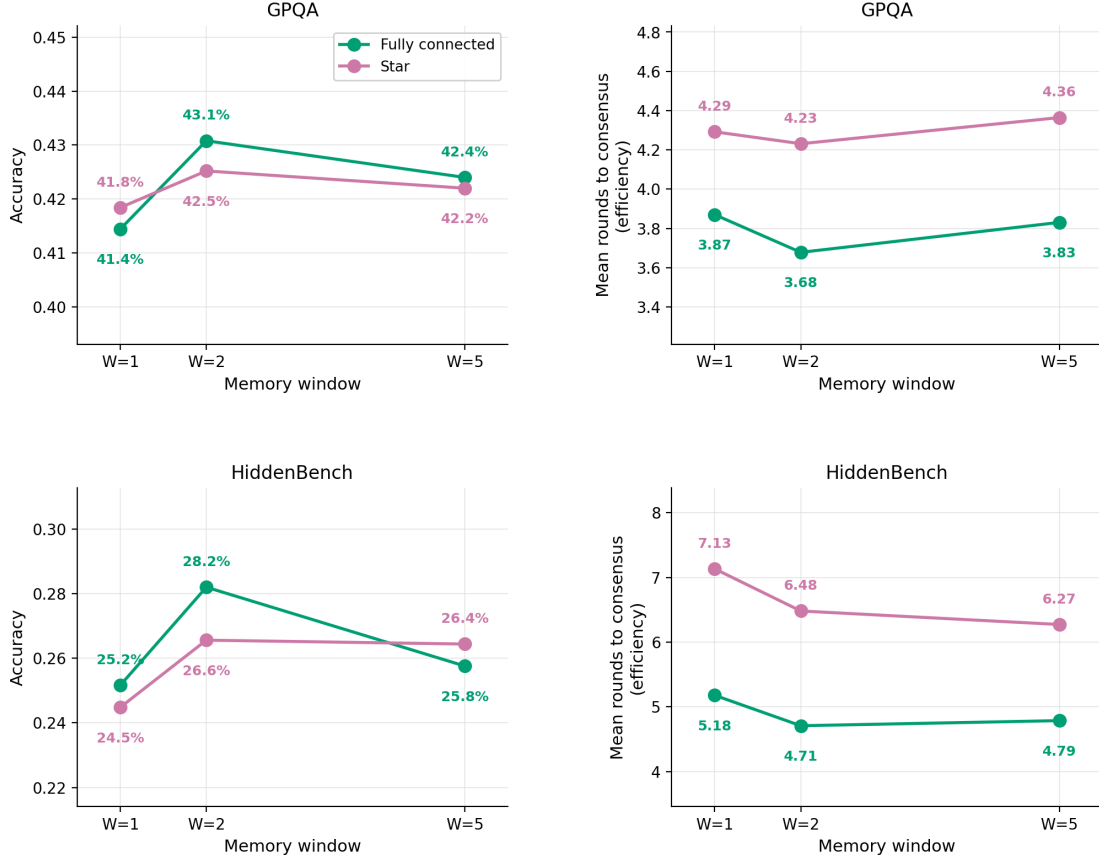


Figure 2: Accuracy (left column) and mean rounds to consensus, i.e. efficiency (right column), as a function of memory window W , for GPQA (top row) and HiddenBench (bottom row). Lines connect the two topologies.

- **Memory window shapes accuracy; $W = 2$ consistently ranks first.** Across both datasets and both topologies, $W = 2$ produces the highest per-task accuracy, outperforming $W = 1$ by up to 3.0pp and $W = 5$ by up to 2.4pp (full per-configuration breakdown in Appendix A.1). We report this as a consistent ordering rather than an established effect. At the task level the differences between memory windows are small and do not hold up under multiple-comparison correction (Section 7.1), so the ordering is suggestive at best. A window of one round gives agents only the most recent exchange, which is too little context to detect persisting disagreement. A window of five carries older, less relevant rounds that dilute the signal from the current debate state. $W = 2$ strikes the balance: enough history to read the trajectory of opinions without burying the most recent exchange.

Effect of topology. Table 4 reports mean accuracy and mean rounds for fully-connected (fc) and star topologies. Rather than pooling across W , the table presents the pooled figure as a summary; the full per- W breakdown is in Appendix A.1, which confirms the same pattern holds in each of the six individual cells.

Dataset	Topology	Accuracy		Mean rounds	
		Mean	Δ	Mean	Δ
GPQA	fc	42.3 %	0.0 pp (ns)	3.79	−0.50 (***)
	star	42.2 %		4.30	
HiddenBench	fc	26.4 %	+0.6 pp (ns)	4.89	−1.74 (***)
	star	25.8 %		6.63	

Table 4: Accuracy and mean rounds to consensus by topology, pooled over $W \in \{1, 2, 5\}$ ($n = 7,500$ repetitions per cell). In the Δ columns, ns denotes not significant and *** denotes $p < 0.001$ (task-paired Wilcoxon over $n = 50$ tasks). Accuracy differences are not significant on either dataset ($p = 0.92$ for GPQA, $p = 0.60$ for HiddenBench); round differences are large and highly significant in both. Pattern holds in each individual W cell; see Appendix A.1.

- **Topology has no detectable effect on accuracy.** Aggregating each task to its mean accuracy and pairing fully connected against star across the 50 tasks, the difference is negligible for both datasets (GPQA $\Delta \approx 0.0$ pp; HiddenBench $\Delta = +0.6$ pp). We find no evidence that the choice of topology determines whether the system reaches the correct answer. Two independent studies report the same null. In a networked convention-formation task, the network type an agent sits in has no significant effect on how well the group finally coordinates (Mehdizadeh and Hilbert, 2026); and in a debate task, giving each agent more visible peers leaves accuracy flat and never above the non-communicative baseline (Bertalanic and Fortuna, 2026).
- **Topology has a large effect on efficiency.** Star topology requires 0.50 more rounds on GPQA and 1.74 more rounds on HiddenBench compared to fully connected. The hub bottleneck in the star slows convergence: every spoke must route information through a single central node, which aggregates conflicting signals and commits more slowly. This is a genuine trade-off rather than a pure cost, however: the star carries only $2(N - 1) = 6$ messages per round against the $N(N - 1) = 12$ of the fully-connected topology (Section 4.1), so each of those extra rounds is roughly half as expensive in tokens, and the two topologies are closer in total communication cost than the round counts alone suggest.

Rounds and accuracy. The relationship between rounds to consensus and accuracy reveals a striking asymmetry (full plot in Appendix A.2). On GPQA, accuracy peaks at rounds 3–4 and is lower both for the earliest stops and for longer debates: neither very fast nor very slow consensus is reliably correct. On HiddenBench the pattern is

opposite: repetitions stopping at round 2 achieve only 2.5 % accuracy, far below random, because early consensus means the group has locked in on the majority-biased wrong answer before any corrective exchange; accuracy then rises sharply over the next few rounds before flattening.

This asymmetry is one instance of a broader point that governs the whole chapter. The two benchmarks are so different in kind, GPQA a knowledge task where debate refines a mostly-correct start and HiddenBench a bias task where it must correct a systematically wrong one, that the motif analyses in the following sections must be read per benchmark rather than pooled. The same measurement routinely points in opposite directions on the two datasets, so the dataset is the first variable to condition on before interpreting any result.

6.2 Self-Reinforcement

Self-reinforcement asks whether an agent that keeps its vote grows steadily more confident in it, measuring the debate’s tendency to entrench a position rather than hold it open to revision. We read it from the slope β_{seg} of confidence against round number within each *vote-stable segment* (a run of at least three consecutive rounds on the same vote), and summarise a cell by the prevalence p_{SR} , the fraction of segments with a positive slope, against a random-walk null of 0.5; the full construction is given in Section 5.2. Pooled across both benchmarks and all configurations, the segment filter leaves 133,696 segments to analyse. Three findings follow, moving from how often self-reinforcement occurs, to what it does to the debate, to what it does to accuracy.

- (i) **It is universal.** p_{SR} never approaches its null, ranging from 0.733 to 0.881 across the 12 (dataset, topology, W) cells, with all 50 tasks individually above 0.5 in every cell (per-configuration values in Appendix A.3). Confidence climbs fastest exactly where the debate has least to resolve, in repetitions that never change their votes and in unanimous starts, while topology and memory move the slope an order of magnitude less than the initial vote composition does. The pattern is entrenchment, not evidence integration.
- (ii) **It drives convergence.** That same entrenchment is what makes the debate settle: tasks with steeper confidence ramps reach consensus in fewer rounds in all 12 cells (task-level Spearman ρ between -0.73 and -0.89 ; Figure 3). This is the clearest and most consistent association in the analysis.
- (iii) **It is blind to correctness.** Faster convergence is not more accurate convergence. Whether self-reinforcement helps depends entirely on *what* it amplifies, which the initial vote distribution fixes in advance: when the correct option is already among the opening votes, stronger reinforcement pushes the group toward it ($r_{pb} = +0.20$ on GPQA, $+0.15$ on HiddenBench); when no agent starts correct, the same force only locks in a wrong answer sooner ($r_{pb} = -0.11$ and -0.07). The accuracy scatter is given in Appendix A.3 (Figure 12).

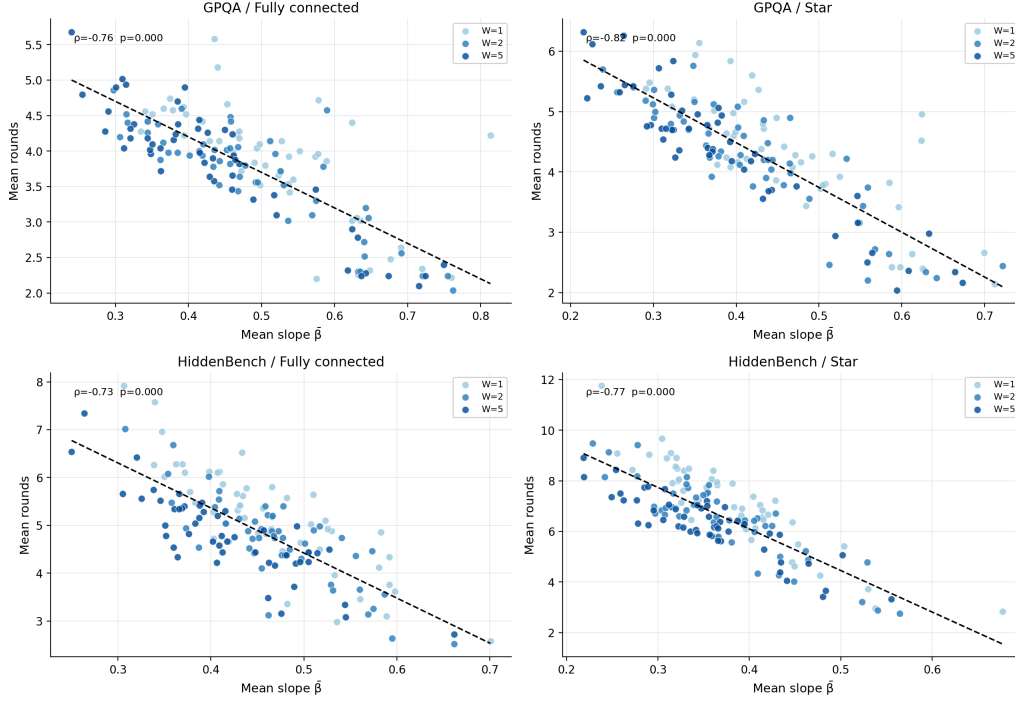


Figure 3: Mean segment slope $\bar{\beta}$ vs. mean rounds to consensus at the task level. Each point is one task; colours denote memory window. Dashed line: pooled OLS fit.

The three findings describe a single behaviour: self-reinforcement is the engine that converges the debate, but it converges on whatever the opening votes already favour rather than on the correct answer, amplifying the initial composition regardless of its quality, and rising confidence is no guide to that quality since the group grows more sure whether its final answer is right or wrong. This is the measured form of the entrenchment mechanism that Oh et al. (2026) identify as a root cause of belief entrenchment, that Estornell and Liu (2024) derive from response homogeneity, and that Cau et al. (2026) describe as a selective amplifier of locally consistent change; it also mirrors reports that debate confidence rises anti-Bayesianly without new evidence (Prasad and Nguyen, 2025, Bertalaníč and Fortuna, 2026). The seeded experiment (Section 6.6) confirms this dependence on the initial composition causally.

6.3 Bistability

Bistability asks whether a task admits more than one stable consensus outcome, and whether the opening vote distribution decides which one the debate reaches. We read it per (task, configuration) cell from two signatures: N_{eff} , the inverse-Simpson diversity of the endpoint attractors (how many stable outcomes coexist across repetitions), and Cramér’s V , which measures how strongly the initial votes select among them (basin sensitivity). A cell is *monostable* when one attractor dominates ($N_{\text{eff}} < 1.5$), *multistable* when several coexist and the initial votes predict the endpoint, and *stochastic* when several

coexist but the endpoint does not follow from the start; the full construction is given in Section 5.3. Repetitions are eligible if they settle on a unanimous endpoint (the 1.97 % that end split are excluded). Three findings follow.

- (i) **Multiple stable outcomes are the norm, and they belong to the task.** Across the 600 cells, 76 % show more than one coexisting attractor ($N_{\text{eff}} > 1.5$), split between deterministic basin selection (multistable) and outcomes that vary independently of the opening votes (stochastic; the per-dataset breakdown is in Appendix A.5). This structure is a property of the question, not the configuration: neither topology nor memory shifts either metric (all task-level confidence intervals contain zero), while per-task N_{eff} is highly consistent across the six configurations (mean pairwise Spearman $\rho \approx 0.90$ on GPQA and 0.84 on HiddenBench). Multiple absorbing states are the expected structure for this class of system, where naming-game dynamics admit one equilibrium per available convention (Baronchelli, 2018, Ashery et al., 2025); our repetition design adds the ability to see which of them a given task actually visits, where prior LLM studies report a single run or run-level means (Cau et al., 2026, Mehdizadeh and Hilbert, 2026).
- (ii) **Bistability helps exactly when the default answer is wrong.** Across tasks, more coexisting attractors go with *higher* accuracy on HiddenBench ($\rho = +0.44$) but slightly *lower*, non-significant accuracy on GPQA ($\rho = -0.16$). The opposition follows from what the dominant attractor is: on GPQA, where agents usually start near the correct answer, a single attractor is already the right one and extra basins only add wrong-answer traps; on HiddenBench, where the majority attractor is the shared wrong answer, competing basins give the correct minority a foothold from which to win. Consistent with this, on HiddenBench the correct attractor is selected more sharply than a wrong one (Cramér’s V of 0.42 versus 0.33): the opening votes predict the outcome more reliably when that outcome is correct.
- (iii) **It is the basin count behind self-reinforcement.** Bistability and self-reinforcement are two views of the same convergence. Monostable cells carry the steepest confidence ramps, and N_{eff} anticorrelates with the mean self-reinforcement slope at the task level ($\rho = -0.75$ on GPQA, -0.44 on HiddenBench): when one attractor dominates, agents lock in early and confidence climbs unchecked, whereas when several coexist the reinforcement signal is split across diverging trajectories. Self-reinforcement is the within-basin amplifier; bistability counts the basins.

Bistability is thus the structural counterpart of the amplifier: each task carries a small set of stable outcomes fixed by the question itself, and the opening vote distribution selects among them, so a diversity of attractors helps only when the single basin it would otherwise collapse into is the wrong one.

6.4 Dominance

Dominance measures how unequally influence is shared. From the confidence-weighted conversion-credit graph, the Herfindahl index D runs from 0 (every agent converts equally many peers) to 1 (a single agent causes every opinion change); the full construction is given in Section 5.4. One structural caveat shapes what can be tested: in the star, the centre is the only agent a leaf can ever convert to, so its raw D is mechanically inflated and its surrogate null degenerate for most repetitions. We therefore report raw D for both topologies but test for *emergent* concentration, above what the wiring alone forces, only in the fully-connected topology, where all agents are structurally equal (per-cell D values in Appendix A.4). Three findings follow, moving from whether a leader emerges, to whether that leader helps, to what its dominance costs.

- (i) **A leader emerges even without a hub.** In the fully-connected topology, where no agent is structurally privileged, conversion credit still concentrates on a de-facto leader above the topology-keyed null: the excess is positive in 49 of 50 GPQA tasks and all 50 HiddenBench tasks, and the fraction of repetitions flagged as significantly concentrated (7.7 % on GPQA, 10.9 % on HiddenBench) runs well above the 5 % chance baseline (Figure 4). On HiddenBench the effect strengthens with larger memory windows, as if more context lets a leader consolidate. The same endogenous asymmetry in nominally symmetric LLM groups is documented elsewhere, where agents form preferential-attachment links (Mehdizadeh and Hilbert, 2025a) and structural position predicts coordination success (Mehdizadeh and Hilbert, 2026).
- (ii) **A leader helps only when the debate chooses it.** Because the group converges onto the dominant agent’s position, that agent’s quality decides the outcome. Where the leader *emerges* from a fully-connected debate, higher dominance goes with higher accuracy on HiddenBench ($\rho = +0.56$), and on GPQA the emergent hub even carries a genuine expertise signal (correct 42 % of the time against a 25 % baseline). Where the leader is instead *imposed* by the wiring, the star hub, higher dominance goes with lower accuracy ($\rho = -0.29$ on HiddenBench), because a pre-assigned and often uninformed centre entrenches the shared initial bias. Dominance is thus not uniformly benign: it amplifies whichever agent leads, and only an endogenously selected leader tends to be the right one.
- (iii) **Concentration costs rounds.** Whatever it does to accuracy, dominance consistently slows convergence. Within each task, configurations with higher mean D take more rounds to settle (per-task Spearman $\rho = +0.55$ on GPQA and $+0.73$ on HiddenBench, positive in nearly every task): a dominant agent must press its position across many rounds before subordinates capitulate.

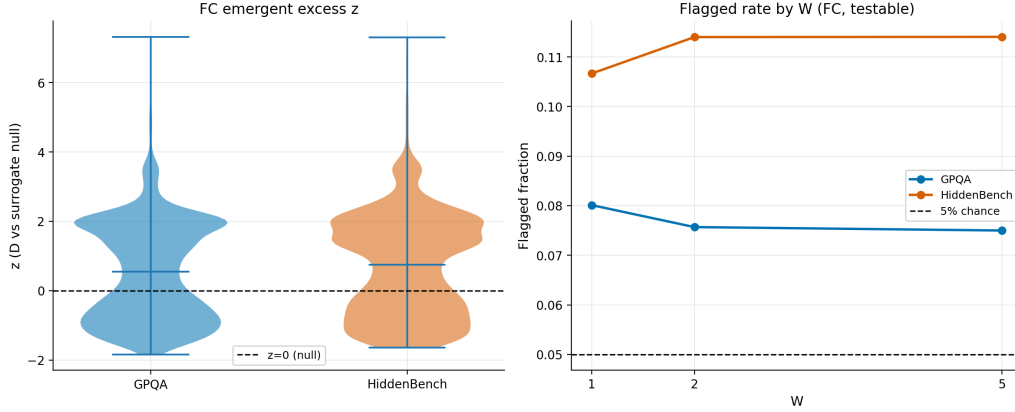


Figure 4: Left: distribution of excess z -scores in fc (testable repetitions only); dashed line: $z = 0$ (null). Right: flagged fraction by memory window; dashed: 5 % chance baseline. HiddenBench consistently exceeds GPQA in both magnitude and flagged rate.

Together these make dominance another face of the amplifier. The debate reliably elects a leader even when its wiring imposes none, and the group’s fate then follows that leader: concentration helps when the leader is endogenously selected and well-informed, and hurts when a fixed hub entrenches a shared error, while in every case a more dominant leader makes consensus slower to reach. The contrast between an emergent and an imposed hub is taken up directly in Section 6.7.

6.5 Limit Cycles

Limit cycles are the rarest motif in this system, and the only one that is a failure rather than a variant of convergence: a limit cycle is a debate that never settles, oscillating between vote configurations instead of reaching a fixed point. We detect them with categorical recurrence quantification, flagging a sequence when its determinism exceeds a shuffle-null floor and its dominant period lies in a genuine oscillatory band ($2 \leq \hat{P} \leq \lfloor L/2 \rfloor$); the full construction is given in Section 5.5. The cooperative design suppresses oscillation almost entirely: 99.5 % of agent vote sequences and 98.7 % of system composition sequences converge to fixed points and never enter the test. Three findings follow.

- (i) **They are rare but genuine.** Of the small non-converged remainder, a significant periodic signature survives in 47 agent-level and 32 system-level cases, both far above the shuffle-null floor (detection funnel in Appendix A.6). Inspected individually, 70 % of the flagged agent cases are clean period-2 orbits whose recurrence matrices show the characteristic checkerboard stripe (Appendix A.6); the rest are near-convergent artefacts. The signal is real, not a test artefact.
- (ii) **They occur only in the star topology.** Almost every genuine cycle sits in a star configuration (45 of 47 agent-level, 26 of 32 system-level); applying the same criteria to the fully-connected cases leaves none that are genuine, all being stable-revisit or

near-convergent artefacts. Flagged cases also concentrate on HiddenBench (34 of 47 agent-level). Figure 5 shows the two clearest examples: a collective cycle in which all four agents alternate between two options in near-perfect synchrony for the full 15 rounds, and a hub-driven cycle in which the group follows a single agent’s $A \leftrightarrow C$ alternation from round 5 onward.

- (iii) **The mechanism is hub-driven majority inversion, purely an efficiency failure.** Every limit cycle runs to the hard round cap without ever reaching consensus, so early stopping never fires. Inspection suggests a consistent, if uncertified, causal chain: agents defer to an observed peer majority, the star hub rebroadcasts that majority for the opposite option after each flip, and with two independently defensible evidence threads and a hub that starts in the minority, each flip recreates the conditions that caused it. The absence of genuine cycles in the fully-connected topology follows directly: with no single node aggregating all peer signals, the inversion loop cannot form. The practical consequence is that a cycle’s final majority vote is a low-confidence artefact of wherever it lands on round 15, so a hard round cap is the only reliable stop and repetitions that reach it should be flagged as unresolved rather than reported as a decision. Failure to recognise termination conditions is itself a catalogued failure mode of deployed multi-agent systems (Cemri et al., 2025), and our result names a specific dynamical route to it.

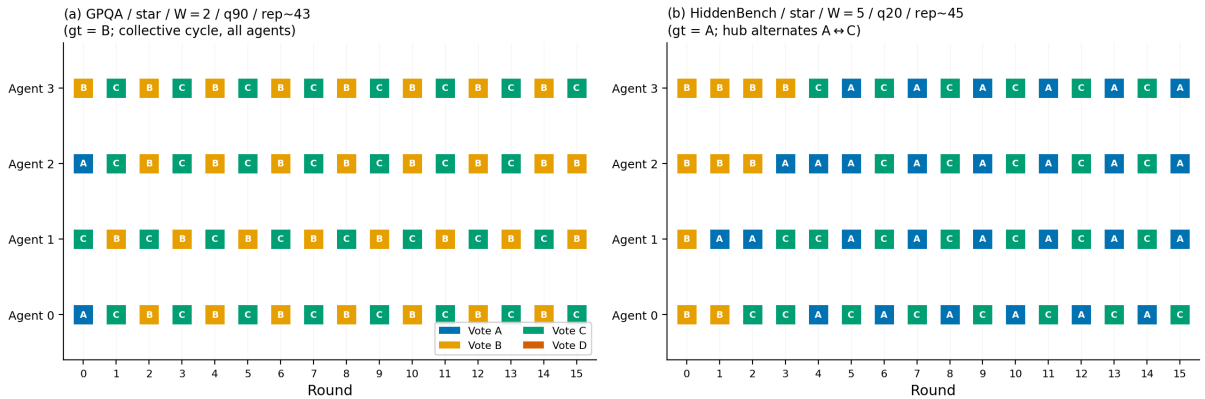


Figure 5: Vote-sequence timelines for two representative limit cycles. Each cell shows the vote (letter) held by an agent (row) at a given round (column); colour encodes the option. Left (a): collective cycle, all four agents alternate together (GPQA / star / $W=2$; ground truth B). Right (b): hub-driven cycle, agents split into an alternating $A \leftrightarrow C$ pattern from round 5 onward (HiddenBench / star / $W=5$; ground truth A). Neither repetition reaches consensus; both run to $T = 15$.

Limit cycles are therefore the failure mode of the same convergence machinery seen elsewhere: when a single aggregator keeps reconstructing the opposite majority, self-reinforcement never gets to lock the group in, and the debate oscillates instead of settling. That this happens only under the star, and never when every agent sees every other, fits the corroborating observation that unanimous states are otherwise effectively absorbing (Bertalaníč and Fortuna, 2026).

6.6 Causal Evidence for the Amplifier Mechanism

The four motif analyses characterise the *dynamics* of debate. This section asks a complementary question: does the opening vote composition causally determine the outcome, or is that association confounded? Three experiments address it in order: a within-task distribution analysis at $R = 1,000$ that establishes the observational pattern, a seeded dose-response that fixes the number of correct agents at $t = 0$ and confirms the causal interpretation, and an adversarial devil’s-advocate probe that tests the amplifier’s robustness.

For the observational analysis, three individual GPQA questions (q84, q125, q144), selected for mid-range accuracy and high initial-vote entropy to ensure genuinely uncertain outcomes, were each run for $R = 1,000$ repetitions at $W = 2$, fully-connected, $N = 4$.

Ground-truth presence in the initial vote distribution. The dominant finding is stark: whether the ground-truth option appears at all in the initial vote distribution is by far the strongest predictor of final accuracy, consistent across all three tasks (Figure 6, Table 5).

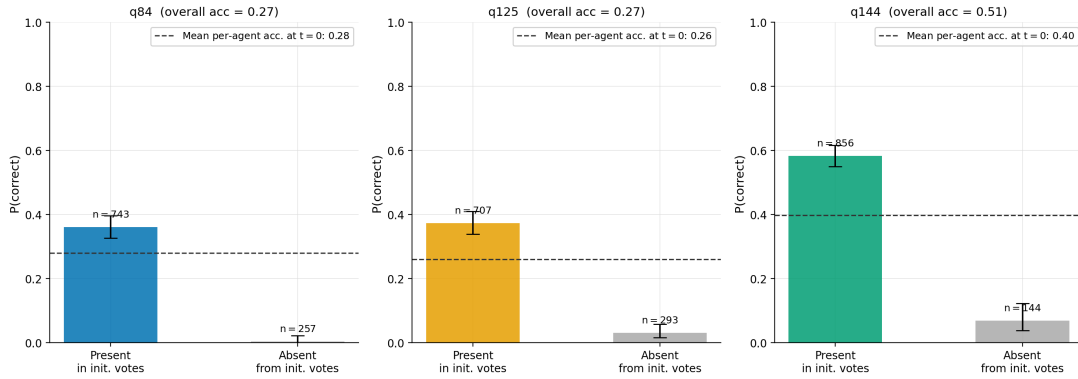


Figure 6: System accuracy conditioned on whether the ground-truth option is present in the initial vote distribution at $t = 0$. Left bars: repetitions where at least one agent voted correctly before debate; right bars: repetitions where no agent did. Dashed line: mean per-agent accuracy at $t = 0$. Wilson 95 % CI (Wilson, 1927). Three GPQA tasks, $R = 1,000$ each, $W = 2$, fully-connected.

Task	Overall acc.	GT absent (acc., n)	GT present (acc., n)	GT is plurality (acc., n)
q84	0.269	0.004 (257)	0.361 (743)	0.620 (213)
q125	0.273	0.031 (293)	0.373 (707)	0.552 (250)
q144	0.509	0.069 (144)	0.583 (856)	0.717 (403)

Table 5: Accuracy by ground-truth presence in the initial vote distribution, $R = 1,000$ per task. GT absent/present: whether the correct option appears in the vote profile at $t = 0$. GT is plurality: subset of present repetitions where the correct option holds the initial majority.

When the correct answer is absent from all initial votes, the system almost never reaches it: accuracy is 0.4 %, 3.1 %, and 6.9 % for q84, q125, and q144 respectively ($\chi^2 > 120$, $p < 10^{-27}$ in all cases). The rare correct answers in the absent condition are consistent with stochastic noise rather than genuine recovery. Within the present condition, accuracy is roughly $2\times$ higher when the correct option also holds the initial plurality than when it is present but not the plurality (e.g. 0.620 vs 0.257 for q84, the latter being the present-but-not-plurality subgroup, i.e. the $743 - 213 = 530$ repetitions where the correct option is in the opening profile but does not lead it). These findings suggest that the system operates primarily as a vote amplifier: debate selectively strengthens options already represented in the initial distribution and overwhelmingly reinforces whichever option holds the initial plurality. The capacity to surface a correct answer that no agent endorsed at the outset is negligible under the present design.

Debate length and confidence do not track correctness. Neither the length of a debate nor its confidence trajectory predicts whether it is right: longer debates are no more accurate, and confidence rises monotonically across rounds whether the final answer is correct or not (full feature-by-outcome correlations in Appendix A.7). What the outcome does track is the opening vote composition, which the next experiment manipulates directly.

Causal test: seeding correct agents. The observational pattern is confounded: whether the ground truth is present at $t = 0$ co-varies with task difficulty, agent confidence, and initial entropy. To isolate the causal contribution of the opening composition, we fix it directly. On 50 GPQA tasks (the same $W = 2$, fully-connected, $N = 4$ configuration, $R = 50$ repetitions each) we build every repetition’s initial four-agent vote profile by sampling from a pool of genuine, independent single-agent responses under a fixed number of correct seeds $k \in \{0, 1, 2, 3\}$: exactly k of the four agents begin on the correct option, the rest on real wrong answers. Only the initial composition is manipulated; every vote is a real response and the debate that follows is unmodified. Sweeping k turns the earlier binary contrast into a dose-response curve.

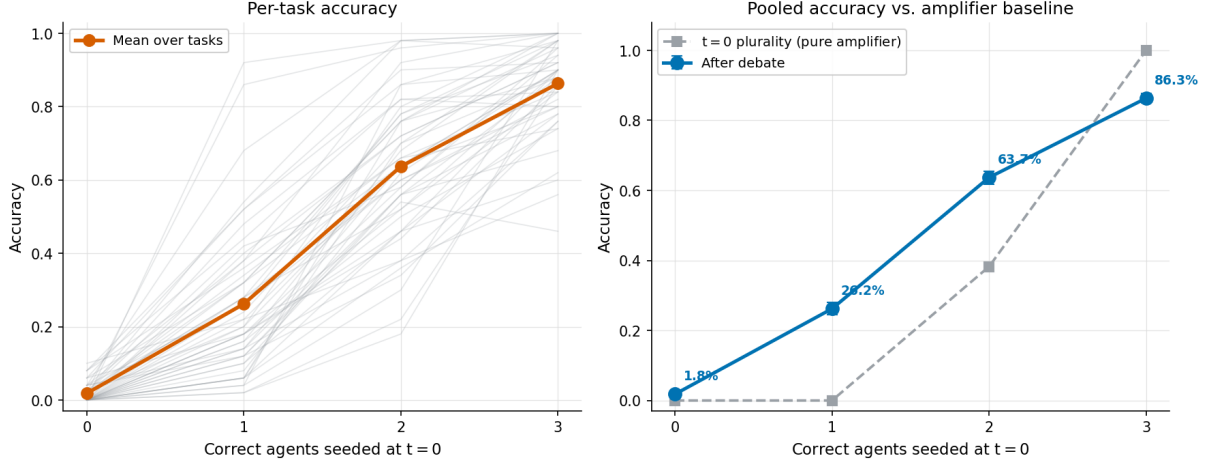


Figure 7: Accuracy as a function of the number of correct agents seeded into the initial vote profile ($k \in \{0, 1, 2, 3\}$) on 50 GPQA tasks ($R = 50$ each, $W = 2$, fully-connected). (a) Per-task accuracy (grey, one line per task) and the mean over tasks (orange). (b) Pooled accuracy with Wilson 95 % CI (blue) against the accuracy a pure plurality vote at $t = 0$ would already achieve (dashed grey). Debate exceeds this amplifier baseline for a correct minority or tie but falls below it for a correct supermajority.

Four findings follow.

- (i) **Accuracy rises monotonically with the number of correct seeds.** Pooled over 2,500 repetitions per condition, accuracy climbs from 1.8 % at $k = 0$ to 26.2 %, 63.7 %, and 86.3 % at $k = 1, 2, 3$ (Figure 7). This is not merely a pooled average: adding one correct seed raises per-task accuracy in 50 of 50 tasks for each of the first two steps and 47 of 50 for the last. Causally, the composition of the opening votes is the dominant lever on the outcome.
- (ii) **Debate is a lossy amplifier, not a reasoner.** Comparing each condition against the accuracy a pure plurality vote at $t = 0$ would already achieve (dashed line in Figure 7) isolates what the debate itself contributes. Where the correct answer starts behind, debate helps: a lone correct minority ($k = 1$, plurality accuracy 0 %) is carried to the right answer 26 % of the time, and an even two–two split ($k = 2$, baseline 38 %) resolves correctly 64 % of the time. Where the correct answer starts ahead, debate *erodes* it: a correct 3/4 supermajority ($k = 3$) that plurality voting would preserve with certainty is talked out of the right answer in roughly one repetition in seven (86 %, not 100 %). And from a standing start it generates almost nothing, recovering the answer de novo in only 1.8 % of $k = 0$ repetitions, confined to 21 of 50 tasks. Debate amplifies the initial signal while blurring it toward a coin flip, rather than reasoning its way to the truth.
- (iii) **Outcomes are all-or-nothing and decided early.** Every repetition in every condition ends fully unanimous: the correct seeds either convert the whole group or capitulate entirely, with no repetition ending on the ground truth as a surviving

minority. When the correct side wins it wins fast, reaching a plurality within the first one to two rounds (mean first round 1.3). Persuasion is front-loaded; the opening exchange, not protracted deliberation, decides the outcome.

- (iv) **A single persistent wrong voice overturns the group.** The amplifier’s indifference to correctness is starkest under adversarial perturbation. Re-running the $k = 2$ condition with one of the two non-correct agents instructed to argue a wrong option throughout ($R = 25$) collapses accuracy from 63.7% to 10.0%, lower in all 50 of 50 tasks and worse than a lone correct minority with no adversary ($k = 1$, 26.2%), even though the adversary starts outnumbered two-to-one. When these runs reach unanimity (87% of them), they converge to the adversary’s wrong answer 99% of the time. An agent that never updates signals conviction under the very rule that lets a correct minority occasionally win, so the group is pulled toward whichever voice is most persistent, not most correct.

The dose-response places the system in a recognisable region of the social-dynamics literature. A single correct agent is a committed 25% minority, and Centola et al. (2018) identify $\approx 25\%$ as the critical mass at which a committed minority overturns an established human convention; our correct minority tips the group about one time in four, short of reliable takeover, plausibly because $N = 4$ admits no finer minority and our agents carry the explicit interaction memory that Centola et al. (2018) find raises the threshold. The adversarial collapse is the same mechanism run in reverse: Agarwal and Khanna (2025) document a confidence-weighted persuasion override, in which a confident-but-wrong interlocutor talks a correct judge out of its answer, and Yao, Shang, Du, He, Lian, Zhang, Su, Swamy and Qi (2025) formalise inter-agent sycophancy as a driver of premature consensus, now targeted by prompt-level mitigations (Pitre et al., 2025).

Taken together, the seeded sweep upgrades the amplifier reading from correlation to intervention: the multi-agent system is a *lossy amplifier of its initial vote distribution*. It reliably strengthens whatever the opening votes favour, more so as that signal grows, yet it neither manufactures a correct answer that no agent held nor protects a correct one that a persistent dissenter attacks. The composition of the opening votes, not the length or intensity of the debate, is the dominant lever on whether collective reasoning succeeds. This intervention was run on GPQA only; the observational ground-truth-presence result holds on both benchmarks (Section 6.2), and extension of the causal claim to HiddenBench is discussed in Section 8.1.

6.7 Effective Topology

The topology parameter fixes who may read whom, but what matters for the outcome is who actually moves whom. We extract the *effective influence topology* of each fully-connected run from its conversion-credit graph: whenever an agent flips its vote, each neighbour that already held the new vote receives credit proportional to the flipping agent’s confidence (the credit function of the dominance metric, Section 6.4). Summing

over rounds and row-normalising assigns each run one of four structural types (Figure 8): *flat* (influence nearly uniform), *hub* (one moderately dominant agent), *two-hub* (two agents of comparable influence), and *star* (one agent responsible for $\geq 75\%$ of all conversions). Three findings follow.

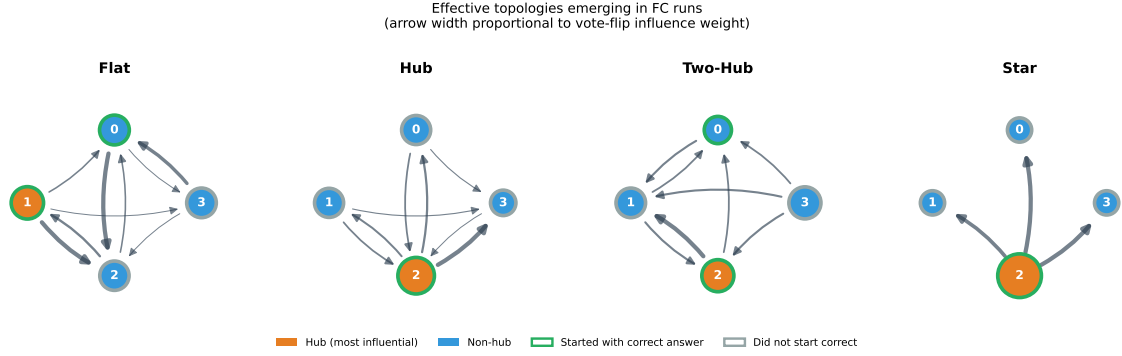


Figure 8: Archetypal examples of the four effective topologies observed in fully-connected runs ($N=4$). Arrow width is proportional to vote-flip influence weight. **Orange fill:** hub (most influential agent). **Green border:** agent that held the correct answer at round 0. The hub-share percentage is shown below each panel.

- (i) **Influence self-organises into a hierarchy, usually a two-hub one.** Even though every agent in a fully-connected run is wired identically, influence is almost never uniform: flat runs are rare ($\approx 5\text{--}7\%$). The dominant type is two-hub (58.7 % of GPQA runs, 59.2 % of HiddenBench), two agents of comparable influence acting as competing faction leaders, and its most common origin is a two-two initial vote split. A clean star, one agent driving three-quarters of all conversions, is rare (2.3 % GPQA, 5.4 % HiddenBench).
- (ii) **The rare star is far more accurate, because its hub is selected, not imposed.** Effective star runs reach by far the highest accuracy on both benchmarks (0.485 on GPQA and 0.562 on HiddenBench, against fully-connected means of 0.423 and 0.264; Figure 9), while the far more common two-hub tracks near or below the mean. The reason is not that a hub structure is intrinsically good but that, when a star forms, the debate has *elected* a hub that disproportionately started correct: 54.5 % of GPQA star hubs and 85.5 % of HiddenBench star hubs held the right answer at round 0, against a 25 % chance baseline. The outcome then follows that hub almost deterministically, correct in 86 % (GPQA) and 92 % (HiddenBench) of runs when the hub started right and collapsing to 16 % and 11 % when it started wrong. The two-hub battle is mediocre precisely because it lacks this selection signal: its correct faction wins 61 % of the time when present, but the group still converges, on a wrong answer, when no agent starts correct. The star is the amplifier at its most concentrated, and it happens to amplify the agent most likely to be right.
- (iii) **Diverse initialisation is the lever that produces stars.** Whether a star forms is driven almost entirely by the opening vote spread: with all four agents starting

on the same option $P(\text{star})$ is 0.3 %, rising monotonically to 3.3 %, 6.7 %, and 22.7 % as the number of distinct starting votes grows to four (Figure 10). This yields a concrete design guideline: prefer a fully-connected topology with deliberately diverse initialisation over a prescribed star. A nominal star fixes its hub before the debate can select one, and on the hard HiddenBench questions that pre-assigned centre holds the correct answer only 15 % of the time, against the 85.5 % of an FC-elected hub. Enforced perspective diversity has been proposed independently as a remedy for belief entrenchment (Oh et al., 2026) and follows from the theoretical analysis of debate homogeneity (Estornell and Liu, 2024); our contribution is the mechanism, that diversity works by raising the chance the debate elects an influence hub, and the elected hub tends to have started correct.

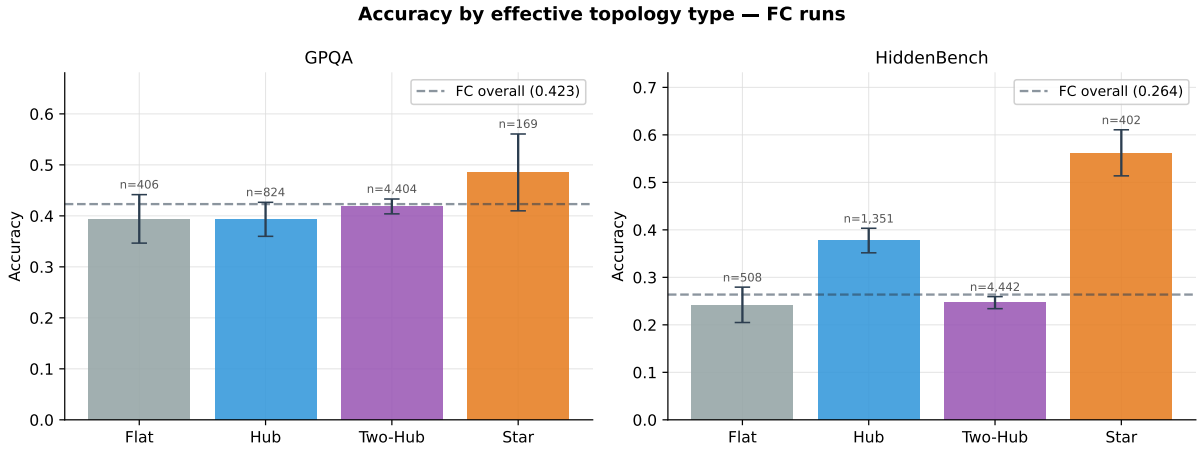


Figure 9: Accuracy by effective topology type for FC runs (error bars: ± 1.96 SE; dashed line: FC overall mean). Star topology is the clear accuracy leader on both benchmarks; the two-hub configuration — by far the most common — performs near or below the overall mean, especially on the harder HiddenBench questions.

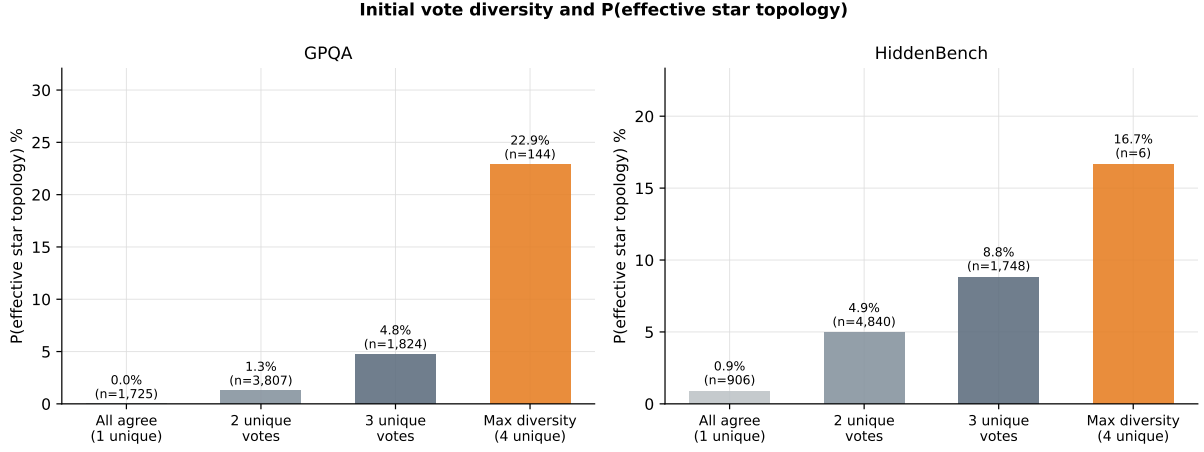


Figure 10: Probability of effective star topology as a function of initial vote diversity (n_{unique} = number of distinct votes held at round 0). Maximally diverse initialisation (one agent per answer option) raises $P(\text{star})$ from 0.3 % to 22.7 %.

Effective topology is the within-run face of the amplifier. The debate reliably elects an influence hierarchy that the wiring never specified, and the group’s answer follows whichever agent wins that election, so accuracy is high exactly when the elected hub is the correct one. This is the structural counterpart to the seeded result (Section 6.6), which establishes the same dependence on the opening composition causally, and to dominance (Section 6.4), which measures the same concentration in aggregate. It also mirrors reports that nominally symmetric LLM populations develop endogenous structural asymmetry, whether through an agent’s betweenness and clustering predicting its coordination success (Mehdizadeh and Hilbert, 2026) or through self-formed preferential-attachment links (Mehdizadeh and Hilbert, 2025a).

7 Discussion: From Dynamics to Diagnostics

This section brings together the empirical results from Section 6 and draws out their interpretation, design implications, and scope. Section 7.1 reads all four motifs as faces of a single amplifier mechanism. Section 7.2 develops the interpretation of that mechanism, its two operating regimes, and the status of null results. Section 7.3 translates the findings into empirically-grounded design principles and the groundwork for a diagnostic suite.

7.1 The Amplifier Mechanism: Reading the Motifs Together

The preceding sections measured four motifs and a high-repetition distribution. Read together, they do not describe four separate phenomena but four views of a single underlying mechanism. This subsection states that mechanism explicitly, shows how each measurement is a signature of it, and separates what the data establish from what they do not.

The convergent finding. Across every analysis in the preceding section, the property that most strongly governs the group’s final answer is the composition of the votes *before any debate takes place*. The high-repetition experiment shows this observationally: whether the ground-truth option is present at $t = 0$ moves per-task accuracy from near zero (0.4–6.9 %) to 36–58 %, and the seeded-initialisation experiment shows it causally: placing a single correct agent into an otherwise-wrong opening profile raises pooled accuracy roughly thirteen-fold (1.8 % to 23.7 %), in all 34 tasks tested (Sections 6.6 and 6.6). The self-reinforcement analysis supplies the mechanism by which the opening profile is converted into the outcome: SR is universal ($p_{\text{SR}} \in [0.73, 0.88]$ in every cell), it is the convergence engine (task-level $\rho(\bar{\beta}, \text{rounds}) \in [-0.89, -0.73]$ everywhere), and its accuracy effect is a pure amplification, positive when the correct option is already present ($r_{pb} = +0.20$ GPQA / $+0.15$ HB) and negative when it is absent (-0.11 / -0.07), cancelling to near zero in the margin. In short, the debate reliably *amplifies* whichever position the opening distribution favours; it only rarely *generates* a position no agent initially held (de-novo recovery in 15 of 34 seeded tasks, but at just 1.8 % of repetitions). We state this as the central empirical claim of the thesis: within the system we tested, deliberation functions predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence. This claim is not novel in kind. Oh et al. (2026) name homogenised dynamics that amplify the majority view regardless of correctness as one of two root causes of belief entrenchment, and Estornell and Liu (2024) derive the same conclusion theoretically from response homogeneity. What our data add is measurement: a per-motif decomposition of how the amplification is realised, conditioned on correctness throughout, and a causal test of the input on which it operates.

How each motif is a signature of the same mechanism. The remaining three motifs describe *how* the amplification plays out, not a separate process. *Bistability* is the cross-repetition face of amplification: because the outcome is selected by the (stochastic)

opening profile, repeated repetitions of the same task land in different basins, so multi-attractor coexistence is common ($N_{\text{eff}} \geq 1.5$ in 78.6 % of cells) and is a stable property of the task rather than of the configuration (cross-config $\rho = 0.78\text{--}0.91$). The within-task accuracy relationship is dataset-specific: on GPQA higher N_{eff} predicts lower accuracy ($\rho = -0.57$), because competing attractors create wrong-answer basins in a setting where agents usually start correct; on HiddenBench it predicts higher accuracy ($\rho = +0.41$), because bistability gives the minority-correct faction a competing stable basin in a setting where the dominant attractor is wrong. The negative coupling between bistability and self-reinforcement (task-level $\rho(N_{\text{eff}}, \bar{\beta}) = -0.79$ on GPQA) is exactly what the amplifier picture predicts: strong within-basin amplification means fast lock-in to one attractor per repetition, whereas high attractor multiplicity dilutes the SR signal across competing basins. *Dominance* is the within-repetition face: amplification is typically driven by a small, concentrated cascade (D correlates ≈ 0.87 on GPQA and ≈ 0.76 on HiddenBench at the task level with the top source’s share and *negatively* with total conversion volume), i.e. one agent tips a near-balanced start rather than the group converging by diffuse attrition. Configurations with higher dominance also require more rounds to converge ($\rho = +0.55$ GPQA, $+0.73$ HiddenBench, within-task), because subordinate agents resist before capitulating. *Limit cycles* are the boundary case in which amplification fails to terminate: when the opening profile is symmetric between two defensible options and the star hub sits in the minority, each flip reconstructs the opposing majority and the group never locks in, running to the hard cap. Finally, the effective topology analysis supplies a within-run complement: even with a fixed FC structure, the emergent influence star, where one agent captures most conversions, achieves near-ceiling accuracy because the correct agent tends to win the influence competition. This is the amplifier operating at a finer grain: the debate amplifies whichever agent the opening competition selects, and diverse initialisation raises the probability that competition is won by the correct one.

The one axis that matters, and the one that does not. Two negative results sharpen the picture. First, the *structural* design axes we varied, communication topology and memory window, do not change *whether* the system reaches the correct answer: topology has no accuracy effect at the task level (the confidence interval for the fc–star gap contains zero on both benchmarks), the memory window has none that survives multiple-comparison correction (a single HiddenBench/fc cell is nominally significant, $q = 0.064$), and bistability is invariant to both. Their influence is confined to the *dynamics*: star topology is slower, is the sole source of genuine limit cycles, and mechanically concentrates influence; larger memory windows speed HiddenBench convergence slightly. The structural topology parameter is distinct from the *emergent* influence topology: conditional on FC runs, whether an effective star forms does predict accuracy substantially, but this emergence is driven by the opening vote composition rather than by the structural configuration, so the same conclusion holds at every level of analysis, the opening vote is the dominant lever. Second, the system’s own internal signals are not indicators of correctness: confidence rises by essentially the same amount on correct and incorrect repetitions ($+1.63$ vs. $+1.64$ on GPQA, with incorrect rising very slightly more), and on HiddenBench early unanimity is anti-correlated with accuracy (round-2 stops reach only 2.5 %). What *does* move accuracy, the opening vote composition, is set before any of these knobs act.

Scope of the claim. These statements are bounded by what we varied and what we held fixed. We manipulated topology, memory window, task difficulty, benchmark, and (in the seeded experiment) the opening vote composition; we did *not* vary the base model, the number of agents ($N = 4$), agent heterogeneity, or the cooperative framing. The amplifier characterisation is therefore established *for a homogeneous, cooperative, 4-agent debate on one model family across two benchmarks*, with one axis (initial composition) tested causally and the rest observationally. Whether the same mechanism dominates for heterogeneous models, larger groups, or adversarial protocols is an open question we return to in Section 8. The value of the motif lens is that each of these signatures is separately measurable from ordinary interaction logs, so a practitioner can check which regime a given system is in rather than having to assume it.

7.2 Interpretation of Findings

The preceding section argued that all four motifs are signatures of the same amplifier mechanism. This section unpacks what that implies: how SR should be read as a dynamical process rather than a benefit, how the two benchmarks place the system in qualitatively different operating regimes, why the structural null results are informative, and what the robustness check does and does not show.

Self-reinforcement as an amplifier, not a benefit. The single most robust behavioural finding is that self-reinforcement is universal and is the engine of convergence, yet is blind to correctness: it is present in every configuration and predicts fewer rounds everywhere, but its relationship to accuracy reverses sign depending on whether the correct option is present in the opening votes. This is the empirical content of describing the system as an amplifier: reinforcement strengthens whatever plurality exists at the outset, which helps when that plurality is right and hurts when it is wrong. The reading aligns with, and gives a dynamical mechanism for, the critical literature that finds debate often fails to beat simple baselines (Zhang et al., 2025) and that majority voting alone accounts for much of the apparent gain (Choi et al., 2025): if the process mostly amplifies the initial vote, then its expected value over a population of repetitions is close to that of the opening vote itself. It is also consistent with the anti-Bayesian confidence drift documented by Prasad and Nguyen (2025), which we observe directly as outcome-independent confidence gain. The reversal of the predicted W -slope ordering on GPQA (§6.2) further supports a peer-conformity reading: shorter memory reduces each agent’s exposure to peer uncertainty, allowing confidence to ratchet more freely regardless of correctness.

One comparison we deliberately do not make is against chain-of-thought or self-consistency: our question is not whether debate is the best inference-time method but what *happens inside* it, so the fair reference point is the round-0 majority vote of the same four agents, and our “gain” figures are stated relative to that pre-deliberation baseline rather than to an external single-agent method.

Two regimes: refinement and correction. The two benchmarks are not merely easy and hard versions of one task; they place the system in two qualitatively different regimes,

and the motif signatures track the difference. On GPQA the agents usually begin with the correct option present (oracle 70.6 %), so amplification is mostly *refinement*: SR consolidates a frequently-correct start, the group ends above the 25 % chance baseline (42.2 %), and higher influence concentration is neutral-to-mildly-helpful. On HiddenBench the agents begin systematically wrong, majority voting at $t = 0$ scores below random (7.6 %) because errors are correlated, the LLM analogue of biased information sampling in human hidden-profile groups (Stasser and Titus, 1985, Li et al., 2025); yet the same amplifier mechanism operates here on a biased starting distribution. The +18.5 pp gain over the majority baseline reflects those repetitions where a correct initial signal is amplified, not a systematic reasoning correction: the oracle bound (46.3 %) confirms that the correct answer is available in roughly half of repetitions, and amplification of those openings drives the recovery. The motif differences follow: bistability is deeper on HiddenBench and, crucially, its relationship to accuracy reverses sign between datasets, helping on HiddenBench (where the dominant attractor is often wrong and a competing basin is beneficial) but hurting on GPQA (where competing attractors create wrong-answer basins that the system can fall into); the group’s final accuracy is no better than chance, limit cycles concentrate there, and influence concentration cuts both ways, helping in the fully-connected topology but *lowering* accuracy when forced through a single hub (star). The same mechanism (amplification of the opening profile) thus produces opposite-looking phenomenology depending on the reliability of that profile, which is precisely why a practitioner cannot read off system health from accuracy alone.

A complementary asymmetry appears at the individual level. On GPQA, agents who vote correctly at $t = 0$ are marginally more confident than those who do not (mean 7.39 vs. 7.25; point-biserial $r = +0.047$, $p < 0.001$): correctness and confidence align, so initially-correct agents carry both a numerical majority and a confidence advantage into deliberation. On HiddenBench the direction reverses ($r = -0.092$, $p < 0.001$; 7.20 correct vs. 7.39 wrong): an agent whose single private fact happens to mislead them is more confident than one whose fact points to the correct answer. The initially-correct minority thus faces a double obstacle—structural outvoting by three peers and a confidence deficit—which accounts for the smaller informed-agent influence advantage on HiddenBench (+1 pp) versus GPQA (+15 pp).

Structural axes shape dynamics, not correctness. Topology and memory window, the two axes most often tuned in the debate literature (Li et al., 2024, Kaesberg et al., 2025), had no accuracy effect that survives multiple-comparison correction, but clear effects on the dynamics: the star topology is slower, concentrates influence roughly three-fold, and is the only configuration that produces genuine (hub-driven) limit cycles, while larger memory windows modestly speed HiddenBench convergence. Both halves of this dissociation are corroborated externally. On the null side, network type has no significant main effect on coordination in an eight-topology factorial (Mehdizadeh and Hilbert, 2026) and communication density does not lift accuracy above a non-communicative baseline at any level tested (Bertalanič and Fortuna, 2026). On the dynamics side, the same eight-topology study finds a large memory-by-topology interaction on settling time, confirming that these axes move the trajectory even where they do not move the outcome. This dissociation between dynamical and epistemic effects is itself a finding: it means these

knobs are levers on *how* the group deliberates (cost, worst-case runtime, entrenchment) but not on *whether* it is right, and it explains why topology-tuning results in the literature are mixed (Yang et al., 2025). The key qualification is that the structural topology parameter is distinct from the *emergent* influence topology. Within FC runs, effective star formation, where one agent captures most of the conversion credit and the correct agent tends to win that competition, predicts above-average accuracy; this structure is inaccessible by assigning the star parameter but arises naturally from diverse opening positions. Prescribing a structural hub fixes the center before the debate can select it, and on hard questions the randomly assigned center started correct only 15.0% of the time versus 85.5% for a debate-elected hub. The mechanism is direct: within effective-star runs, when the hub votes correctly at $t = 0$ group accuracy reaches 86.1% on GPQA and 92.4% on HiddenBench; when it votes incorrectly, accuracy collapses to 15.6% and 10.7% respectively. The hub’s opening vote—a function of its randomly assigned private information—is the single largest predictor of group outcome in the star configuration, which explains why prescribing the star parameter is unreliable without also controlling which agent occupies the center.

Null and near-null results as findings. Several results are informative precisely because they are negative, and the study was designed to treat them as such (Section 1.2). Genuine limit cycles are essentially absent in the fully-connected topology; bistability is invariant to topology and memory; memory window does not move accuracy; and the marginal SR–accuracy correlation is near zero. The last is not a true null but a cancellation of two opposite conditional effects, which we could only see by conditioning on ground-truth presence, a reminder that an aggregate null can hide structured, offsetting effects. Reporting these against principled null models (shuffle surrogates, permutation tests, sign tests) is what separates “no effect” from “not enough power to see one,” and we have tried to keep that distinction explicit throughout.

Robustness to the system prompt. The agent persona used throughout includes an explicit anti-conformity instruction, so we can ask whether the motif structure is an artefact of that particular wording. A paired comparison against an earlier repetition with a milder persona, on the same HiddenBench questions and configurations (Appendix A.8), shows that the more critical prompt changes the *dynamics*, faster fully-connected convergence, fewer cap-hits, and a higher unanimous-convergence rate, but does not change accuracy (0 of 6 configurations) and does not suppress self-reinforcement (0 of 6). The apparent shift toward more multistable labels is a by-product of more repetitions converging rather than a genuine change in attractor multiplicity. The amplifier mechanism thus survives a deliberate attempt to make agents less conformist, which is mild evidence that it is a property of the interaction structure rather than of a single prompt. This is a single HiddenBench comparison, and is not evidence of robustness across models or group sizes, which we did not test. Bertalaníč and Fortuna (2026) argue on structural grounds that prompt-level remedies are unlikely to resolve these dynamics: adversarial prompting risks widening the oracle gap by increasing the rate at which valid reasoning is attacked, and distinct prompts across a homogeneous parameter space produce stylistic variance without decoupling the underlying error distributions. Our null result on accuracy is consistent with that expectation.

7.3 From Dynamics to Diagnostics: Design Implications

The motivation for this thesis was that outcome metrics tell a practitioner *whether* a configuration worked but never *why*, and give no way to tell a healthy deliberation from a pathological one (Miehling et al., 2025). The SOMA detectors are a first, partial answer: each is computed from ordinary interaction logs and each reports on a distinct aspect of the deliberative dynamics. We do not claim to deliver a validated diagnostic tool. What the evidence supports is more modest and, we think, more honest: a set of *empirically-grounded design principles*, tiered by how strongly our data support them, together with the observation that the very quantities used to detect each motif can serve as the instruments a future diagnostic suite would need. This subsection states the principles and that mapping.

A tiered reading of the evidence. We separate the principles by evidential strength so that the confident claims are not diluted by the speculative ones. *Tier 1 (strong: universal across our cells, and for the first, causal).* (1) The opening vote composition is the dominant lever on the outcome; investment in the process that generates round-0 votes dominates tuning of the debate itself. (2) The system amplifies far more readily than it generates, so it is best used as a consensus filter over candidate answers rather than as a mechanism expected to discover an answer no agent holds. (3) The system’s own confidence and unanimity are not signals of correctness and must not be used to gate trust (corroborated independently by Bertalaníč and Fortuna 2026, who report near-zero confidence–correctness correlations and show that peer exchange further degrades confidence discrimination). *Tier 2 (directional: consistent in sign, but dataset-dependent in size, so “verify on your own setup”).* (4) Communication topology is a lever on efficiency and worst-case behaviour, not accuracy; the fully-connected topology is the safer default and the star should be used only when a designated aggregator is wanted. (5) A hard round cap with explicit “unresolved” handling is necessary, because only some topologies admit non-terminating debates and every observed limit cycle ran to the cap (failure to recognise termination conditions accounts for 12.4% of annotated multi-agent failures in deployed frameworks; Cemri et al. 2025). (6) Decisive leadership helps when all agents can see one another (fully-connected) but *hurts* when influence is forced through a single hub (on HiddenBench/star, higher dominance lowers accuracy), so its value depends both on the topology and on whether the leader is more likely correct. (7) Diverse initialisation on multi-option tasks raises the probability of effective star formation from under 1 % to 23 %; since the effective hub disproportionately starts correct, this is a low-cost lever for above-average accuracy on questions with ≥ 4 answer options (consistent with enforced perspective diversity as an entrenchment remedy; Oh et al. 2026). (8) Persistent non-updating agents are a vulnerability: a single devil’s advocate reduces accuracy from 63.7 % to 10.0 % even when outnumbered two-to-one, so non-updating behaviour should be monitored and handled explicitly (the convention-formation analogue is minority-driven norm change; Ashery et al. 2025; see also Yao, Shang, Du, He, Lian, Zhang, Su, Swamy and Qi 2025 and Pitre et al. 2025 for mitigation). *Tier 3 (diagnostic / minor knobs).* (9) Bistability is a stable property of the task, so tasks can be profiled offline, and stochastic tasks, where convergence is effectively random and accuracy tends to be lowest, can be

flagged for routing elsewhere. (10) The memory window is a minor efficiency knob, not an accuracy lever. (11) Prompt changes (e.g. anti-conformity instructions) buy dynamical robustness, not accuracy.

Each detector is an instrument. The practical bridge from these principles to a future tool is that every principle above is checkable with a quantity we already compute (Table 6). This is what we mean by “groundwork for a diagnostic suite” rather than the suite itself: the measurements exist and have been validated against null models on 30,000 repetitions; what remains, packaging them into an online monitor, and validating the recommendations on models and group sizes beyond ours, is future work, not a claim we make here.

Principle	Observable (this thesis)	How a builder checks it
Opening composition is the master lever (T1)	$P(\text{GT present})$, initial plurality; seeded contrast	Log the round-0 vote profile; relate to outcome across repetitions
Amplifies, does not generate (T1)	De-novo recovery rate	Fraction of repetitions whose final answer was absent at $t = 0$
Confidence / unanimity $\neq$ quality (T1)	Outcome-independent confidence gain; early-stop accuracy	Check self-confidence calibration against held-out truth
Topology sets dynamics, not accuracy (T2)	Rounds, cap-rate, dominance D by topology	Compare efficiency / worst-case across topologies, not accuracy
Cap + unresolved-flagging is required (T2)	RQA determinism + period guard	Flag any repetition reaching the round cap as unresolved
Profile tasks; route stochastic ones (T3)	N_{eff} , Cramér’s V , cell label	Pre-classify tasks offline; treat stochastic tasks with caution
Diverse init raises $P(\text{effective star})$ (T2)	Initial vote entropy; effective topology type	On ≥ 4 -option tasks, seed distinct starting positions to raise $P(\text{star})$ toward 23 %
Non-updating agents are a vulnerability (T2)	Per-agent vote entropy across rounds	Flag agents with near-zero vote entropy; treat as suspect

Table 6: Mapping from design principle to the measurement that checks it. Every observable is computed in this thesis from ordinary interaction logs; the right column states how a practitioner would apply it to their own system. The suite itself is not built here, this is the set of instruments it would use.

Relation to enterprise deployment. The three Tier-1 principles are, in effect, safety principles for deploying such systems in consequential settings. A system that confidently reaches unanimity on a wrong answer, that cannot manufacture a correct answer absent

from its inputs, and whose outcome is largely fixed before deliberation begins, is one whose reliability must be established *upstream* (in the quality and diversity of the agents that produce the opening positions) and *externally* (through verification that does not trust the system’s own confidence), not through longer or more elaborate debate. This is a deflationary message, but a useful one: it directs effort toward the levers that our data show actually move the outcome and away from those that do not. We emphasise once more that these recommendations are grounded in a single model family, four agents, and two benchmarks, and should be treated as hypotheses to be re-checked, using exactly the instruments in Table 6, on any new deployment.

8 Limitations and Future Work

The claims in this thesis are bounded, and stating those bounds precisely is essential to not overstating them. This section groups the limitations into those of *scope* (what was varied) and those of *method* (how the motifs were measured), and then maps them onto a prioritised research agenda.

8.1 Limitations

Scope: what we did and did not vary. The experimental grid varies communication topology, memory window, task difficulty, and benchmark, and the seeded experiment additionally varies the opening vote composition causally. It does *not* vary the base model (a single model family throughout), the number of agents ($N = 4$), agent heterogeneity (all agents share priors and competence), or the interaction framing (cooperative, synchronous, two-phase). Every quantitative claim should be read as conditional on these fixed choices. This matters most for the results that already differ between our two benchmarks: if a single model on two datasets produces opposite signs for the value of dominance and of stylistic diversity, then the effect of changing the model, or introducing capability asymmetry, is likely to be at least as large. The heterogeneity axis is especially consequential because the critical literature identifies model heterogeneity, not debate per se, as the reliable source of gains (Zhang et al., 2025, Chen, Saha and Bansal, 2024), and capability asymmetry is what drives the accuracy-decreasing herding reported by Wynn et al. (2025). Our homogeneous design isolates the dynamics of a uniform collective by construction, but it cannot speak to heterogeneous ones. Recent evidence tempers this expectation in one direction. In a controlled comparison of mixed against single-model teams within the 7-8B class, heterogeneous compositions underperformed their best homogeneous member in six of eight condition-task pairs, and modal sycophancy in mixed teams remained comparable to homogeneous teams (Bertalaníć and Fortuna, 2026). Model diversity therefore does not appear to be sufficient on its own to dissolve the dynamics we characterise, which strengthens rather than weakens the case for first characterising the homogeneous baseline. Whether the same holds across capability tiers rather than within one remains untested.

A related scope boundary concerns the memory window. We tested $W \in \{1, 2, 5\}$, a range that keeps the history short relative to the round cap. At sufficiently large W , every agent can see the complete history of every other agent’s messages regardless of topology: a spoke in a star can reconstruct the full exchange through accumulated hub summaries, and the structural difference between fully-connected and star communication becomes negligible. This suggests that the topology effects we report—faster convergence under FC, hub-driven limit cycles under star—could diminish or vanish if W were large enough to bridge the information gap between topologies. Our results characterise the low-to-moderate memory regime; whether the topology effect survives large memory windows is an open question.

Decoding stochasticity. Every result in this thesis is conditioned on a single decoding temperature ($T = 0.4$). Temperature controls the entropy of the output distribution: lower temperatures concentrate probability mass on the model’s modal completion, higher temperatures spread it across alternatives. In the context of debate dynamics this is a first-order parameter: at high temperature agents start with more diverse positions and are harder to lock into a fixed point; at low temperature they converge faster but may become overconfident. A recent physics-inspired account models the biased consensus problem as a Potts spin system and shows analytically that the system undergoes a phase transition as temperature increases from a consensus phase to a disordered phase (Okawa, 2026). Empirically, ablating temperature from $T = 0.4$ to $T = 0.7$ in a comparable protocol reduced the modal sycophancy rate by 6.2 percentage points (Bertalaníć and Fortuna, 2026), suggesting that our findings may be conservative: the self-reinforcement and convergence rates we report are likely higher than they would be at typical deployment temperatures. Replicating the main analyses at higher temperatures is a natural sensitivity check we did not perform.

Task-selection bias from Tier-3 screening. All quantitative results are conditioned on Tier-3 tasks, those where the outcome varies across repetitions under the screening configuration. By construction, this excludes tasks that are trivially easy (unanimous at $t = 0$), nearly deterministic (the group always converges to the same answer), or too hard for any signal to emerge. The prevalence rates reported here, how often each motif appears, are therefore rates *within the contested regime*, not across the full task distribution. Extrapolating them to the broader population of questions, or to operating conditions where most tasks fall outside Tier 3, would overstate motif frequency.

The causal evidence is GPQA-only. The seeded-initialisation experiment, our only intervention and the strongest support for the amplifier claim, was run exclusively on GPQA (paragraph 6.6). This is a real restriction on the causal claim, and in one respect the less favourable case is the one left untested: on HiddenBench, where agents begin systematically wrong, amplification of the opening distribution should if anything be more consequential (and more often harmful), so the mechanism is plausibly at least as strong there. But that remains an expectation, not a measurement. The causal statement should therefore be read as established for GPQA and as a well-supported but observational hypothesis for HiddenBench; a seeded HiddenBench arm is the most valuable single extension of this work.

Adversarial agents and deployment risk. The experiments in this thesis use cooperative, homogeneous agents: every agent is instructed to reason honestly and update in good faith. A devil’s advocate probe (paragraph 6.6) suggests this is a consequential assumption. A single agent instructed to advocate persistently for a wrong answer reduces system accuracy from 63.7% to 10.0% even when two of four agents start correct, falling below the 26.2% obtained with a natural 1/4-correct minority and no adversary. This is a deployment risk: a consistently biased agent—whether from a misconfigured prompt, a compromised model, or adversarial injection—exploits the same front-loaded update dynamics that allow a lone correct minority to occasionally flip the group. Because the protocol has no mechanism for distinguishing persistent conviction from persistent misin-

formation, the amplifier can be hijacked asymmetrically: truth-seeding produces a $13.0\times$ lift relative to no ground truth; adversarial persistence erases that lift and then some. Quantifying this vulnerability across group sizes and topologies, and evaluating structural defences—outlier detection via vote entropy, trust discounting for non-updating agents—is an important direction that this thesis leaves open.

Effective topology analysis is FC-only and threshold-based. The effective influence topology is extracted only from FC runs; the star topology’s influence graph is structurally prescribed and the classification is not meaningful there. Within FC runs, the four-type taxonomy (flat, hub, two-hub, star) is defined by threshold rules on the share of total influence captured by the top one or two agents. The star boundary ($\geq 75\%$ for a single agent) and the sample sizes for the rare types, 2.3% and 5.8% star rates, mean that type-level estimates carry substantial uncertainty. The design-guideline claim (diverse init raises $P(\text{star})$) is observational: we measure that $P(\text{star})$ co-varies with initial vote diversity across naturally-occurring runs but do not directly manipulate diversity.

Method: signatures, not certified mechanisms. All four SOMA detectors identify *signatures*, not certified dynamical objects. The self-reinforcement slope is a phenomenological positive trend, not a proof of a positive-feedback loop; the RQA test flags periodic recurrence, not a mathematically certified limit cycle; N_{eff} measures endpoint coexistence, not a certified multistable attractor landscape; and the dominance index measures conversion concentration, not certified causal influence. With the single exception of the seeded-initialisation experiment, which is a genuine intervention, every association reported here is observational. Establishing that, say, hub identity *causes* the outcome, rather than co-varying with a question-specific expertise signal, would require perturbation experiments we did not run.

Specific measurement caveats. Four narrower caveats bound specific results. (i) The self-reinforcement null may be subject to survivorship bias: agents whose confidence drifts downward are more likely to flip their vote and terminate a vote-stable segment, which can censor negative slopes and inflate p_{SR} above the 0.5 baseline; the prevalence result is therefore conservative in direction but not bias-free in magnitude. (ii) The diversity $\rightarrow$ SR $\rightarrow$ rounds mediation is partly mechanical, since longer segments afford more rounds over which a slope can accumulate; we report it as a decomposition, not as evidence that SR temporally precedes convergence. (iii) The bistability N_{eff} measure can in principle exceed the option count when split (non-unanimous) endpoints are admitted; because we restrict attractors to unanimous endpoints (a 2.1% loss of repetitions), N_{eff} is bounded by the option count by construction, and we verified $N_{\text{eff}} \leq M$ in all 420 cells. (v) The dominance influence graph is built from vote adoptions during the *Update* phase, which is precisely where the protocol instructs agents to update after reading peers’ *Argue* drafts. Conversion edges therefore partly reflect the structural fact that agents are *supposed* to be influenced in this phase; a high dominance score on a hub-to-spoke edge may indicate genuine persuasive leadership or simply that the protocol routes all information through the hub. We treat dominance as a concentration *signature* and never as a certified causal influence for this reason.

Statistical multiplicity. This is an exploratory study reporting many tests across twelve configuration cells and several motifs. Inference is conducted at the task level (the unit of analysis, $n \approx 35$; Section 5), and we apply corrections within families of tests where appropriate (Benjamini–Hochberg false-discovery-rate control across configuration cells and outcome variables, permutation and surrogate nulls per detector), but we do not impose a single study-wide correction, and some false positives are expected across the full battery. We have tried to lean on effect sizes and on consistency across cells rather than on isolated p -values, and the central claims (universality of SR, the amplifier effect, the topology/accuracy dissociation) rest on effects that are large and repeated across many cells rather than on any single borderline test.

8.2 Future Work

The limitations above map directly onto a prioritised research agenda.

1. **From signatures to causes.** With the exception of the seeded experiment, our findings are observational. Targeted perturbation experiments, seeding basins to certify the bistability landscape, perturbing the hub to test whether dominance is causal or an expertise proxy, and an asynchronous protocol variant to isolate synchrony-induced cycles, would upgrade the remaining associations toward causal claims.
2. **Effective topology under heterogeneity and adversarial conditions.** The effective topology analysis is confined to FC runs with homogeneous agents. Whether effective star formation rates change under model heterogeneity, capability asymmetry, or partial non-cooperation is unknown; the devil’s advocate result suggests the formation mechanism is exploitable, so characterising it under adversarial conditions is a high-priority extension. A targeted diverse-initialisation intervention that deliberately seeds each agent with a distinct starting answer would also test the $P(\text{star})$ lever causally rather than observationally.
3. **Heterogeneity and scale.** The most consequential untested axes are model heterogeneity and group size. The critical literature identifies heterogeneity as the reliable source of debate gains (Zhang et al., 2025, Chen, Saha and Bansal, 2024) and capability asymmetry as a driver of harmful herding (Wynn et al., 2025); testing whether the amplifier characterisation survives heterogeneous agents and larger N is the natural next step, and one our homogeneous design cannot address.
4. **Correcting the self-reinforcement null.** The survivorship bias in the SR prevalence null (downward-drifting confidence censored by vote flips) should be addressed with a matched null that models the flip hazard, to establish how much of the above-baseline p_{SR} is genuine.
5. **Toward the diagnostic suite.** Finally, the SOMA instruments in Table 6 could be packaged into an online monitor and the Tier-2 recommendations validated across

models, group sizes, and task domains beyond the two studied here. This is the step that would turn the groundwork of this thesis into the practical tool it points toward.

6. **A generative model of debate dynamics.** The motif taxonomy is descriptive. A forward model of how agents update—whether as Fictitious Play, gradient-based reinforcement, or a social-learning rule—would make testable predictions about which motifs arise under which conditions and would connect the empirical patterns to game-theoretic equilibrium concepts (Mehdizadeh and Hilbert, 2026). This is the missing link between the diagnostic layer and a prescriptive theory of multi-agent deliberation.
7. **Removal-based influence attribution.** The dominance metric identifies concentration by tallying vote adoptions, a phenomenological measure. Complementing it with removal-based attribution—estimating each agent’s marginal causal contribution to the final outcome by masking or ablating their messages—would give a sharper, interventional measure of leadership (Anonymous, 2026a, Cui et al., 2025, Anonymous, 2026d). Applied to hub agents, this would test whether structural dominance translates to causal influence or whether dominance nodes are proxies for question-specific expertise.

9 Conclusion

This thesis asked whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and what such a treatment reveals that outcome metrics cannot. We built a controlled cooperative debate system, developed four literature-grounded motif detectors, and ran the full grid on two deliberately contrasting benchmarks, comprising 30,000 debates. We answer the three research questions in turn.

RQ1: measurement. The motifs are detectable, at very different prevalences, and each detector distinguishes genuine structure from a principled null. Self-reinforcement is *universal*: positive confidence trends dominate in every configuration ($p_{\text{SR}} \in [0.73, 0.88]$, all cluster-bootstrap intervals excluding the 0.5 baseline). Bistability is *common and task-intrinsic*: most task-config cells show genuine multi-attractor coexistence ($N_{\text{eff}} \geq 1.5$ in 76.3%), and this is a stable property of the task rather than the configuration. Dominance appears as *behavioural influence concentration* above the topology-keyed null even in fully-connected debate. Limit cycles are *rare but real*: 99% of sequences converge, and the small residue that does not yields 47 agent-level and 32 system-level statistically flagged cases, of which 33 and 7 respectively are genuine period-2 oscillations on manual inspection, arising almost exclusively under the star topology through a hub-driven mechanism. The absence of genuine cycles in fully-connected debate is itself a clean result rather than a measurement failure.

RQ2: characterisation. The two structural axes shape the *dynamics* but not the *correctness* of debate. Topology is the dominant structural lever: the star is slower, concentrates influence roughly threefold, and is the sole source of genuine limit cycles, while the fully-connected topology produces stronger self-reinforcement and faster convergence. The memory window has only secondary effects on convergence speed and none on accuracy, and bistability is invariant to both axes. What governs outcomes is the *composition of the opening votes*: motifs are informative as diagnostics of a dynamical process, not as guarantees of quality. Self-reinforcement is the convergence mechanism everywhere (task-level $\rho(\bar{\beta}, \text{rounds}) \in [-0.89, -0.73]$), yet it amplifies the initial vote distribution rather than selecting for the correct answer. Dominance is helpful when every agent sees every other but harmful when influence is forced through a single hub. Bistability marks the tasks where convergence is least reliable, and limit cycles mark pure efficiency failures. The motifs thus form an explanatory layer beneath accuracy, and are the basis for the tiered, empirically-grounded design principles of Section 7.3.

RQ3: causation. The outcome is largely determined before deliberation begins. Observationally, whether the ground-truth option is present in the opening votes is the strongest predictor of accuracy; causally, the seeded-initialisation experiment shows that placing one correct agent into an otherwise-wrong opening raises accuracy roughly thirteen-fold and in every task tested, while the system almost never manufactures a correct answer no agent initially held. The overarching finding, stated with its scope, is that *within a homogeneous, cooperative, 4-agent debate on one model family across two benchmarks*, deliberation functions predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence.

Multi-agent debate is often presented as a way to make language models reason better together (Du et al., 2024, Liang et al., 2024). Our results suggest a more precise and more cautious picture: in the homogeneous cooperative setting we studied, the debate mostly amplifies what its members already believe. The SOMA suite operationalises this insight into four measurable detectors, each computable from ordinary interaction logs; its broader value is as groundwork for a diagnostic instrument that would let a practitioner check, before trusting the group’s output, which dynamical regime the deliberation is actually in.

A Appendix

A.1 Accuracy and Efficiency: Full Configuration Table

Table 7 reports accuracy, mean rounds to consensus, fraction of repetitions hitting the hard cap $T = 15$, and mean vote flips for all 12 (dataset, topology, W) configurations ($R = 50$ repetitions per task, $n = 2,500$ repetitions per cell).

Dataset	Config	Accuracy	Mean rounds	p_{cap}	Mean flips
GPQA	fc / $W = 1$	41.4 %	3.87	0.2 %	2.98
	fc / $W = 2$	43.1 %	3.68	0.2 %	2.90
	fc / $W = 5$	42.4 %	3.83	0.4 %	2.91
	star / $W = 1$	41.8 %	4.29	1.0 %	3.11
	star / $W = 2$	42.5 %	4.23	1.0 %	3.77
	star / $W = 5$	42.2 %	4.36	1.6 %	3.43
HiddenBench	fc / $W = 1$	25.2 %	5.18	2.5 %	4.27
	fc / $W = 2$	28.2 %	4.71	1.0 %	3.92
	fc / $W = 5$	25.8 %	4.79	2.0 %	3.86
	star / $W = 1$	24.5 %	7.13	13.3 %	4.82
	star / $W = 2$	26.6 %	6.48	11.2 %	5.22
	star / $W = 5$	26.4 %	6.27	10.7 %	5.12

Table 7: Full system performance per configuration ($R = 50$, 50 tasks, $n = 2,500$ repetitions per cell).

A.2 Accuracy and Efficiency: Rounds vs. Accuracy

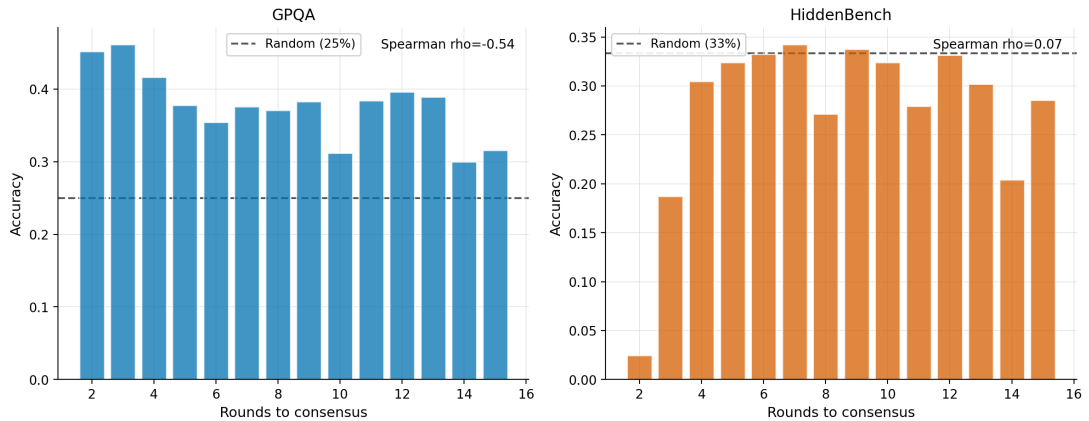


Figure 11: Mean accuracy as a function of rounds to consensus, for GPQA (left) and HiddenBench (right). Points show means over repetitions reaching each round count (minimum $n = 20$). All configurations pooled ($N = 15,000$ per dataset).

A.3 Self-Reinforcement: Supplementary Tables and Figure

Table 8 reports p_{SR} and mean slope $\bar{\beta}$ for all 12 (dataset, topology, W) cells; Table 9 breaks the segment-level slope down by structural condition; Figure 12 shows the marginal (pooled) relationship between $\bar{\beta}$ and task accuracy, whose near-zero correlation is discussed in Section 6.2.

Dataset	Topology	p_{SR} (range over W)	$\bar{\beta}$ (range over W)
GPQA	fc	0.763–0.825	0.441–0.515
GPQA	star	0.733–0.809	0.390–0.463
HiddenBench	fc	0.839–0.881	0.415–0.467
HiddenBench	star	0.804–0.844	0.350–0.373

Table 8: p_{SR} and mean segment slope $\bar{\beta}$ (confidence gain / round) per (dataset, topology) configuration, range reported over $W \in \{1, 2, 5\}$. $N_{\text{tasks}} = 50$, $R = 50$.

Condition	Group	n_{segs}	$\bar{\beta}$	p_{SR}
Ceiling	No ceiling	115,137	0.404	0.798
	Ceiling hit	18,559	0.564	0.910
Terminality	Non-terminal	14,341	0.262	0.682
	Terminal	119,355	0.445	0.829
Initial correctness	Init wrong majority	116,400	0.410	0.809
	Init correct majority	17,296	0.535	0.840
Vote stability	Some agent flipped	113,660	0.380	0.796
	Fully static rep.	20,036	0.685	0.908
Init vote dist.	4-0-0-0	21,435	0.663	0.897
	3-1-0-0	50,286	0.412	0.824
	2-2-0-0	26,629	0.370	0.791
	2-1-1-0	34,035	0.345	0.767
	1-1-1-1	1,311	0.308	0.656

Table 9: Mean segment slope $\bar{\beta}$ and p_{SR} by subgroup (all segments pooled across configs, $N = 133,696$). Pooled segment counts are descriptive; the subgroup orderings are confirmed at the task level.

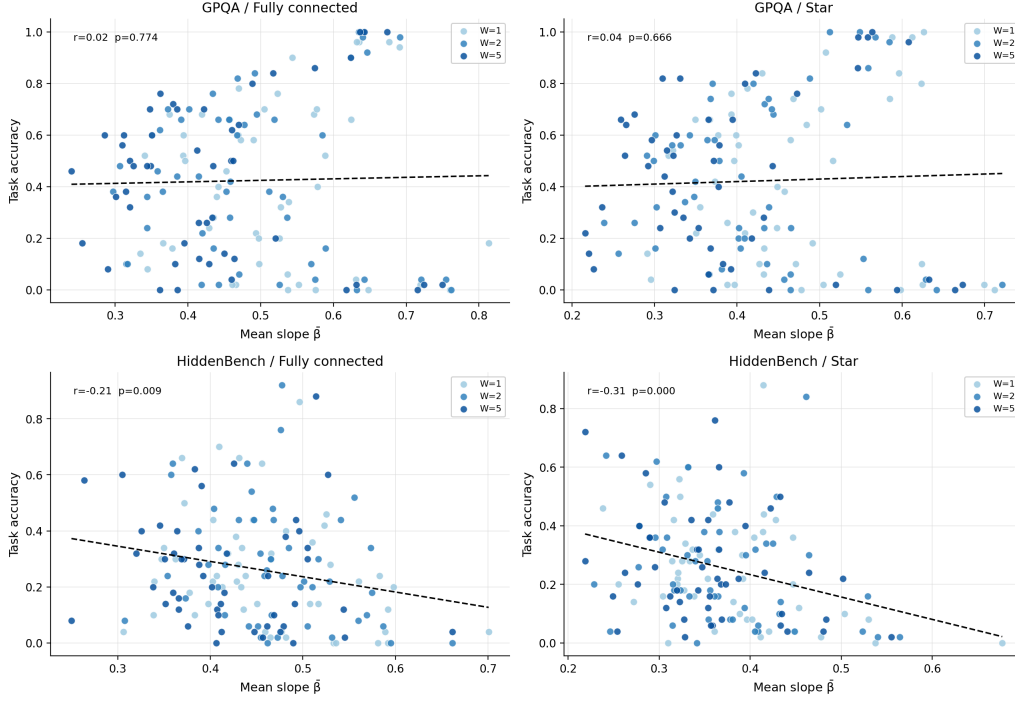


Figure 12: Mean segment slope $\bar{\beta}$ vs. task accuracy at the task level (pooled W). Dashed line: pooled OLS fit. Pearson r shown in each panel. The near-zero marginal correlations reflect a cancellation between opposite effects in the GT-present and GT-absent subpopulations (see Section 6.2).

A.4 Dominance: Full Configuration Table

Table 10 reports mean raw D per (dataset, topology, W) cell. Star D is $3.3\text{--}4.3\times$ higher than fully-connected across all cells, but this gap is largely mechanical: the star centre is the sole eligible conversion source for every leaf, so its influence share is structurally inflated and its surrogate null degenerate. Emergent excess ($z > 0$, flagged fraction) is therefore identified only for the fully-connected topology, as discussed in Section 6.4.

Dataset	Topology	Mean D			fc excess (pooled W)	
		$W = 1$	$W = 2$	$W = 5$	$\bar{z}$	Flagged
GPQA	fc	0.154	0.154	0.155	+0.53	7.7 %
GPQA	star	0.601	0.601	0.610	—	—
HiddenBench	fc	0.157	0.178	0.179	+0.73	10.9 %
HiddenBench	star	0.540	0.574	0.578	—	—

Table 10: Mean raw D per (dataset, topology, W) cell. fc emergent excess ($z > 0$) reported as mean z and flagged fraction (star null degenerate, not reported there). $N_{\text{tasks}} = 50$, $R = 50$.

A.5 Bistability: Label Distribution and Supplementary Figures

Table 11 gives the per-dataset breakdown of the three bistability classes referenced in Section 6.3. Of the 600 cells, 23.7% are monostable, 41.5% multistable, and 34.8% stochastic; HiddenBench is far more stochastic than GPQA.

Dataset	Monostable	Multistable	Stochastic	N_{eff} mean	V mean
GPQA ($M = 4$)	86	151	63	1.97	0.45
HiddenBench ($M \in \{3, 4\}$)	56	98	146	2.06	0.35

Table 11: Bistability label distribution across all 600 (task, config) cells ($N_{\text{tasks}} = 50$, $R = 50$). The label counts pool all tasks per benchmark; HiddenBench mixes 46 three-option and 4 four-option tasks. The N_{eff} column is a within-benchmark mean and should be read per option-count stratum, since N_{eff} is bounded by M and is not directly comparable across option counts.

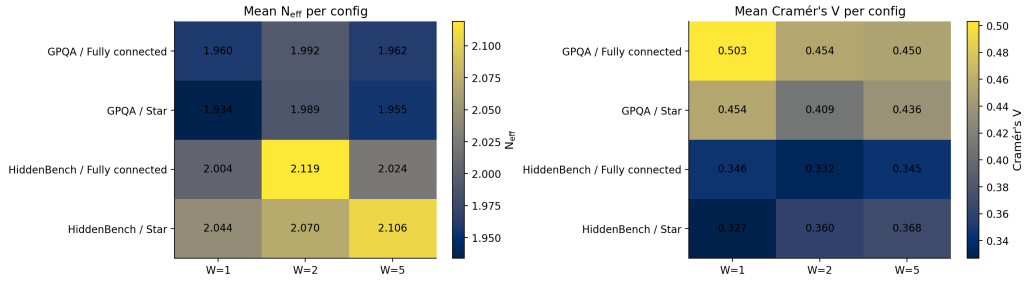


Figure 13: Mean N_{eff} (left) and Cramér's V (right) across all 12 (dataset, topology, W) configurations. The absence of a consistent gradient confirms that topology and memory window do not drive bistability.

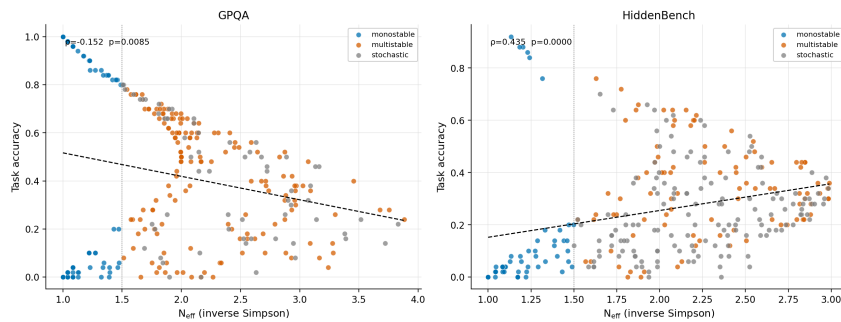


Figure 14: N_{eff} plotted against per-task accuracy for all (task, config) cells, with cells coloured by bistability label. The near-zero correlation confirms that the number of attractors does not predict accuracy.

A.6 Limit Cycles: Detection Funnel and Recurrence Matrices

Table 12 gives the full limit-cycle detection funnel referenced in Section 6.5. Assessed at the repetition level (the unit that is independent under the null, since star spokes driven by a common hub are not independent successes), both the agent-level and system-level flagged counts fall far above the shuffle-null floor, confirming that the flagged cases are genuine periodic structure rather than test artefacts.

Level	Total	Fixed points	Candidates	Flagged	p_{LC} (cond.)
Agent-level	120,000	119,355 (99.5 %)	633 (0.5 %)	47	0.074
System-level	30,000	29,609 (98.7 %)	388 (1.3 %)	32	0.083

Table 12: Limit-cycle detection funnel. *Fixed points*: trailing constant segment $k \geq 3$, excluded from the test. *Candidates*: non-converged sequences with $L \geq 4$. *Flagged*: DET above the 95th percentile of 1,000 shuffle surrogates and dominant period $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$, $p < 0.05$. The fixed-point and candidate counts do not sum exactly to the total because a small residue of sequences are non-converged yet too short to test ($L < 4$): 12 at the agent level ($119,355 + 633 + 12 = 120,000$) and 3 at the system level ($29,609 + 388 + 3 = 30,000$). These are excluded as untestable.

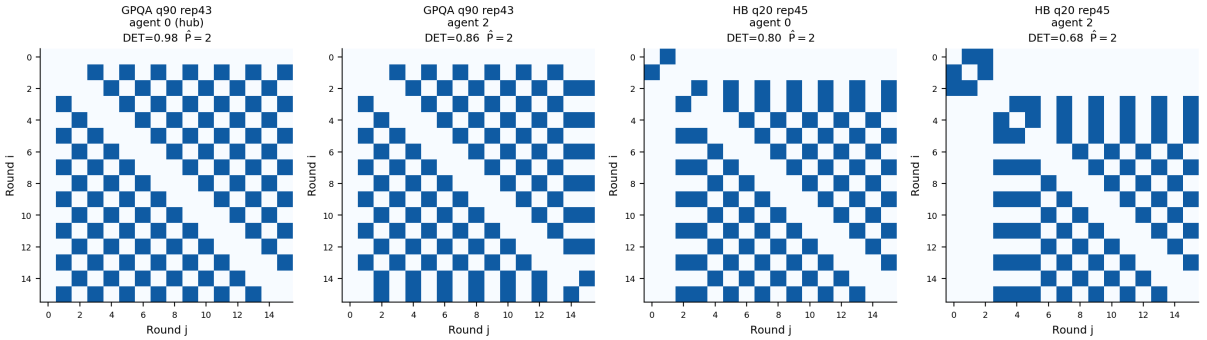


Figure 15: Recurrence matrices for two agents from each of the cases in Figure 5. The alternating checkerboard diagonal stripe pattern (spacing 2) is the characteristic RQA signature of a period-2 orbit. DET (determinism) and dominant period $\hat{P}$ are shown above each matrix.

A.7 High-Repetition Analysis: Additional Figures

This appendix provides the full feature-by-outcome breakdown and correlation matrix for the high-repetition experiment described in Section 6.6 ($R = 1,000$, three GPQA tasks, $W = 2$, fully-connected topology, $N = 4$ agents).

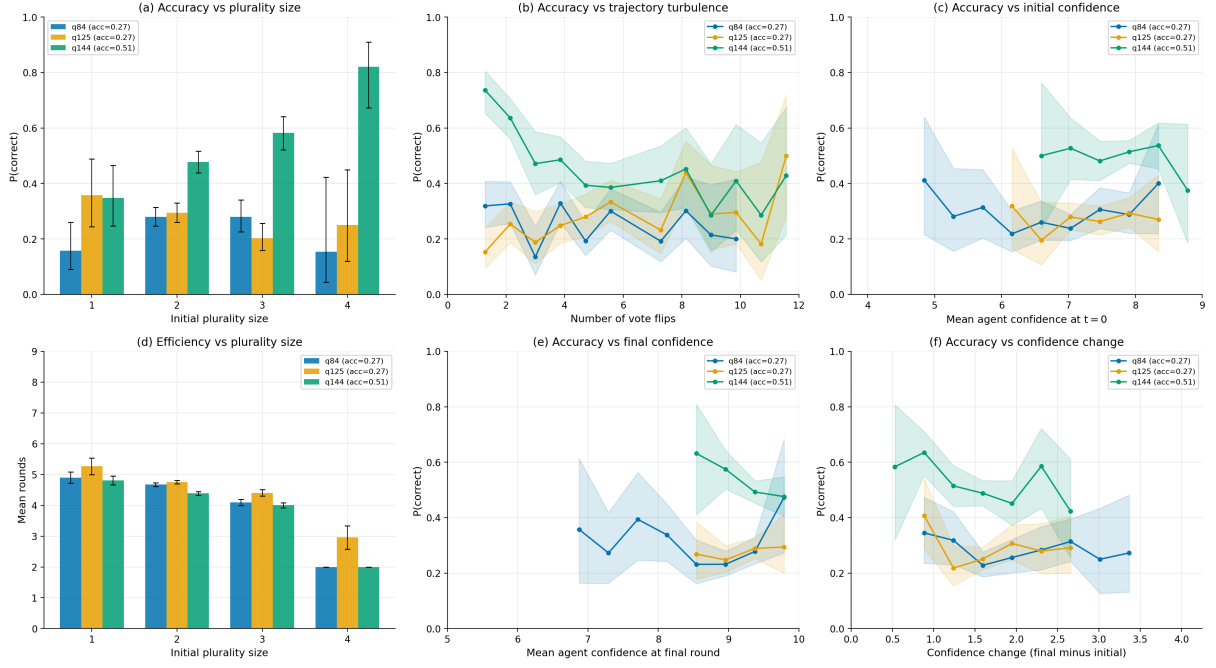


Figure 16: Full breakdown of accuracy and efficiency as a function of six repetition-level features. The top row (a–c) reports accuracy; the bottom row reports efficiency (d) and accuracy (e, f). (a) Accuracy by initial plurality size. (b) Accuracy by vote-flip count (binned, Wilson 95 % CI, $n \geq 10$). (c) Accuracy by mean initial confidence. (d) Efficiency by initial plurality size. (e) Accuracy by mean final confidence. (f) Accuracy by confidence change (final minus initial); dotted line at zero. $R = 1,000$ per task, $W = 2$, fully-connected.

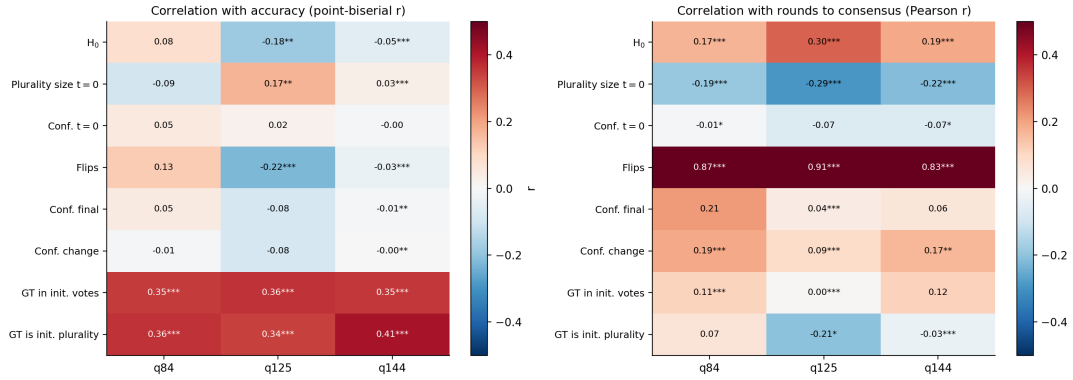


Figure 17: Feature-outcome correlation matrix for the high-repetition experiment. Left: point-biserial correlation r between each feature and binary accuracy (correct/incorrect). Right: Pearson r between each feature and rounds to consensus. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

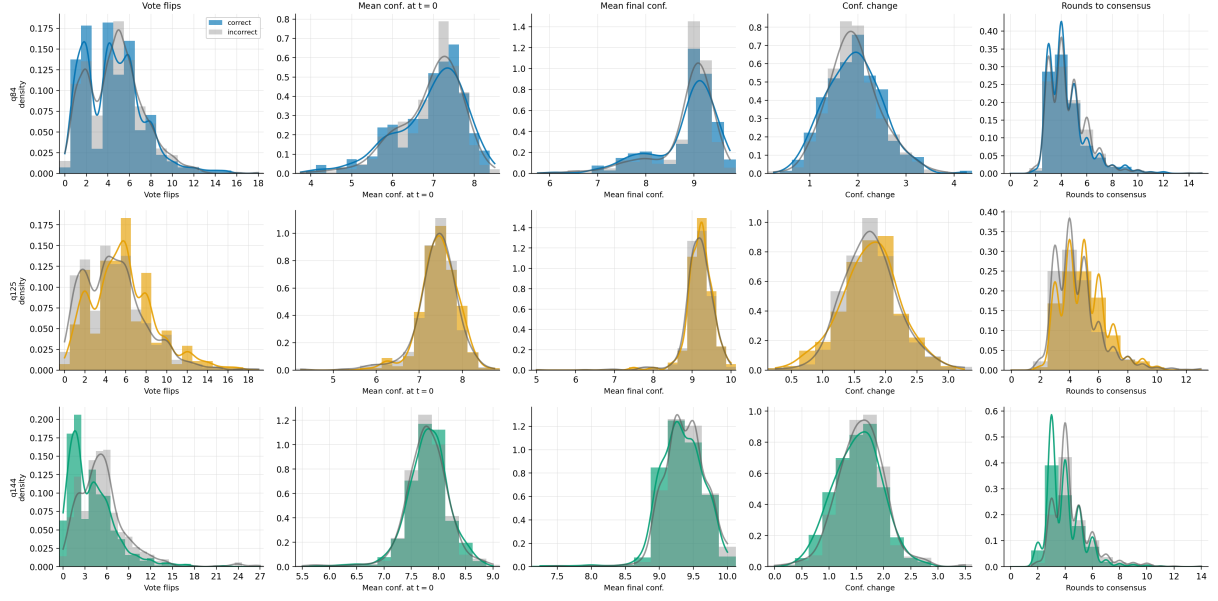


Figure 18: Feature distributions split by outcome (correct: colour; incorrect: grey) for three GPQA tasks, $R = 1,000$ each, $W = 2$, fully-connected. Rows: q84, q125, q144. Columns: vote flips, mean initial confidence, mean final confidence, confidence change (final minus initial), rounds to consensus. Histograms show empirical density; curves are kernel density estimates. Confidence change distributions for correct and incorrect repetitions overlap almost perfectly in all three tasks, indicating that agents gain confidence at the same rate regardless of whether the system reaches the correct answer. Vote flip and round distributions separate in task-dependent directions (see Section 6.6).

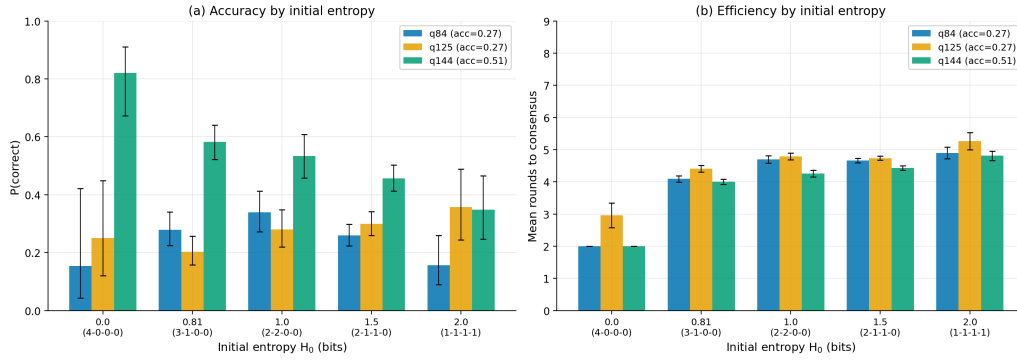


Figure 19: Accuracy (left) and mean rounds to consensus (right) as a function of initial vote entropy H_0 , for the three high-repetition tasks (q84, q125, q144). Accuracy is not monotone in H_0 ; the apparent trend is largely a confound with the probability that the ground truth is present in the initial votes. Entropy bins with fewer than 20 repetitions are omitted.

High-repetition analysis: entropy effects.

A.8 System-Prompt Robustness Check

The agent persona used throughout this thesis includes an explicit anti-conformity instruction. To check whether the motif structure is an artefact of that particular wording, we compare the final dataset against an earlier experiment that used a milder persona, on the same 35 HiddenBench questions and six configurations (GPQA was not part of the earlier experiment, so this check is HiddenBench-only). The two personas differ in one appended sentence: the milder version ends at “Be critical, your neighbours may be wrong, and so may you,” while the version used here adds “Do not change your position simply because others disagree; only update if you encounter a specific argument you cannot counter.”

The earlier experiment used $R = 30$ repetitions and the final one $R = 50$; we therefore pair by question within each configuration and test the paired difference (Wilcoxon signed-rank, Holm-corrected (Holm, 1979) across the six configurations), so the differing repetition counts affect only per-task precision, not the pairing. Table 13 reports the pooled means and the number of the six configurations with a Holm-significant paired shift.

Measure	Milder	Critical	Sig. configs
Accuracy	0.316	0.261	0/6
Mean rounds	6.85	5.76	3/6 (−)
Ceiling-hit fraction	0.146	0.068	3/6 (−)
Convergence rate	0.862	0.940	3/6 (+)
Dominance D	0.375	0.371	0/6
Hub capitulation	0.146	0.157	4/6 (+)
Self-reinf. p_{SR}	0.830	0.844	0/6
Self-reinf. slope	0.372	0.409	1/6 (+)
Limit cycle (agent)	0.019	0.043	0/6
Limit cycle (system)	0.040	0.096	0/6
Bistability N_{eff}	2.04	2.06	1/6 (+)
Bistability Cramér’s V	0.394	0.346	0/6

Table 13: Effect of the anti-conformity system prompt on each motif measure, HiddenBench only. “Milder” is the earlier persona, “critical” the one used throughout this thesis. *Sig. configs*: number of the six (topology, W) configurations showing a Holm-significant paired shift, with the shared sign. The prompt changes convergence dynamics but not accuracy, and does not suppress self-reinforcement.

The pattern is consistent: the more critical prompt changes the *dynamics*, faster fully-connected convergence (rounds down by roughly one to one and a half, ceiling-hits roughly halved) and a higher unanimous-convergence rate, but leaves accuracy unchanged (0/6) and does not reduce self-reinforcement prevalence (0/6). The near-flat N_{eff} (2.04 vs 2.06) and more flagged limit cycles are consistent with more repetitions converging (more unanimous endpoints to count) rather than a reduction in multistability, and Cramér’s V if

anything trends down. The amplifier mechanism therefore survives a deliberate attempt to make the agents less conformist, which is mild evidence that it reflects the interaction structure rather than one prompt. This is a single HiddenBench comparison and is not evidence of robustness across models or group sizes.

A.9 Prompt Templates

For reproducibility we reproduce the core prompt components verbatim. The shared *persona* (with {name} the agent identifier) frames the task and the two-phase protocol and carries the anti-conformity instruction analysed in Appendix A.8:

```
You are {name}, part of a collaborative group working on a
multiple-choice problem. Your goal is to help the group find the
correct answer, agreement should follow from better arguments, not
the other way around.
```

```
Each round (after round 0) has two phases:
```

```
Phase A: output three structured fields, defense (a new concrete
reason your current vote is correct), challenge (dispute a
specific current peer claim, or concede it and name any remaining
disagreement), question (a targeted question to a named peer
referencing a concrete claim). Do not change your vote in this
phase.
```

```
Phase B: you see your neighbors' Phase A outputs. Produce your
updated vote, private reasoning, and public message.
```

```
Be critical, your neighbors may be wrong, and so may you. Do not
change your position simply because others disagree; only update if
you encounter a specific argument you cannot counter.
```

Prompt 1: Shared agent persona (system prompt): the two-phase protocol and the anti-conformity instruction, with {name} the agent identifier.

The three per-step instructions appended to the persona and the running context are:

Round 0: Carefully read the question and options. Produce your initial vote, private reasoning, and public message.

Phase A: Produce your Phase A structured output. Do not change your vote in this phase.

Phase B: Update your belief, private reasoning, and public message. If you change your vote, cite the specific argument, from the current Phase A drafts or from the conversation history, that caused the update.

Prompt 2: Per-step instructions appended to the persona (round 0, *Argue* phase, and *Update* phase), controlling what each agent is asked to produce at each step.

In the *Update* phase each agent additionally receives its own *Argue* draft and its neighbours' *Argue* drafts in randomised order, truncated to the memory window W ; confidence is elicited as an integer from 1 to 10. The complete prompt-assembly code is in the electronic appendix.

A.10 Motifs Beyond the Present Selection

Section 5.1 restricts the study to four motifs. The same structural/dynamical lens admits several others that we leave to future work. On the structural side, in the emergent influence graph:

- **Mutual-reinforcement dyad** (a two-node positive loop): two agents that repeatedly push each other the same way.
- **Influence feed-forward loop**: A sways B , B sways C , and A also sways C directly, predicting delayed or robust lock-in.
- **Cascade / chain**: sequential adoption $A \rightarrow B \rightarrow C$, the structural signature of herding.

On the dynamical side, further regimes of the joint trajectory:

- **Consensus / fixed point**: the baseline regime (a point attractor) against which the others are defined.
- **Polarisation / stable split**: persistent within-repetition disagreement into stable subgroups, distinct from across-repetition bistability.
- **Negative autoregulation / self-correction**: the sign-flipped twin of self-reinforcement, an agent that damps its own confidence.

These span a coherent picture of the regimes a small interacting system can fall into; a study must start somewhere, so we treat them as directions for future investigation.

A.11 Benchmark Survey

This appendix records the full survey of candidate benchmarks considered for motif detection and the reason each was retained or rejected against the four selection criteria of Section 4.3 (discrete shared decision variable, verifiable ground truth, calibrated difficulty, sufficient initial disagreement).

A.11.1 Native Multi-Agent Benchmarks

These were designed to evaluate multi-agent systems but are mostly structurally incompatible with motif detection. They fall into three families.

Family 1: Task-decomposed / role-specialised. Agents are bound to specific sub-tasks rather than deliberating over a shared decision, so there is no shared position space across agents.

- **REALM-Bench** (Geng and Chang, 2025): planning/scheduling problems; agents bound to specific machines, “communication” is notification cascades rather than deliberation.
- **MultiAgentBench / MARBLE** (Zhu et al., 2025): role-specialised agents (e.g. five database agents each diagnosing a different root cause). Its Bargaining and Werewolf sub-environments *do* have shared decision variables and could serve as secondary datasets.
- **PARTNR** (Chang et al., 2024): embodied household tasks with hard-coded heterogeneous capabilities; dyadic by design, ruling out group-level motifs.
- **Collab-Overcooked** (Sun et al., 2025): chef and assistant operate in distinct action spaces.

Family 2: Competition / cooperation without a shared decision. Agents have positions in physical or tactical space rather than a shared decision space, so position metrics cannot be defined directly.

- **BattleAgentBench** (Wang et al., 2024): tank combat; each agent’s state is a continuous physical configuration (x, y, θ, h) , with no shared categorical variable.
- **SPIN-Bench** (Yao, Wang, Hsieh, Zhou, Zou, Cheng, Wang and Viswanath, 2025): planning and board/strategy games. Only Diplomacy is close to deliberation, but its “position” is a spatial supply-centre configuration, and the authors report that coordination quality degrades sharply beyond two to three agents.

Family 3: Social / deliberative interaction. Closer in spirit to our goal, with a mixed picture.

- **SOTOPIA** (Zhou et al., 2024): social scenarios with holistic multi-dimensional scoring, but $N = 2$ by design, which is structurally fatal here: dominance and mutual inhibition require $N \geq 3$, and bistability requires multiple factions.
- **LLM-Deliberation** (Abdelnabi et al., 2024): a genuinely multi-agent ($N = 6$) negotiation environment with public scratchpads and native adversarial variants. Its deal space is too sparse for equality-based agreement metrics and it has no per-agent correctness signal, so it is the strongest candidate for a *future* mixed-motive extension rather than a correctness benchmark here.
- **HiddenBench** (Li et al., 2025): the hidden-profile paradigm (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012) formalised as a benchmark, with validity criteria guaranteeing that information pooling is required. Adopted as our secondary substrate (Section 4.3.2).

Conceptually adjacent (frameworks and reference corpora). Cited but not used as substrates: **AgentVerse** (Chen, Su, Zuo, Yang, Yuan, Chan, Yu, Lu, Hung, Qian, Qin, Cong, Xie, Liu, Sun and Zhou, 2024), a framework rather than a benchmark; and the **DEBATE corpus** (Chuang et al., 2025), a human public/private opinion-dynamics corpus used as a reference point rather than an LLM task environment.

A.11.2 Single-Agent Reasoning Benchmarks Adopted by MAD

The foundational MAD work (Du et al., 2024) established the methodology on GSM8K (Cobbe et al., 2021), MMLU (Hendrycks, Burns, Basart, Zou, Mazeika, Song and Steinhardt, 2021), and similar sets, later extended to MATH (Hendrycks, Burns, Kadavath, Arora, Basart, Tang, Song and Steinhardt, 2021) and StrategyQA (Geva et al., 2021). These are operationally attractive (discrete answers, clean ground truth, cheap, well-precedented) but fail our use case through **saturation**: GSM8K and MMLU are effectively solved by frontier models (Achiam et al., 2023), so the typical trajectory is immediate unanimous agreement with no dynamical signal. Critical reassessments confirm that on saturated benchmarks debate frequently fails to beat single-agent baselines (Zhang et al., 2025, Choi et al., 2025). MATH retains higher difficulty but requires non-trivial answer normalisation and has long solutions that inflate token cost, making it worse on operational tractability. GPQA-Diamond (Rein et al., 2024) is the current frontier instance of the same lineage that still leaves headroom, which is why we adopt it as primary (Section 4.3.1).

A.12 Per-Knob Justification

This appendix expands the classification of tunable components summarised in Table 2, giving the full rationale for each scan, fix, or defer decision.

Memory window and contents (scan). Zargari Marandi (2026) systematically compare within-round, cross-round, and rank-adaptive protocols and find a real tradeoff between convergence speed, peer-referencing, and argument diversity. Wynn et al. (2025) document failure modes in which erroneous contributions persist and propagate across debate rounds, and language models are known to use long contexts unevenly (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024), so the amount and content of retained memory is itself a design choice rather than neutral bookkeeping. The convergence-time control of Du et al. (2024) is functionally memory-driven. We scan $W \in \{1, 2, 5\}$ with $W = 1$ as the near-Markov baseline. A factorial memory $\times$ topology study finds memory window is the dominant main effect on both accuracy and convergence rate, outweighing topology in that design (Mehdizadeh and Hilbert, 2026), which provides further motivation for treating W as a first-class scan axis.

Communication topology (scan). Topology is the canonical lever for network-motif analysis and an established first-class design choice in MAD (Li et al., 2024). Regan et al. (2024) report that hub placement in network topologies substantially shifts performance, and Chen et al. (2023) observe emergent leader-follower structure in directed graphs, a dominance motif under another name. We scan $\{\text{FC}, \text{STAR}\}$.

Number of agents (fix, benchmark-dependent). Du et al. (2024) use $N = 3$ as the canonical baseline; Liang et al. (2024) report only modest degradation up to $N = 4$. The joint answer-profile state space grows as M^N , so $N = 4$ is the smallest value at which the two canonical topologies are structurally distinct while the state space stays tractable for trajectory statistics. $N = 4$ throughout both benchmarks.

Number of rounds (fix at $T = 15$). T behaves as an observation budget rather than a structural perturbation, since trajectories can be cropped post hoc but not extended. Accuracy-level convergence is reported within a few rounds (Li et al., 2024), but motif detection needs longer windows, since limit cycles in particular require $T \geq 2 \cdot \text{period}$ and the period is unknown a priori. We fix $T = 15$ as a hard cap with early stopping $u = 3$ to recover efficiency. Figure 20 shows the efficiency–reliability trade-off across thresholds.

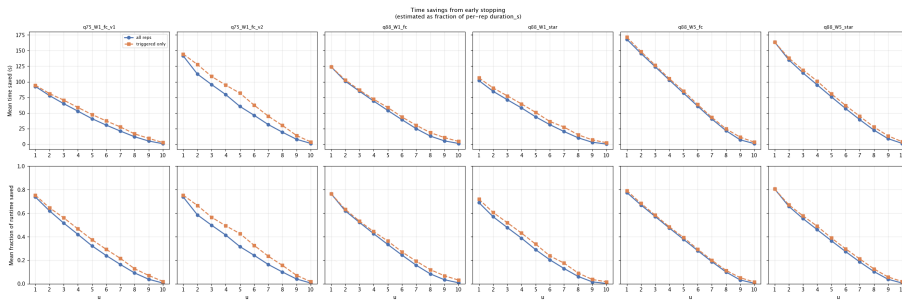


Figure 20: Mean fraction of runtime saved by early stopping as a function of the threshold u . Solid line: all repetitions. Dashed line: only repetitions that triggered stopping. At $u = 3$ the saving is roughly 50 % per repetition.

Output structure (fix, full *Argue/Update*). The structured output is the measurement apparatus, not engineering taste. Private reasoning and confidence versus the public

message enable the public/private dissociation; the *Argue* fields structure the deliberation (the influence graph for dominance detection is built from vote adoptions, not the draft text; Section 5); and the vote-stability rule separates argumentation from belief update so they can be measured independently. Structured outputs are the norm in serious MAD work (Liang et al., 2024, He and Feng, 2025).

Cooperativeness (fix, cooperative). Liang et al. (2024) identify the operational sweet spot: a modest level of tit-for-tat is required, with pure cooperation collapsing into Degeneration-of-Thought and pure adversarial framing preventing convergence. Our defense/challenge/question scaffold implements modest tit-for-tat while *Update* phase belief update leaves room for convergence.

Synchrony (fix, parallel-synchronous). Du et al. (2024) establish synchronous simultaneous-talk as the canonical MAD protocol, also adopted by Liu, Wang, Huang, Xu, Zeng, Jiang, Yang and Li (2024). Synchronous execution with randomised peer-message order also removes agent-order position effects that would otherwise confound dominance detection.

Model selection (fix, homogeneous mid-tier). We fix a single homogeneous model, `mistral-medium-instruct`, to isolate topology and memory effects from capability-asymmetry effects: Wynn et al. (2025) document that models shift from correct to incorrect answers under peer pressure, producing dominance-like dynamics attributable to conformity. A homogeneous baseline ensures any observed dominance is structural rather than capability-driven; heterogeneous setups are reserved as an ablation. This choice is further supported by evidence that mixed-model teams within the same capability tier do not systematically outperform the best homogeneous member and exhibit comparable modal sycophancy rates, suggesting heterogeneity alone does not dissolve the dynamics under study (Bertalaníć and Fortuna, 2026).

Sampling temperature (fix at 0.7, reasoning disabled). Deterministic decoding collapses MAD into voting among identical samples, and Zhu et al. (2026) show the homogeneous deterministic case cannot improve over a single agent. We use temperature 0.7 with reasoning disabled to obtain diverse yet focused outputs without chain-of-thought token overhead. An ablation comparing $T = 0.4$ against $T = 0.7$ in a comparable debate protocol found that the lower temperature increased the modal sycophancy rate by 6.2 percentage points (Bertalaníć and Fortuna, 2026), providing external support for the higher-temperature choice.

Tool access (defer). For GPQA-Diamond, tools would push single-agent accuracy toward the ceiling and remove dynamical headroom, introduce non-reproducibility through web search, and violate the Google-proof design rationale of Rein et al. (2024).

Compute budget. Reliable motif detection requires $R = 30$ – 50 repetitions per configuration: bistability needs enough samples to surface both attractor basins, limit cycles enough trajectories to confirm period- ≥ 2 structure, and dominance estimators statistical power. At ≈ 25 s per repetition, the naive cost of the full grid on all 198 GPQA-Diamond tasks would be $198 \times 6 \times R \times 25$ s, i.e. ≈ 248 h (≈ 10.3 days) at $R = 30$ and ≈ 413 h (≈ 17.2 days) at $R = 50$. The two-phase difficulty-based procedure of Section 4.4 reduces

this to ≈ 29 h by running the full grid only on promoted Tier-3 tasks, while the screening pass still yields prevalence estimates across the broader task set.

B Electronic appendix

The complete source code, including all scripts required to reproduce the figures and results, is publicly available in the following GitHub repository:

<https://github.com/echtermeyer/LLM-Based-MAS>

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AI Tools

I hereby confirm that I have written all parts of this work myself. The following AI tools were used during the creation of this thesis and the presentations. All AI-generated suggestions and outputs were critically reviewed, tested where applicable, and independently revised by me. No AI tool was used for the main analysis or interpretation of experimental results.

- **GitHub Copilot** (Inline Completion): Used for inline code completion suggestions during development. All outputs were reviewed and modified as appropriate.
- **Anthropic Claude Sonnet 5** (AI as Tutor): Used to explore and understand core concepts in complex-systems theory and multi-agent debate.
- **Anthropic Claude Opus 5** (Performance and Resource Optimization): Used to parallelize the evaluation pipeline for the experiments described in Section 4 and reported in Section 6. The generated code was reviewed and tested carefully.
- **Anthropic Claude Opus 5** (Improve Writing Style, Paraphrasing and Reformulation): Used for proofreading, linguistic refinement, and paraphrasing. Suggestions were not adopted verbatim but were manually reviewed, adapted, and reformulated.

All AI outputs and suggestions were independently evaluated and selectively adopted after critical review.

München, 30.09.2026

Eric Echtermeyer